

AN ANALYTIC APPROACH TO
ORNSTEIN-UHLENBECK PROCESSES WITH
FLUCTUATING PARAMETERS
AND
APPLICATIONS IN THE MODELING OF
FIXED INCOME SECURITIES

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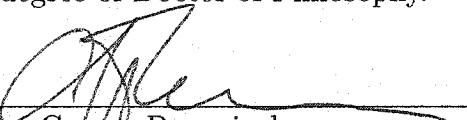
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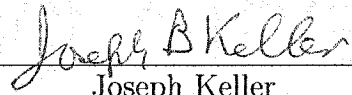
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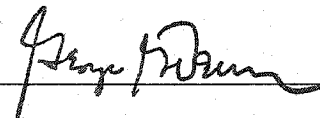
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Abstract

We generalize the Vasicek model for bond prices, in which the instantaneous rate of interest follows an Ornstein-Uhlenbeck process with positive mean, to accommodate randomly fluctuating mean and volatility. Fluctuations are introduced by allowing the Ornstein-Uhlenbeck parameters governing the long run mean and instantaneous variance of this primary interest rate process to depend on a secondary mean reverting stochastic process. The time scale over which oscillations in the secondary process occur is assumed to be short compared with the longer mean reversion time of the primary process and this separation of time scales is exploited in our asymptotic approach to the solution of certain partial differential equations relevant to pricing and estimation.

Both continuous and discretely valued secondary stochastic processes are treated. The former gives rise to a family of stochastic volatility models for which we supply explicit formulas correcting the original Vasicek model - the corrections being parsimonious and robust to the exact specification of volatility process. The latter finds interpretation in the regime switching literature and here too, for the first time in this context, accurate closed form bond prices are derived. The domain of applicability of our results, bolstered by numerical tests of the pricing formulas, is discussed with particular emphasis placed on the modeling of default intensities and credit derivatives. We supply evidence for bimodality and predictability in US corporate spreads consistent with regime switches, and evidence of rapid fluctuations in US short term interest rates consistent with our treatment of stochastic volatility.

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Chapter 1

Main Results and Applications

In Vasicek's seminal paper "An Equilibrium characterization of the term structure" 1977 [57], a stochastic evolution of the instantaneous rate of interest arises endogenously in an economy where investors share a common plausible preference for certainty over risk. This **short rate** follows an Ornstein-Uhlenbeck process

$$dx_t = \kappa (\theta - x_t) ds + \sigma dW_t. \quad (1.1)$$

with positive mean. Vasicek calculated quantities essential for the modeling of bonds and related securities, notably the expectation of the exponentiated negative integrated x_t process:

$$B(t, x; T) = \mathbb{E}^{x,t} \left[e^{-\int_t^T x_s ds} \right], \quad (1.2)$$

whose analytic solution implies a **term structure of interest rates**, or equivalently, prices for **discount bonds** (those paying no coupons) of all maturities. Given the mathematical similarity between such "discounting" and randomly killed diffusions it comes as no surprise that (1.2) is also useful in the modeling of financial products sensitive to credit events (such as the sudden default on obligations by the issuer of a corporate bond).

The primary focus of our work is a calculation of (1.2) performed in more general settings where the process x_t is similar to that used by Vasicek, but accommodates

randomly fluctuating parameters θ and σ . We model fluctuations by introducing a process x_t satisfying

$$dx_t = \kappa(\theta(y_t) - x_t)ds + \sigma(y_t)dW_t \quad (1.3)$$

where the parameters $\theta(y_t)$ and $\sigma(y_t)$ are themselves stochastic and depend on a second process y_t . Our work branches at this point according to whether the y_t process is continuous or discretely valued.

In the first half of the thesis we treat a class of processes where y_t is Ornstein-Uhlenbeck and correlated with x_t . The joint process is described:

$$\begin{aligned} dx_t &= \kappa(\theta - x_t)dt + f(y_t)dW_t \\ dy_t &= \alpha(m - y_t)dt + \beta \left(\rho dW_t + \sqrt{1 - \rho^2} dZ_t \right), \end{aligned}$$

We permit the diffusion parameter $\sigma(y_t) = f(y_t)$ to take almost any form, yielding a broad class of **stochastic volatility** term structure models.

In the latter half of our work we examine a process

$$dx_t = \kappa(\theta(y_t) - x_t)ds + \sigma(y_t)dW_t$$

where y_t is a Markov process taking two values with generator:

$$\lambda Q = -\lambda \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

The y_t process switches between the two states with instantaneous probability proportional to λ . The immediate application is a **regime switching** short rate model (generalizing Hansen and Poulsen, 2000 [37]) for which we supply accurate analytic bond pricing formulas. We also find empirical evidence supporting the use of regime switching models in credit derivative applications, and in this context illustrate a genuine practical advantage of analytic formulas over numerical methods.

1.1 Exploiting Differences in Time Scales

Our approach follows Sircar, Fouque, Papanicolaou [31], [28], [33], [32], [12], [29], [50] closely in chapter (2) and in spirit throughout. When we describe the parameters $\theta(y_t)$ and $\sigma(y_t)$ as *fluctuating* we envisage a functional of a stable *rapidly oscillating* ergodic process y_t . In the case where y_t is also an Ornstein-Uhlenbeck process this literally means that the mean reversion parameter α is large *compared with the mean reversion parameter κ of the x_t process*. In the case where y_t is a two state process we envisage state switching occurring relatively rapidly ($1/\lambda$ is small) *compared with the typical mean reversion speed (proportional to $1/\kappa$) of the x_t process*. These notions are not precise, and are complicated by the fact that we are interested in expectations of processes integrated over finite time horizons: a third time scale.

1.1.1 Limitations

As stated, the parameter restrictions, $1/\alpha \ll 1/\kappa$ and $1/\lambda \ll 1/\kappa$, may appear to represent, a priori, a rather harsh limitation on the scope of our results and the ability of the process (1.3) to match empirical time series. However, whilst these assumptions *do* restrict the scope of application, the limitations turn out to be far less severe than our mathematical ansatz would suggest. Of course, where the approach breaks down, existing well-developed methods of modeling should be used instead. For example, we may downgrade the primary process x_t and upgrade the secondary process y_t in keeping with the philosophy of a more egalitarian multifactor model.

Therefore, we introduce the separation of time scales as a very deliberate modeling and pricing technique. We are well aware that the burden of demonstrating a significant domain of applicability within finance lies squarely on our shoulders and this is particularly true for the family of stochastic volatility models we introduce. In this stochastic volatility context our corrections capture the first order effects of introducing fluctuations in volatility (or more precisely $\sigma(y_t)$) which are correlated with movements in the primary process x_t . To demonstrate the relevance of the rapid fluctuation analysis use variogram techniques in section 2.3.2 to demonstrate, in a

very direct way, that there is indeed a fast mean reverting component in the volatility of time series proxying the short rate. This work was assisted by Knut Solna and draws on techniques in Fouque, Papanicolaou, Sircar, Solna [33].

In contrast to fast mean reverting volatility, regime switches in financial time series (in particular interest rates) have attracted considerable attention (for example [36],[5], [4], [48], [37]) and represent a well established modeling tool. In section 3.5 we add to this literature by examining a data set consisting of thousands of time series of US corporate spreads. We find new indirect evidence for regime switches using tests for multimodality seldom applied in the financial literature. We do not claim that these regime switches occur rapidly in nature.

The possibility of rapid regime switches in nature or as a modeling device (discussion in section 3.1) is interesting, but, as we now explain, by no means crucial to establishing a broad domain of application for the results. The fluctuations appearing in the regime switching process are treated somewhat differently from the fluctuations in the stochastic volatility model. We are able to push the bond price formulas we derive well beyond the limitations implied by the assumption $1/\lambda \ll 1/\kappa$ (corresponding to relatively fast regime switches) and show that the rapid oscillation asymptotics are useful even when $1/\lambda \approx 1/\kappa$ (corresponding to moderately frequent regime switches). For example, in section 3.4.5 we test numerically a model where the mean short rate θ switches dramatically between five per cent and fifteen percent, the diffusive parameter is also stochastic and a large short rate mean reversion parameter $\kappa = 2$ is used in contrast to a relatively small regime switching intensity $\lambda = 0.5$. The regime switches occur every two years on average, *the same or slightly longer* time scale than the typical mean reversion time of the short rate process. Our analytic results stand up quite well nonetheless, accurately estimating two key quantities. The first quantity is the difference between the simulated bond price one obtains when starting y_t in the high interest rate regime and the bond price one obtains when starting y_t in the low interest regime. The second is the regime independent correction to bond prices accounting for the error one would make if one used the standard Vasicek model, an average θ of ten percent and a root mean square average σ .

1.1.2 Advantages

To summarize, our enthusiasm for the asymptotic techniques exploiting differences of time scales comes from two sources. The first source is direct empirical evidence for rapid mean reversion in short rate volatility supplied in section 2.3.2. The second is the success of a collection of numerical tests of the regime switching model demonstrating a broader domain of applicability than might be initially supposed.

The assumed separation of scales enables us to peer into the pricing partial differential equations and solve them (or approximately solve them) in an asymptotic sense. In particular we are able to derive what we believe to be the first analytic formula for bond prices in the regime switching literature. The representation for the bond price given in section 3.4.3 isolates the effect of regime switches into two informative components. The “symmetric” adjustment to bond prices accounts for the effect of fluctuations of the parameters about their means and is *regime independent*. The “antisymmetric” correction to bond prices can be interpreted as a regime delta, or the jump in bond prices which occurs as we move from one regime to another.

The ability to isolate the effects of fluctuations in x_t independent of the initial value of y_t is a key advantage of the asymptotic approach and suggests methods for stable pricing and estimation. In chapter 2, we illustrate that the first order effects of stochastic volatility can be captured in a parsimonious manner and robust to the exact specification of both the y_t process and the functional relationship between y_t and the volatility of x_t . The advantage of the asymptotic approach is not so much that we can approximately solve the PDE analytically - which is in itself interesting - but that we can, as far as first order effects are concerned, eliminate the unobservable y_t process from direct consideration in pricing and estimation. Stable and rapid estimation of the primary process x_t , avoiding the need to estimate jointly the observable x_t and unobservable y_t , is a promising possibility for which we provide tools in section 2.2.7.

In this manner a two factor model (x_t, y_t) can be considered to be a one factor model x_t corrected for the effect of y_t fluctuations, perhaps the closest thing in pricing and estimation with diffusion processes to a free lunch. Put another way, we demonstrate a class of one dimensional diffusions which are, in the context of pricing and estimation of fixed income securities, tractable enough to complement candidates

such as Ornstein-Uhlenbeck, CIR and more generally affine square root processes in an “almost affine setting” for which there is much existing technology.

1.2 Theoretical Results

Our pricing formulas build on those for the original Vasicek model, discussed in more detail in appendix A. Equation (1.2)

$$B_{vas}(t, x; T) = \mathbb{E}^{x,t} \left[e^{-\int_t^T x_s ds} \right],$$

with an appropriate interpretation of parameters, represents the price of a bond and has analytic solution

$$B_{vas}(x; t, T) = \exp \left\{ - \left[\tilde{\theta}(T - t) - (\tilde{\theta} - x) \frac{1 - e^{-\kappa(T-t)}}{\kappa} + \frac{\sigma^2}{4\kappa^3} (1 - e^{-\kappa(T-t)})^2 \right] \right\}, \quad (1.4)$$

where $\tilde{\theta} = \theta - \sigma^2/2\kappa^2$. The bond price is the exponential of a constant plus linear function of the **state variable** x_t , placing it in the **affine** term structure models as first classified by Brown and Schwartz [10], developed by Duffie, Kan, Singleton and others (including [19],[20]) and recently characterized cleanly by Filipovic¹. Bond option prices can also be computed analytically, as in appendix A.

Our bond pricing results for Vasicek models with fluctuating parameters take the form of asymptotic series in a parameter which corresponds to the rapidity of fluctuations of the secondary process y_t about its mean. The first term in the expansion has the same functional form as the bond price solution for the original Vasicek model. This term represents a somewhat naive attempt to approximate the bond price by averaging y_t -dependent parameters over the invariant distribution of the secondary process y_t .

We treat higher order terms in the bond price asymptotic series as corrections to the standard constant parameter Vasicek model. These corrections arise from a

¹Paper shortly to be released: <http://www.math.ethz.ch/filipo/>

variety of complex interactions and convexity effects, many of which are ignored by averaging θ and σ^2 .

1.2.1 Processes with Fluctuating Volatility, General Results

In chapter 2 we assume a separation of time scales in the pricing PDE

$$\begin{aligned} \frac{\partial B}{\partial t} + \frac{1}{2}f(y)^2 \frac{\partial^2 B}{\partial x^2} + (\kappa(x_\infty - x) - \lambda_1(y)f(y)) \frac{\partial B}{\partial x} - xB \\ + \beta \rho f(y) \frac{\partial^2 B}{\partial x \partial y} + \frac{1}{2}\beta^2 \frac{\partial^2 B}{\partial y^2} \\ + \left(\alpha(m - y) - \beta \left[\rho \lambda_1(y) - \lambda_2(y) \sqrt{1 - \rho^2} \right] \right) \frac{\partial B}{\partial y} = 0, \end{aligned}$$

for the price at time t of a T maturity discount bond $B(t, x, y; T)$ when $x_t = x$ and $y_t = y$. Namely, we assume $\epsilon := 1/\alpha$ is small. The pricing PDE is well understood and, with different boundary conditions, applies to securities other than bonds. We sketch a derivation of the bond pricing PDE in appendix B.2, including a plausibility argument for the appearance of functions $\lambda_1(y_t)$ and $\lambda_2(y_t)$ representing market prices of risk.

Bond Pricing

We show that in the rapid oscillation limit (as $\epsilon \rightarrow 0$ and fluctuations in the secondary process y_t become feverish) bond prices $B(t, x; T)$ take the same form as (A.8),

$$B(t, x; T) \rightarrow B_{vas}(t, x; T; \kappa; \bar{\theta}, \sqrt{\sigma^2})$$

with averaged parameters

$$\begin{aligned} \overline{\sigma^2} &= \langle f^2(y) \rangle_{OU} \\ \bar{\theta} &= x_\infty - \frac{1}{\kappa} \langle \lambda_1(y) f(y) \rangle_{OU} \end{aligned}$$

where the notation $\langle \cdot \rangle_{OU}$ indicates an average with respect to the invariant distribution of the y_t process. For example

$$\langle f^2(y) \rangle_{OU} = \frac{1}{\sqrt{2\pi\nu^2}} \int_{-\infty}^{\infty} f^2(y) e^{-y^2/2\nu} dy$$

where $\nu^2 = \beta^2/2\alpha$. This limiting formula is a plausible approximation to the bond price but fails to take into account the interplay between volatility uncertainty, short rate uncertainty, correlation between the short rate and volatility, and market premia demanded for the taking of associated risks. The real effects of fluctuating volatility are considerably more complicated than those predicted by naive averaging.

Exploiting the assumption that volatility reverts rapidly to its mean we derive explicit formulas which capture those first order effects of stochastic volatility that the “averaging” limiting case misses. We show that in fact the bond price is better approximated by multiplying the fast oscillation limit $B_{vas}(t, x, T; \kappa, \theta, \sigma)$ by a maturity dependent correction factor:

$$B(t, x, y, T) \approx B_{vas}(t, x, T; \kappa, \theta, \sigma) (1 + \epsilon V(T - t))$$

where

$$\begin{aligned} V(u) &= \frac{V_3}{\kappa^3} \left(u - \tau(u; \kappa) - \frac{1}{2} \kappa \tau(u; \kappa)^2 - \frac{1}{3} \kappa^2 \tau(u; \kappa)^3 \right) \\ &\quad - \frac{V_2}{\kappa^2} \left(u - \tau(u; \kappa) - \frac{1}{2} \kappa \tau(u; \kappa)^2 \right) + \frac{V_1}{\kappa} (u - \tau(u; \kappa)) \\ \tau(u; \kappa) &= \frac{1 - e^{-\kappa u}}{\kappa}, \end{aligned}$$

and the constants V_1 , V_2 , and V_3 are small. We supply formulas for calculating these numbers when the function $f(y)$ is known.

Bond Option Pricing

Similar results are obtained for the rapid fluctuation limit when we analyze bond option prices. The correction to account for stochastic volatility for a European call

on a discount bond can be expressed

$$C(t, x; T, T_0) = C_{vas}(t, x; T; T_0) + V(T - T_0)B_{vas}(t, x; T)N(h_1) - \int_t^{T_0} \mathbb{E} \left[e^{-\int_t^u \bar{x}_s ds} \mathcal{A}C_{vas}(u, \bar{x}_u) | \bar{x}_t = x \right] du,$$

where all notation carries over from section 1.2.1 and the operator $\mathcal{A}$ can be explicitly calculated once the function $f(y_t)$ is known. Here $C_{vas}(t, x; T, T_0)$ is the price of a European call on a discount bond as calculated using the constant parameter Vasicek model. The bond matures at time T . The owner of the option may call at time T_0 .

1.2.2 Results for a Particular Stochastic Volatility Model

We apply the theory to a specific model with risk neutral short rate x_t and volatility driving process y_t obeying

$$\begin{aligned} dx_t &= \left(\kappa(x_\infty - x_t) - \tilde{\lambda}_1 e^{y_t} \right) dt + e^{y_t} dW_t \\ dy_t &= \left(\frac{1}{\tilde{\epsilon}}(m - y_t) - \tilde{\lambda}_1 \frac{\nu}{2} \rho - \tilde{\lambda}_2 \frac{\nu}{2} \sqrt{1 - \rho^2} \right) dt + \frac{\nu}{\sqrt{2\tilde{\epsilon}}} \left(\rho dW_t + \sqrt{1 - \rho^2} dZ_t \right) \end{aligned}$$

The parameters with tildes arise from a specification of risk preferences and preempt a transformation of parameters in section 2.2.5 which cleans up the analysis somewhat. The choice of the exponential function, taking the role of $f(y_t)$ in the general theory, is made for tractability and to exaggerate excursions of the y_t process. The scale separation is characterized by the size of $\tilde{\epsilon}$, assumed small. We call this the exponential Ornstein-Uhlenbeck stochastic volatility (**ExpOU SV**) model. The bond price correction can now be given very explicitly because the averaged parameters

$$\begin{aligned} \sigma &= e^{m+\nu^2} \\ \theta &= x_\infty + \frac{\sigma \tilde{\lambda}_1}{\kappa} e^{-\nu^2/2} \end{aligned}$$

are easily calculated from integrals with respect to the (gaussian) invariant distribution of y_t . The group parameters

$$\begin{aligned}\tilde{V}_1 &= -\frac{\tilde{\lambda}_1}{\sqrt{2}} \left(\rho \tilde{\lambda}_1 + \sqrt{1 - \rho^2} \tilde{\lambda}_2 \right) e^{m + \frac{1}{2}\nu^2} \\ \tilde{V}_2 &= -\frac{1}{\sqrt{2}} \left(\rho \tilde{\lambda}_1 + \sqrt{1 - \rho^2} \tilde{\lambda}_2 \right) \sigma^2 - \frac{e^{3m} \rho \tilde{\lambda}_1 \sqrt{2}}{\nu} \left(e^{\nu^2} - e^{\frac{1}{2}\nu^2} \right) \\ \tilde{V}_3 &= -\frac{e^{3m} \rho}{\sqrt{2}} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right).\end{aligned}$$

are also available in closed form and enable us to write down the maturity dependent function

$$\tilde{V}(t) = \tilde{V}^{(1)} \int_0^t \tau(s; \kappa) ds - \tilde{V}^{(2)} \int_0^t \tau^2(s; \kappa) ds + \tilde{V}^{(3)} \int_0^t \tau^3(s; \kappa) ds$$

as an analytic formula. Again $\tau(s; \kappa) = \frac{1 - e^{-\kappa t}}{\kappa}$ and the integrals $\int_0^t \tau^n(s; \kappa) ds$ present no difficulty. The corrected bond price

$$B(t, x, T) = \left(1 - \sqrt{\epsilon} \nu \tilde{V}(T - t) \right) B_{vas}(x, t, T; \kappa, \theta, \sigma).$$

is thus obtained.

Comparison with Jump Diffusions

In section 2.2.6 we illustrate one use of the new analytic bond prices. We add exponentially distributed jumps to the Exponential Ornstein-Uhlenbeck Stochastic Volatility model. We hypothesize that excursions in y_t , exaggerated by the exponential function and governing the short time horizon variance of x_t , would have similar effects on pricing as the addition of small jumps to the x_t process. A clean expansion of the yield

$$y(t, x; T) = -\frac{\log B(t, x; T)}{T - t}$$

on discount bonds maturing at time T can be achieved using the convenient function $\tau(u; \kappa) = \frac{1 - e^{-\kappa u}}{\kappa}$:

$$y(t, x; T) \approx y_0 + y_1 \frac{\tau(T-t)}{T-t} + y_2 \frac{\tau^2(T-t)}{T-t} + y_3 \frac{\tau^3(T-t)}{T-t} + y_4 \frac{\tau^4(T-t)}{T-t}.$$

For small maturities $\tau(T-t; \kappa) \approx T-t$. For large $T-t$, $\tau(T-t; \kappa) \rightarrow 1/\kappa$. The expansion becomes less informative as $T-t$ tends to infinity, but for maturities less than several multiples of the characteristic mean reversion time $1/\kappa$ the series illuminates the contributions from jumps and stochastic volatility. The coefficients y_k can be written in terms of transformed parameters

$$\Omega = \{\kappa, \theta, \sigma, \xi, \lambda_1, \lambda_2, \epsilon, \nu, \mu\}$$

where ν, μ parameterize jumps and $\kappa, \theta, \sigma, \xi, \lambda_1, \lambda_2$ are related to the original model parameters for the process (1.5) in section 2.2.5. The coefficients in the yield expansion are

$$\begin{aligned} y_0 &= \frac{\kappa \theta - \epsilon \lambda_1 (\lambda_1 + \lambda_2) \sigma}{\kappa} - (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-2} \\ &\quad - \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-3} + \frac{\eta}{1 + \kappa} \\ y_1 &= -\frac{\kappa \theta - \epsilon \lambda_1 (\lambda_1 + \lambda_2) \sigma}{\kappa} + (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-2} \\ &\quad + \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-3} - \frac{\eta \kappa \mu}{\kappa + \mu} \\ y_2 &= 2 (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-1} \\ &\quad + 1/2 \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-2} + \frac{\eta \kappa \mu^2}{2 \kappa + 2 \mu} \\ y_3 &= 1/3 \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-1} - 1/3 \frac{\eta \kappa \mu^3}{\kappa + \mu} \\ y_4 &= 1/4 \frac{\eta \kappa \mu^4}{\kappa + \mu} \end{aligned}$$

Though not terribly intuitive, these expressions indicate, under the assumption of small jumps (small μ), to what degree jumps and bursts of volatility impact the yield

curve and moreover to what degree they have similar or overlapping effects. The stochastic volatility corrections and jumps modify the first four coefficients, with the coefficient y_4 being of order μ^3 at most. Put another way, there is significantly more freedom in the parameters when we use *both* small jumps and bursty volatility than we really need. If stochastic volatility corrections are used then the marginal increase in yield curve flexibility obtained by adding small jumps of mean size μ is only of order μ^3 . If stochastic volatility corrections are not used then the marginal increase in yield curve flexibility obtained by adding small jumps of mean size μ is of order μ^2 .

Likelihood Estimation

In section 2.2.7 we consider the transition probability $\tilde{p}(z_1, y_1, t_1; z_2, y_2, t_2)$ for the process

$$\begin{aligned} dz_t^\varepsilon &= -\kappa z_t^\varepsilon dt + \sigma e^{y_t^\varepsilon} dZ_t \\ dy_t^\varepsilon &= -\frac{1}{\varepsilon} y_t^\varepsilon dt + \sqrt{2} \frac{\nu}{\sqrt{\varepsilon}} \left(\rho dZ_t + \sqrt{1 - \rho^2} dB_t \right) \end{aligned}$$

This is desired for likelihood estimation of the ExpOU SV short rate model. Fixing terminal values z_2, y_2, t_2 , reversing time and writing $p(z, y, t) = p(z, y, 0; z_2, y_2, t)$ we examine the backward Kolmogorov equation

$$\left(\frac{\partial}{\partial t} + \mathcal{L}^\varepsilon \right) p(z, y, t) = \left\{ \frac{\partial}{\partial t} + \frac{1}{\varepsilon} Q + \frac{1}{\sqrt{\varepsilon}} \mathcal{L}_1 + \mathcal{L}_2 \right\} p(z, y, t) = 0 \quad (1.5)$$

where

$$\begin{aligned} Q &= \frac{1}{2} 2\nu^2 \frac{\partial^2}{\partial y^2} - y \frac{\partial}{\partial y}, \\ \mathcal{L}_1 &= \sqrt{2} \nu \rho e^{kt+y} \frac{\partial^2}{\partial y \partial z} \\ \mathcal{L}_2 &= \frac{1}{2} e^{2kt+2y} \frac{\partial^2}{\partial z^2}. \end{aligned}$$

We consider the marginal $\bar{p}(z_1, y_1, t_1; z_2, t_2)$ where y_2 is integrated out. In other words, we wish to find the probability density of terminal x_t states given initial knowledge of both x_t and y_t . We show that under the assumption of fast mean reversion this transition probability is, up to first order in ϵ , independent of the initial state of the y_t process. This is an intuitive result. The initial value of y_t is quickly forgotten amidst the rapid fluctuations and therefore has little impact on the probability of arriving at a given x_t value.

Two terms in an asymptotic approximation of the transition density are derived. The first term takes on the same simple gaussian form as the well known transition probability for an Ornstein-Uhlenbeck process with constant parameters. This is easily understood. The rapid fluctuations of y_t are in the limit $\epsilon \rightarrow 0$ a device for replacing a constant diffusion parameter σ with one drawn each instant from the unconditional distribution of $\sigma(y_t)$. However this is equal in law to an analogous Ornstein-Uhlenbeck process with constant σ equal to the square root of the average of the squared diffusion parameter $\sigma^2(y_t) = e^{-2y_t}$ over the unconditional distribution of y_t .

The second term in the expansion takes into account effects missed by this averaging trick. Correlation between the x_t and y_t processes becomes important when ϵ is small (but not infinitesimal). We show that the correction required can be deduced from the solution $p_1(z, t)$ to a nonhomogeneous PDE

$$\left\{ \frac{\partial}{\partial t} + \langle \mathcal{L}_2 \rangle_{OU} \right\} p_1(z, t) = -a e^{3kt} \frac{\partial^3}{\partial z^3} p_0(z, t). \quad (1.6)$$

where $\langle \mathcal{L}_2 \rangle_{OU}$ is a y_t averaged operator

$$\langle \mathcal{L}_2 \rangle_{OU} = \frac{1}{2} e^{2kt} \langle e^{2y} \rangle_{OU} \frac{\partial^2}{\partial z^2}$$

and

$$p_0(z, t) = \frac{1}{\sqrt{2\pi \frac{1-e^{-2\kappa t}}{\kappa} \bar{\sigma}^2}} \exp \left(\frac{-z^2}{2 \frac{1-e^{-2\kappa t}}{\kappa} \bar{\sigma}^2} \right).$$

We show that equation (1.6) can be solved by Duhamel's principle and the expression

for $p_1(z, t)$ reduced to a one dimensional quadrature. In this way, the likelihood function for the original ExpOU SV x_t process can be rapidly approximated. We hope to illustrate how this may be used in future work.

1.2.3 Processes with Regime Switching Mean and Volatility

In the latter half of the thesis we move from a continuous y_t process to one with discrete jumps, aligning us with some regime switching literature. Here both θ and σ are allowed to depend on y_t which follows a two state Markov process with Poisson switching times independent of the x_t process. Models in this genre have been treated in the literature and perform well in empirical tests such as those carried out by Ang and Bekaert [4] and Basal and Zhou [5]. Our contribution is to demonstrate, under some restrictive conditions, the existence of accurate analytic approximations to the bond price, and to isolate the terms which depend on knowledge of the initial regime.

These formulas apply only to a certain simple regime switching models, and not to the full variety in the literature. However, the short rate process we consider does generalize the one considered by Hansen and Poulsen 2000 [37]. We assume a risk neutral short rate process

$$dx_t = \kappa (\theta(y) - x) dt + \sigma(y) dW.$$

whose qualitative features are discussed in section 3.1. Bimodal unconditional distributions, predictability, trending and jump like phenomena can be accommodated depending on the choice of parameters. We also illustrate in 3.1.4 how the likelihood of negative x_t values can, if so desired, be severely reduced without unduly curtailing volatility. This may be interpreted as a delayed response to an objection which has haunted the Vasicek model since its publication. The problem of negative interest rates (and negative spreads) is tempered by the introduction of parameter fluctuations which *increase* parameter uncertainty - a surprising turn of events perhaps but one whose mystery evaporates rather rapidly upon inspection.

Bond Pricing

The calculation of

$$u(t, x) = \mathbb{E}_0 \left[\exp \left(- \int_0^t x_s ds \right) \right],$$

the exponentiated negative integrated regime switching x_t process under the law of the same is again the focus of our attention in chapter 3, although we have applications other than bond pricing in mind. Here t represents time to maturity. We wish to solve the partial differential equation

$$-\frac{\partial}{\partial t}u + \kappa(\theta(y) - x)\frac{\partial}{\partial x}u + \frac{1}{2}\sigma^2(y)\frac{\partial^2}{\partial x^2}u + \beta xu + \lambda Qu = 0 \quad (1.7)$$

with initial values $u(0, x) = 1$. Here λ parameterizes the arrival intensity of regime switches through the generator

$$\lambda Q = -\lambda \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}. \quad (1.8)$$

of the y_t process. We derive an asymptotic solution in powers of $\epsilon := 1/\lambda$ and demonstrate the surprising fact that *all terms in the asymptotic solution for the bond price of all orders can be explicitly calculated.*

Balancing and Solubility Conditions

We examine requirements (arising from solubility of Poisson equations and averaging considerations) placed on terms $u_k(x, y, t)$ under the assumption that the expansion

$$u^\epsilon(x, y, t) = u_0(x, y, t) + \epsilon u_1(x, y, t) + \epsilon^2 u_2(x, y, t) + \dots \quad (1.9)$$

solves the PDE (1.7) in an asymptotic sense. As y_t takes only two values, it is convenient to represent the functions $u_k(x, y, t)$ as vectors

$$u_k(x, y, t) = \begin{bmatrix} u_{k1}(x, t) \\ u_{k2}(x, y) \end{bmatrix}.$$

Viewed this way, the matching of coefficients of ϵ in the expansion (1.9) substituted in the the PDE (1.7) yields a pair of equations for each order in k . For a general two state Markov process y_k with generator Q , these pairs of equations

$$\begin{aligned} 0 = & -\frac{\partial}{\partial t}u_{k,1} + \kappa(\theta_1 - x)\frac{\partial}{\partial x}u_{k,1} + \frac{1}{2}\sigma_1^2\frac{\partial^2}{\partial x^2}u_{k,1} - xu_{k,1} \\ & + \frac{1}{2}(u_{k+1,1} + u_{k+1,2}) \end{aligned} \quad (1.10)$$

$$\begin{aligned} 0 = & -\frac{\partial}{\partial t}u_{k,2} + \kappa(\theta_2 - x)\frac{\partial}{\partial x}u_{k,2} + \frac{1}{2}\sigma_2^2\frac{\partial^2}{\partial x^2}u_{k,2} - xu_{k,2} \\ & + \frac{1}{2}(u_{k+1,1} + u_{k+1,2}) \end{aligned} \quad (1.11)$$

are coupled by terms coming from the generator of the regime switching process. We show that with our simple choice of generator (1.8) there exists a change of basis which partially decouples the PDE's (1.10). This change of basis is precisely the decomposition of y valued functions $g(y)$ into **symmetric** (or regime independent)

$$\bar{g} = \frac{g(y_1) + g(y_2)}{2}$$

and **antisymmetric**

$$\underline{g} = \frac{g(y_1) - g(y_2)}{2}.$$

components. In particular the bond price function $u(t, x, y)$, (with t representing time to maturity), can be best analyzed as

$$u(x, y, t) = \begin{cases} \bar{u}(x, t) + \underline{u}(x, t) & , y = 1 \\ \bar{u}(x, t) - \underline{u}(x, t) & , y = 2 \end{cases}$$

The symmetric component $\bar{u}(x, t)$ is an average across regimes, or the price one would assign to a bond in a situation of regime uncertainty where the market prices indicate the neither regime is more likely than the other (if it subsequently becomes clear to all market participants that we are in fact in the first regime, bond prices will jump by an amount equal to the antisymmetric part).

Using the change of basis we derive equations

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \overline{u}_k(x, t) = -\underline{\mathcal{D}} \underline{u}_k(x, t) \quad (1.12)$$

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \underline{u}_k(x, t) = -\underline{\mathcal{D}} \overline{u}_k(x, t) - \underline{u}_{k+1}(x, t) \quad (1.13)$$

where the equation (1.12) is analogous to solubility conditions (2.8), (2.11), (2.24), (2.21) in chapter 2 and we term (1.13) a **balancing** condition. The operators

$$\begin{aligned} \overline{\mathcal{D}} &= \kappa(\overline{\theta} - x) \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial x^2} \\ \underline{\mathcal{D}} &= \kappa \underline{\theta} \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial x^2} \end{aligned}$$

contain the symmetric $\overline{\theta} = \frac{\theta_1 + \theta_2}{2}$ and antisymmetric $\underline{\theta} = \frac{\theta_1 - \theta_2}{2}$ parts of θ (and similarly σ^2). In particular $\overline{\theta}$ is the average level of mean reversion and $\underline{\theta}$ is half the difference between levels to which the short rate reverts.

Formal Algorithm for Calculating the Bond Price

The symmetric and antisymmetric parts of the bond price can be computed to all orders using the following formally simple algorithm:

1. Let $k = 0$ and $\underline{u}_0(x, t) = 0$.
2. Solve the nonhomogeneous PDE

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \overline{u}_k(x, t) = -\underline{\mathcal{D}} \underline{u}_k(x, t)$$

with initial condition given by $\overline{u}_k(0, x) = 1$ for $k = 0$ and $\overline{u}_k(0, x) = 0$ for $k > 0$.

3. Let

$$\underline{u}_{k+1}(x, t) = \left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \underline{u}_k(x, t) + \underline{\mathcal{D}} \overline{u}_k(x, t).$$

4. Increment k and return to step 2.

Explicit Representation of the Bond Price

As explained in section 3.4.2, the nonhomogeneous equation in step two of the above algorithm can be solved cleanly irrespective of the number of times we have looped through step three. Our task is simplified because all symmetric and antisymmetric correction terms, $\overline{u}_k(x, t)$ and $\underline{u}_k(x, t)$ respectively, are multiples of the rapid fluctuation limit $\overline{u}_0(t, x)$ which takes on the same form as the bond price for the constant parameter Vasicek model. We write

$$\overline{u}_k(t, x) = \overline{v}_k(t) \overline{u}_0(t, x)$$

$$\underline{u}_k(t, x) = \underline{v}_k(t) \overline{u}_0(t, x)$$

where $\overline{v}_k(t)$ and $\underline{v}_k(t)$ depend only on the time to maturity t . We derive representations of $\overline{v}_k(t)$ and $\underline{v}_k(t)$ as the result of successive operations with certain differential, multiplicative and integral operators:

$$\underline{v}_k = \left(-\frac{\partial}{\partial t} + \underline{\phi} \cdot \int \cdot \underline{\phi} \right)^{k-1} \underline{\phi} \cdot 1 \quad (1.14)$$

$$\overline{v}_k = \int \cdot \underline{\phi} \left(-\frac{\partial}{\partial t} + \underline{\phi} \cdot \int \cdot \underline{\phi} \right)^{k-1} \underline{\phi} \cdot 1 \quad (1.15)$$

where 1 is the function identically equal to one, $\underline{\phi}$ is the operator multiplying by

$$\kappa \underline{\theta} \tau(t) + \frac{1}{2} \underline{\sigma}^2 \tau^2(t),$$

and $\int$ is the integration operator mapping a function $g(s)$ to its semi-definite integral

$$\left(\int \cdot g \right) (t) = \int_0^t g(s) ds.$$

The symmetric (1.15) and antisymmetric (1.14) terms are readily calculated up to ten orders or so using a symbolic package, the first antisymmetric correction to the

bond price is specified by

$$\underline{v}_1 = \frac{1}{4\kappa^2}(1 - \exp(-\kappa t))(2\underline{\theta}\kappa^2 + \sigma^2 - \sigma^2 \exp(-\kappa t))$$

and the first symmetric correction is specified by

$$\begin{aligned} \overline{v}_1 = & \frac{1}{96\kappa^5} \{ -3 \exp(-4\kappa t) \underline{\sigma}^4 + 16 \exp(-3\kappa t) \underline{\theta} \underline{\sigma}^2 \kappa^2 \\ & + 16 \exp(-3\kappa t) \underline{\sigma}^4 - 24 \exp(-2\kappa t) \underline{\theta}^2 \kappa^4 - 72 \exp(-2\kappa t) \underline{\theta} \underline{\sigma}^2 \kappa^2 \\ & - 36 \underline{\sigma}^4 \exp(-2\kappa t) + 96 \exp(-\kappa t) \underline{\theta}^2 \kappa^4 \\ & + 144 \underline{\theta} \underline{\sigma}^2 \exp(-\kappa t) \kappa^2 + 48 \underline{\sigma}^4 \exp(-\kappa t) + 48 \kappa^5 t \underline{\theta}^2 + 48 \kappa^3 t \underline{\theta} \underline{\sigma}^2 \\ & + 12 \kappa t \underline{\sigma}^4 - 25 \underline{\sigma}^4 - 88 \underline{\theta} \kappa^2 \underline{\sigma}^2 - 72 \underline{\theta}^2 \kappa^4 \} . \end{aligned}$$

Furthermore, the complexity of terms in the asymptotic series and their computational time required to derive them is easily bounded. We show that the calculation of all terms in the expansion can be reduced to matrix multiplication.

In chapter 3 we provide special cases of these formulas for “stochastic mean” models (obtained by setting $\underline{\sigma} = 0$) and stochastic volatility (obtained by setting $\underline{\theta} = 0$). The stochastic mean formulas complement the use of numerical schemes (for example Hansen and Poulsen[37]). Numerical schemes are fast when regime switches occur rarely - precisely when our formulas should not be applied.

The stochastic volatility formulas complement our work in chapter 2 - the major difference being that we do not introduce correlation between the x_t and y_t processes *directly* (correlation *is* present in fact, as explained in section 3.1.2). We are skipping the “correlation correction” and proceeding to higher order terms.

Numerical Tests of the Rapid Fluctuation Asymptotics

In section 3.4.5 we test numerically the accuracy of our bond pricing formulas. An exaggerated example, focusing on the effect of oscillating θ , is obtained by assuming regime dependent parameters as reported in table 1.1. The regime switching intensity $\lambda = 5$, mean reversion rate $\kappa = 2$ and bond maturity $T = 15$ govern the important time scales summarized in table 1.2.3

in a manner consistent with our asymptotic approach. The typical time between regime switches is slightly shorter than the typical mean reversion time of the x_t process and both are shorter than the contract maturity. We therefore expect the rapid fluctuation asymptotics to accurately match the values obtained by simulation and indeed, as can be seen from the numbers in bold in table 1.3, the agreement is excellent.

To benchmark the rapid fluctuation asymptotics we consider other logically motivated analytic approximations, discussed in section 3.3. These produce reasonable approximations to the bond price reported in table 1.3. The first two rows of this table report the bond price calculated using sensible averages of the parameters θ and σ^2 . Two bond prices, corresponding to the two possible initial regimes, are given. The first row, called the “Mean Vasicek” approximation, uses the unconditional mean of θ which in our example is twenty percent. The second row, termed the “Dynamic Mean Vasicek” approximation, is obtained by assuming a deterministic time dependent $\theta(T - t)$ equal to the conditional mean of $\theta(y_t)$ given a particular initial regime.

Parameter	Regime 1	Regime 2
κ	2	2
θ	0.35	0.15
σ	0.0001	0.0001

Table 1.1: Parameters for a regime switching model. We would not anticipate such dramatic regime switches - from fifteen to thirty five percent - in short rates. The example is chosen to challenge the regime switching asymptotics. Results are reported in table 1.3.

	Process	Timescale	Years
Regime switches	y_t	$1/\lambda$	0.2
Short rate fluctuations	x_t	$1/\kappa$	0.5
Contract maturity		T	15

Table 1.2: Characteristic time scales for an example regime switching model used to numerically test the asymptotic approach. Times are in years. The shortest time scale is the mean time between regime switches. The intermediate time scale is the typical time of oscillation or mean reversion of the short rate. The longest time scale is the time to maturity of the bond.

Method	Bond 1	Bond 2	Antisymmetric	Symmetric
Mean Vasicek	0.023517	0.023517	0	0
Dynamic mean Vasicek	0.023383	0.023652	-0.0057143	1.6308e-05
rapid fluctuation (approx)	0.023368	0.023734	-0.0077734	0.00142
rapid fluctuation (exact)	0.023611	0.024067	-0.010281	0.01427
Simulation (200,000)	0.023622	0.024095	-0.010020	0.01447
Sampling Error	0.4%	0.4%	0.0003	0.0003

Table 1.3: Analytic bond price estimates compared with simulation for a regime switching short rate model. The first two rows of this table report the bond price calculated using sensible averages of the parameters θ and σ^2 . We compare these benchmarks with the rapid fluctuation asymptotics, calculated in two ways. The first (approximate) asymptotic series uses matrix calculations in a space of polynomials $P(\tau)$ described in section 3.4.3. The exact series was also calculated using a symbolic package. Terms in either series beyond the sixth or seventh were negligible. We also estimated bond prices using the technique of averaging over arrival time simulations as described in section 3.3.3. For each of the five methods, two bond prices, corresponding to the two possible initial regimes, are reported. The final columns are the antisymmetric and symmetric components of the bond price as defined in the text.

The bond prices obtained using these averaging ideas are, as one would expect, not horribly far from the “true” prices obtained by simulation, given our rather extreme regime switches. However, a closer examination reveals that important effects are being missed. If we examine sums and differences of the bond prices the symmetric (regime independent) and antisymmetric (regime dependent) effects are isolated. As can be seen from the last two columns of table 1.3, only the formulas derived from the rapid fluctuation asymptotics perform well. Our formulas, written directly in terms of antisymmetric and symmetric corrections, match the simulated antisymmetric and symmetric effects (corresponding to $\underline{v}_k$ and $\overline{v}_k$ terms) within the sampling error of the simulation.

These numerical results are encouraging because in spite of unrealistic large jumps in θ of twenty percent we are able to instantly price more accurately than a simulation technique described as “smart” by Hansen and Poulsen [37]. We say more accurately because far less than two hundred thousand simulations would be used in any practical application.

However, acknowledging that our asymptotic technique presumes a rapid fluctuation in y_t , we push the limits of the formulas in section 3.4.5. We establish heuristics leading to accurate approximations. We show that quite good results are obtained even when our starting assumptions relating to time scale separation are strained considerably. Whether or not extrapolation schemes can be profitably combined with the asymptotics to further enlarge the domain of applicability is a question for future research.

1.3 Applications

1.3.1 Credit Derivatives

We envisage application of our work in the modeling of credit derivatives where the primary process x_t represents a **default intensity**. A default intensity, also known as a hazard rate, is the idealized instantaneous probability of an issuer of a bond or cashflow defaulting on payment obligations due to bankruptcy or other reasons.

Many US and foreign corporations and government agencies issue debt which (expressed as a yield on a couponless bond) trades above or below standard benchmarks such as the London Interbank Offering Rate (**LIBOR**). Typically, a greater rate of interest is paid to the corporate bond holder to compensate the owner against risk of default. The additional rate of interest above the (virtually) risk free LIBOR investment or treasury is referred to as a **spread**. Default swaps and similar highly spread sensitive products are reasonably liquid and reflect (modulo changing recovery rates, liquidity premia and other issues) a continuous market assessment of default probabilities of corporations and government agencies.

In section 3.2.2 we describe, in broad terms, the basic default swap deal. As default swaps, also called credit swaps, are sensitive to the creditworthiness of a single asset, the prices of these benchmark products can be used to calibrate risk neutral models for pricing more complex credit sensitive deals which might depend on five, twenty or hundreds of **reference assets**. For many reasons, among them the simple desire for diversification, these products are a huge growth industry. They are also a substantial

modeling challenge.

The buyer and seller of a credit derivative faces several sources of uncertainty, at least two directly related to the creditworthiness of reference assets. The first source of uncertainty is the possibility of one or multiple defaults. The second source of uncertainty is change in the market perceived probabilities of defaults. In addition there may be many other complications including but not restricted to changes in interest rates, exchange rates, credit ratings or regulations and counterparty risk. Therefore a comprehensive system for pricing, hedging, calibration, estimation and risk management represents a considerable undertaking and we address only a small component in this thesis. In particular, the non trivial algorithmic and mathematical questions related to the estimation and simulation of correlated default events fall well outside the present scope.

Instead, we focus on the stochastic process representing the dynamics of an individual spread or default intensity. Our empirical analysis in section 3.5 suggests that the dynamics are complex and not well approximated by Ornstein-Uhlenbeck or square root diffusions. We argue that regime switching processes, possessing as they do the ability to match interesting qualitative features such those described in section 3.1, constitute a better attempt at modeling the real world and risk neutral dynamics. This suggestion is far from original and indeed the quantized nature of credit ratings (“AAA”, “AA”, and so forth) almost screams for the same.

Regime switching models, however, are not simple to price with and herein lies our main contribution. For the more complex credit products thousands of calculations, each one mathematically similar to the bond pricing problem, must be carried out just to arrive at a price. Hedging and risk management impose even heavier computational burdens, and for this reason *instantly* computable bond prices are more necessity than convenience. In terms of tractability, the bar is raised much higher in the credit world than in the modeling of most interest rate products. Our work is important because we provide the first example in the literature of an analytic approach to pricing with a regime switching process x_t . Analytic formulas for pricing are useful for reasons other than speed, but an idea of the potential benefits may be from table 1.4. The formulas, as verified by numerical experiments in section 3.4.5, are very accurate

Bond calculations	1	100	1,000	100,000	10,000,000
Application	Single bond	Default swap	Credit derivative	Complex synthetic	Hedging and risk mgmt
Naive simulation	1 min	$1\frac{1}{2}$ hrs	16 hrs	2 months	20 years
Smart simulation	1.7 sec	3 min	30 min	2 days	6 months
Backward induction	0.1 sec	10 sec	$1\frac{1}{2}$ min	$2\frac{1}{2}$ hrs	$1\frac{1}{2}$ weeks
Analytic formulas	0.001 sec	0.1 sec	1 sec	$1\frac{1}{2}$ min	$2\frac{1}{2}$ hours

Table 1.4: Feeling the need for speed. Very rough, indicative, relative comparison of computation times for numerical and analytic methods of pricing and hedging when the primary process θ is subject to regime switches. The relative performance of backward induction, as detailed by Hansen and Poulsen[37], over simulation is reported in [37] and reproduced in table 3.1. This ratio of 17/1, sensitive to the accuracy required for a given application, has been used as a guide in constructing the table. The times for a particular application are not to be taken to literally and are intended only to provide an approximate indication of the relative scope and practicality of numerical and analytic methods. This table assumes that pricing is carried out to within a basis point on a single machine with specifications similar to those given in [37]. We have attempted to be conservative in relating the computational complexity of applications to the complexity of pricing a individual bond. These ratios may be wrong by an order of magnitude or more depending on the application. We thank Shinghoi Lee for assistance.

and capture the effects of *reasonably rapid* regime switches in more detail than other logically motivated analytic approximations. The results are arrived at by looking at the rapid regime switching asymptotics of the pricing PDE. For this reason, as explained in section 3.3, the formulas work best when numerical schemes are slowest, and vice versa. Therefore our analytic work complements simulation and numerical techniques sketched in section 3.3. In future work we hope to determine whether the slow fluctuation asymptotics be used to improve the accuracy of backward induction schemes.

1.3.2 Interest Rate Modeling

The analytic bond prices derived in this thesis are exponential affine functions of the state variable x_t , a fact of substantial practical significance. As detailed in appendix B, a motivation for staying close to the family of affine models is the wealth of existing techniques which facilitate the pricing of many common fixed income securities. In a sense made precise by our asymptotic approach, we are able to build models which are quite removed from the affine class in terms of their dynamics (incorporating regime switches or complex volatility specifications) yet **almost affine** when it comes to pricing. The affine technology facilitates a product independent approach to modeling which focuses on fundamental quantities described in appendix B.3. Principal among these is the bond price itself and similarly derived expectations of the x_t and integrated x_t process. Our work treats these quantities directly. The pricing of many important securities, which we now describe, follows automatically.

Common Interest Rate Products

The essentials of a basic interest rate swap in section 3.2.2, and show in appendix B.4 how they may be priced using the result in this thesis. Essentially a swap is an agreement between two parties to exchange a “fixed” cashflow (for example, fixed coupon payments from a bond) for a “floating” cashflow (variable interest payments calculated using short term yields, LIBOR forward rates or some other reference mutually agreed upon). A **cap** or **floor** deal is very similar, but awards one party

optionality on the “floating” payments. For example, a minimum yield might be guaranteed so as to provide protection against falls in LIBOR. This optionality can be split up into individual options on each payment date and the pricing carried out as in appendix B.4 using transform methods. We also review the pricing of **forward** and **future** contracts which are especially relevant to European markets.

This by no means exhausts the possibilities. The asymptotic techniques can almost certainly be applied in the pricing of products which are heavily path dependent, including American bond options. Analogous work in the context of equities and stochastic volatility was carried out by Sircar, Fouque and Papanicolaou [31].

Simplicity and Parsimony

Simplicity endows survival advantages to a model over and above that which the casual Wall Street cynic would suggest. If we adopt the position that overnight interest rates mean-revert (albeit over fairly long time intervals) to a nonzero mean then the Vasicek model is perhaps the simplest possible continuous time short rate model. Traders must combine models with intuition, experience, and market information less amenable to modeling than published prices. For this reason the rigidity - or more neutrally, parsimony - of simple models can be a good thing. Rigidity facilitates the inspection and analysis of market peculiarities and endows parameters with a utility beyond what might be theoretically justified from an econometric time series standpoint. No model can outperform the market and the better understanding is arrived at, as in chess, not by human or machine, but by a combination of both.

A classic example is the use of the somewhat unrealistic Black Scholes option pricing formula in the examination of option prices. There is no single fixed income equivalent whose popularity or scope rivals Black Scholes or supplies as convincing a low (or one) dimensional intuitive grasp of the market - such may be too much to ask for given the complexity of term structure movements. As even the constant parameter Vasicek or CIR models have a three dimensional parameter space - two more than the ever popular Black Scholes model for stock options - the difficulty of the task is apparent. We are certainly not in a position to claim a clear quantifiable advantage of the asymptotic corrections in a trading context over a broad range of

competitors.

However, in a sense relative to the dimensionality and flexibility of the two dimensional process (x_t, y_t) with which we begin, and also relative to more complex models inside and outside the affine class, the end result of our analysis is, we would argue, simplicity rather than complexity. We focus on obtaining those effects which can be ascertained independent of difficult to estimate or unobservable secondary quantities (the y_t process) and maintain the intuition gleaned from the primary observed quantity (the x_t process). We hope that in addition to econometric use, the Vasicek model with parameter fluctuations may prove its worth in a less formal context and provide another lens for effectively examining markets. In this regard there is much work to be done.

Econometric Models of the Term Structure

As far as econometric models of the term structure are concerned, the traditional Vasicek model with constant parameters is too simple and restrictive to contend with its preferred affine colleagues (Dai and Singleton [14]). We believe that adding fluctuations is unlikely to resurrect this (or any single factor model) in an econometric sense. This being the case the application of our results *within the context of econometric interest rate modeling* needs to be framed appropriately. We are proposing not a particular model for short rates in the style of Vasicek, but an adding to the list of “very tractable one dimensional processes” appropriate for use in the construction of multifactor models. Our approach also suggests ideas for estimation of these models and in section 2.2.7 we estimate corrections to transition probabilities of an Ornstein-Uhlenbeck process accounting for short term fluctuations in volatility.

The ability to account for certain factors without explicit knowledge of their parameters or even functional form is yet to be exploited in a multifactor setting and represents an interesting possibility for future research. In carrying out the detailed calculations for special cases we push a more general philosophy. Dai and Singleton[14], among others, suggest a picture where the short rate itself rapidly mean reverts to a more slowly varying long-run mean. It is likely, though we have not performed the analysis, that time scale separation here can directly exploited for estimation and

pricing in this context.

1.4 Empirical Results

Our empirical results are intended to provide support for the intended applications in interest rate and credit modeling. Empirical work in these areas, particularly the former, is extensive and we do not attempt a survey here. Our techniques are somewhat unusual and rarely found within the financial literature, but they are direct, specific to our overall modeling philosophy, and have proven useful in other applied mathematical contexts.

1.4.1 Evidence for Fast Mean Reverting Volatility in US Short Rates

As our bond pricing corrections in the case of stochastic volatility assume a fast mean reverting component we examine in a straight forward manner a time series for a proxied short rate, and test for the existence of fast mean reversion. The analysis focuses on the structure function specific to the exponential Ornstein-Uhlenbeck stochastic volatility model. The test is a direct and reasonably robust measurement of the decorrelation time of the (unobservable) volatility driving process, and verifies anecdotal evidence of a rapidly mean reverting component of volatility. We are not able to measure the elusive mean reversion time in a precise manner, but this is unimportant to our analysis. The very existence of mean reversion on the order of weeks rather than years is all that we require in order to validate our asymptotic approach.

Applying the same tests to simulations from a variety of models, we demonstrate that incorporating rapid volatility fluctuations into the model assists our ability to match features of the data. Without a direct modeling of these fluctuations, as provided for example in our family of stochastic volatility short rates, qualitative features of the empirical structure function are not replicated. In this work we are grateful for the assistance and expertise of Knut Solna.

1.4.2 Bimodality and Predictability of Spread Distributions

With applications to credit derivatives in mind, we apply critical smoothing techniques developed by Silverman[56] and others to examine spread time series often modeled in the literature (for example [46]) as CIR processes. These “bump hunting” tests are designed to count the number of modes in the “true” distribution generating the data. We applied the tests to hundreds of time series of US corporate spreads. These time series, consisting of credit spreads of various maturities and lengths, were provided to us by Robert Fuhrman of the Morgan Stanley Dean Witter Fixed Income Index Group.

The critical smoothing tests are variance sensitive and somewhat open to interpretation when applied to individual data sets. However across hundreds of spread time series we show that they provide significant results. We find strong evidence for bimodality in spread distributions which cannot be explained by sampling effects and thoroughly rejects the hypothesis that the spread time series are generated by a simple one dimensional Markovian process such as an Ornstein-Uhlenbeck or square root diffusion. By contrast, with appropriate choice of parameters regime switching processes provide an excellent fit to the data.

Our work, in which we were advised by Carrie Grimes, is largely orthogonal to previous empirical work on regime switching models insofar as we focus on global qualitative aspects of the dynamics not easily identified by examination of families of autoregressive models or differenced data. They also represent a tentative step in the direction of estimation techniques which treat thousands of time series simultaneously. The results are intuitive and in agreement with the intuition of at least one market participant (we thank Zhifeng Zhang for suggesting the qualitative fact we have verified).

A more local property of the spread dynamics, trending, can be quantified as the proportion of consecutive monthly spread moves in the same direction. We were surprised by the degree of predictability in the data. Spreads were approximately twice as likely to move the same direction in a given month as they had the previous month than they were to move in the opposite direction. This empirical curiosity does not

violate any “laws” of finance, but does represent a serious challenge to diffusion models (often used to model spreads) whose increments are almost independent. Many (but by no means all) regime switching models display the same degree of one step ahead bias.

1.5 Caveat Emptor

At the outset of calculations central to this thesis we assume a separation of time scales which may find empirical support in some markets and not in others. In particular, the secondary stochastic process y_t is generally assumed to mean revert on a time scale shorter than the typical mean reversion time of the x_t process. In the first half of thesis we are concerned with a correction to the Vasicek formula which accounts for the first order effect of adding stochastic volatility correlated with the short rate process. The correction is independent of the initial value of the y_t process and as such represents largely the contribution to the bond price from high frequency fluctuations in volatility. The correction *does not* account for volatility variation over longer periods of time that would be better accounted for in an egalitarian two factor model.

Our analysis strongly suggests existence of a high frequency component (a component which rapid mean reverts) in volatility of a swaps implied (USD-LIBOR) short rate. Similar fast mean reversion has been found in equity market indices [33] and, consistent with heresay, is likely to exist in many markets. Where no such component exists however our corrections are likely to be misleading.

The second half of the thesis in which we treat a simple two state regime switching model has a slightly different emphasis due to the fortuitous availability of explicitly computable higher order correction terms. Here the approximate bond pricing formulas are in fact (for all practical purposes) exact formulas *provided regime switches occur frequently*. They are also pretty good when regime switches occur quite frequently and numerical simulations carried out towards the end of the thesis demonstrate that the formulas derived may sometimes be used *with care* even when regime switches occur relatively rarely. At this point however the asymptotic series certainly require

truncation.

Thus in both chapters 2 and 3 the accuracy of the formulas we derive depend roughly on how rapidly the secondary process returns to its mean or, in the case of discretely valued y_t , switches states. The expansions represent an attempt to effectively solve, within a limited domain of applicability, a partial differential equation which doesn't especially want to be solved. This domain of applicability is not easily specified in a rigorous manner as it probably exceeds whatever boundaries could be so imposed and extends considerably beyond what is strictly implied by a literal interpretation of our ansatz. Not wishing to err too far on the side of caution we can only suggest that the reader not try to wag the dog by its tail and remember our starting point: an examination of perturbations of a Vasicek model where the parameters *fluctuate* about their means over time scales which are relatively short. Should the time scale of fluctuation of the variables θ and σ be long an analysis of time scales may be beneficial (a *slow fluctuation* approach is the subject of future work) but the *rapid fluctuation* results presented in this thesis are inapplicable.

Chapter 2

Vasicek Models with Stochastic Volatility

In this chapter we treat an Ornstein-Uhlenbeck process

$$dx_t = \kappa (x_\infty - x_t) dt + \sigma(y_t) dW_t^*$$

whose diffusion parameter σ fluctuates rapidly. The source of fluctuations is a second process y_t , also Ornstein-Uhlenbeck, correlated with the primary process x_t . We examine implications for the term structure of interest rates when the short rate x_t is so modeled. An asymptotic analysis of bond and bond option pricing within this broad **stochastic volatility** [34], [7], framework is presented and we calculate explicit analytic formulas which account for the uncertain volatility in a robust manner, independent of the exact volatility specification. The correction arises from an interplay between market risk premia (arising from inability to hedge short rate and short rate volatility movements), and the correlation between the short rate and short rate volatility. The key assumption is that volatility is bursty and mean-reverts *fast* with respect to the longer timescale of derivative contracts.

The pricing results are then specialised to an example of a stochastic volatility model where the diffusion of the primary process x_t is an exponential function of the secondary process y_t . The exponential function exaggerates excursions by the

volatility driving y_t process, leading to short periods where the volatility is significantly larger than usual. We speculate that this behaviour may have a similar effect on yields as the addition of small jumps and verify this suspicion by decomposing bond yields in a particular non-orthogonal basis. For this particular model we are also able to approximate the conditional transition density for the short rate, useful in likelihood estimation. Specifically, we reduce to a one dimensional quadrature that part of the likelihood function which is independent of the initial and final states of the secondary y_t process.

Our results are limited in scope insofar as they account for pricing discrepancies arising from rapid fluctuations in volatility, corresponding to rapid mean reversion of the y_t process. We examine a time series of empirical short rates, or rather proxies for this idealised instantaneous rates implied by USD-LIBOR swaps for evidence of a fast mean reverting component in volatility. We use variogram analysis to confirm that this is indeed the case.

As the pricing formulas we derive are independent of initial values of the secondary volatility driving process y_t , we are able to treat the two factor model as a one factor model for the purpose of fitting yield curves. As a rough guide to the flexibility and stability of the stochastic volatility model we compare its ability to fit contemporaneous snapshots of the term structure of interest rates against single factor jump diffusions. This analysis is preliminary but promising. The separation of time scales implicit in our approach - and manifest in the correction formulas - seems to help us match simultaneously the long and short end behaviour of the term structure of interest rates, using only a few parameters.

2.1 Results for a Class of Models with Rapidly Fluctuating Volatility

2.1.1 A Family of Stochastic Volatility Vasicek Type Models

We extend the Vasicek model described in appendix A by introducing stochastic volatility σ_t given by a positive function $f(y_t)$ of an ergodic Markov diffusion process

y_t with a unique invariant distribution, for example an Ornstein-Uhlenbeck process. The Gaussian distribution of this last choice is not crucial for the asymptotic results, but we write a specific driving process to illustrate the methodology.

In the real world $\mathbb{P}$ the processes (x_t, y_t) satisfy

$$\begin{aligned} dx_t &= \kappa(x_\infty - x_t)dt + f(y_t)dW_t^* \\ dy_t &= \alpha(m - y_t)dt + \beta \left(\rho dW_t^* + \sqrt{1 - \rho^2} dZ_t^* \right), \end{aligned}$$

where W_t^* and Z_t^* are independent standard Brownian motions. The parameter ρ with $|\rho| < 1$ allows a correlation between the Brownian motion W_t^* driving the short rate and its volatility. Typically, we would expect $\rho > 0$ as rising volatility tends to push bond prices down and yields up. This is confirmed from empirical evidence. We note that the invariant distribution of y_t is the normal distribution $\mathcal{N}(m, \beta^2/2\alpha)$.

In the risk-neutral world $\mathbb{P}^{(\lambda_1, \lambda_2)}$, the model for the short rate x_t becomes

$$dx_t = (\kappa(x_\infty - x_t) - \lambda_1(y_t)f(y_t))dt + f(y_t)dW_t \quad (2.1)$$

$$\begin{aligned} dy_t &= \left(\alpha(m - y_t) - \beta \left[\rho \lambda_1(y_t) - \lambda_2(y_t) \sqrt{1 - \rho^2} \right] \right) dt \\ &\quad + \beta \left(\rho dW_t + \sqrt{1 - \rho^2} dZ_t \right), \end{aligned} \quad (2.2)$$

where W_t and Z_t are two independent standard $\mathbb{P}^{(\lambda_1, \lambda_2)}$ -Brownian motions. A plausibility argument for the change of measure implied here is given in appendix B. The market price of risk $\lambda_1(y_t)$ may depend on y_t but we assume that it does not depend on the short rate x_t . Similarly the *market price of volatility risk* $\lambda_2(y_t)$ associated with the second source of randomness Z_t may depend on y_t or simply be a constant. The effect is to preserve under the pricing measure the Markovian structure of the model and the autonomy of the volatility-driving process y_t , so that (2.1-2.3) remains a genuine stochastic volatility model. The assumption is that the market does not price term-structure derivatives as though the volatility is significantly affected by the level of the short-rate, or non-Markovian phenomena.

For general functions f these models do not belong to the affine family because yields will not be affine in the second factor y_t . However, they may be considered

“almost affine” because yields are well approximated by an affine function of the short-rate x_t alone.

2.1.2 The Partial Differential Equation Satisfied by Bond Prices

The no-arbitrage price $B(t, x, y; T)$ at time t of a zero-coupon bond maturing at time T is given by

$$B(t, x, y; T) = \mathbb{E}^{\lambda_1, \lambda_2} \left[e^{-\int_t^T x_s ds} | x_t = x, y_t = y \right], \quad (2.3)$$

where the expectation $\mathbb{E}^{\lambda_1, \lambda_2}$ is taken with respect to the distribution of the joint process (x_t, y_t) solving (2.1-2.3) and starting at time t from (x, y) . Using a two-dimensional Feynman-Kac formula, we deduce that $B(t, x, y; T)$ is the solution of the partial differential equation

$$\begin{aligned} \frac{\partial B}{\partial t} + \frac{1}{2} f(y)^2 \frac{\partial^2 B}{\partial x^2} + (\kappa(x_\infty - x) - \lambda_1(y) f(y)) \frac{\partial B}{\partial x} - xB \\ + \beta \rho f(y) \frac{\partial^2 B}{\partial x \partial y} + \frac{1}{2} \beta^2 \frac{\partial^2 B}{\partial y^2} \\ + \left(\alpha(m - y) - \beta \left[\rho \lambda_1(y) - \lambda_2(y) \sqrt{1 - \rho^2} \right] \right) \frac{\partial B}{\partial y} = 0, \end{aligned} \quad (2.4)$$

with the terminal condition $B(T, x, y; T) = 1$ for every x and y . The terminal condition expresses the tautology that an bond as we have defined it is worth unity at maturity.

2.1.3 Fast Mixing Limit, a Naive Estimate of the Bond Price

In the context of fast mean-reverting stochastic volatility, the rate of mean-reversion α in the OU process y_t driving the volatility is large and $\nu = \beta/\sqrt{2\alpha}$ remains of order one. In the limit as α tends to infinity the volatility fluctuates so rapidly that for all intents and purposes we are effectively sampling, at each instant, a value for y_t plucked from the gaussian invariant distribution of the Ornstein-Uhlenbeck process. One would expect the process to behave in identical fashion to a constant parameter

Ornstein-Uhlenbeck process

$$dx_t = \kappa(x_\infty - x_t) dt + \sqrt{\langle f^2(y_t) \rangle} dW_t^*$$

where the constant $\sqrt{\langle f^2(y_t) \rangle}$ is the root mean square diffusion parameter averaged with respect to the invariant distribution of y_t . We will show that this is indeed the case, thus deriving the fast mixing limit and demonstrating that the stochastic volatility model can be treated as a perturbation of the traditional constant parameter Vasicek model.

2.1.4 Asymptotic Approach

Using the notation and the method developed in the context of equity markets in [30] the asymptotic analysis is carried out as $\epsilon = 1/\alpha$ becomes small. We set

$$\beta = \frac{\sqrt{2}\nu}{\sqrt{\epsilon}}$$

intending to take $\epsilon \rightarrow 0$ while keeping ν , and hence the unconditional variance of y_t , constant. Emphasizing the scale separation we write

$$\mathcal{L}^\epsilon = \frac{1}{\epsilon}Q + \frac{1}{\sqrt{\epsilon}}\mathcal{L}_1 + \mathcal{L}_2,$$

where

$$Q = \nu^2 \frac{\partial^2}{\partial y^2} + (m - y) \frac{\partial}{\partial y},$$

is the infinitesimal generator of the mean-reverting OU process y_t scaled by $1/\alpha$, and

$$\mathcal{L}_2 = \frac{\partial}{\partial t} + \frac{1}{2}f(y)^2 \frac{\partial^2}{\partial x^2} + (\kappa(x_\infty - x) - \lambda_1 f(y)) \frac{\partial}{\partial x} - x1,$$

is the Vasicek operator with volatility $f(y)$ and long run mean $\frac{1}{\kappa}(x_\infty - \lambda_1 f(y))$. The operator

$$\mathcal{L}_1 = \sqrt{2\nu}\rho f(y) \frac{\partial^2}{\partial x \partial y} - \sqrt{2\nu} \left(\lambda_1(y)\rho + \lambda_2(y)\sqrt{1-\rho^2} \right) \frac{\partial}{\partial y},$$

contains the two market prices of risk and singular terms due to the skew.

The partial differential equation (2.4) can be written

$$\mathcal{L}^\varepsilon B = 0 \tag{2.5}$$

with the terminal condition $B(T, x, y; T) = 1$. We look for an asymptotic solution of the form

$$B(t, x, y; T) = B_0(t, x, y; T) + \sqrt{\varepsilon} B_1(t, x, y; T) + \varepsilon B_2(t, x, y; T) + \dots, \tag{2.6}$$

with the terminal conditions $B_0(T, x, y; T) = 1$ and $B_1(T, x, y; T) = B_2(T, x, y; T) = \dots = 0$. Our strategy is to expand equation (2.5) in powers of the small parameter ε by substituting (2.1.4) and the expansion (2.6). We then look at the equations at each order starting with the highest order $\mathcal{O}(\varepsilon^{-1})$ which reads

$$QB_0(t, x, y; T) = 0,$$

from which we deduce that $B_0 = B_0(t, x; T)$ does not depend on y since Q is the generator of an ergodic Markov process operating on the variable y only. The $\mathcal{O}(\varepsilon^{-1/2})$ terms give

$$QB_1(t, x, y; T) + \mathcal{L}_1 B_0(t, x; T) = 0,$$

which reduces to

$$QB_1 = 0,$$

since $\mathcal{L}_1$ takes derivatives with respect to y and B_0 does not depend on y from the

previous order. We conclude that also $B_1 = B_1(t, x; T)$ does not depend on y . This is extremely important since it means that the present value $y_t = y$ of the unobservable volatility driving process will not be needed for the first correction to B_0 .

2.1.5 A Solubility Condition giving rise to the Fast Mixing Limit

The $\mathcal{O}(1)$ terms lead to

$$QB_2(t, x, y; T) + \mathcal{L}_1 B_1(t, x; T) + \mathcal{L}_2 B_0(t, x; T) = 0,$$

which reduces to

$$QB_2(t, x, y; T) + \mathcal{L}_2 B_0(t, x; T) = 0, \quad (2.7)$$

since B_1 does not depend on y and consequently $\mathcal{L}_1 B_1 = 0$. This is a Poisson equation for B_2 which cannot be solved unless QB_2 is centered with respect to the invariant distribution of the OU process.

In other words the **solubility condition** is

$$\langle \mathcal{L}_2 B_0 \rangle = 0 \quad (2.8)$$

where $\langle \cdot \rangle$ denotes the averaging with respect to the Gaussian density $\mathcal{N}(m, \nu^2)$. As B_0 does not depend on y , it can be pulled out of the brackets to obtain

$$\langle \mathcal{L}_2 \rangle B_0 = 0,$$

which is exactly the partial differential equation (A.4) with the constant coefficients

$$\bar{\sigma}^2 = \langle f^2 \rangle \text{ and } \bar{\theta} = x_\infty - \langle \lambda_1 f \rangle / \kappa, \quad (2.9)$$

and the same terminal condition $B_0(T, x; T) = 1$. We obtain

$$B_0(t, x; T) = B_{vas}(t, x; T, \overline{\sigma^2}, \bar{\theta}) = e^{-a_{vas}(T-t) - \tau(T-t; \kappa)x}, \quad (2.10)$$

where, analogous to the Vasicek bond price in appendix A, the functions τ and a_{vas} are given by

$$\begin{aligned} \tau(T-t; \kappa) &= \frac{1 - e^{-\kappa(T-t)}}{\kappa} \\ a_{vas}(T-t) &= \left(\bar{\theta} - \frac{1}{2} \langle \sigma^2(y) \rangle \right) ((T-t) - \tau(T-t; \kappa)) + \frac{\langle \sigma^2(y) \rangle}{4\kappa} \tau(T-t; \kappa)^2 \end{aligned}$$

To recap, this formula is exactly the Vasicek one-factor bond pricing function

$$B_{vas}(t, x, T; \kappa, \bar{\theta}, \bar{\sigma})$$

with the averaged parameters $(\kappa, \bar{\theta}, \bar{\sigma})$ related to the stochastic volatility model parameters in (2.1-2.3) by (2.9).

2.1.6 Correcting the Bond Price

The $\mathcal{O}(\sqrt{\varepsilon})$ terms give a Poisson equation in B_3 whose solubility condition is

$$\langle \mathcal{L}_2 B_1 + \mathcal{L}_1 B_2 \rangle = 0. \quad (2.11)$$

From the previous Poisson equation (2.7) we deduce that

$$B_2 = -Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle) B_0 + k(t, x)$$

where $k(t, x)$ does not depend on y . The condition (2.11) gives the following equation for $B_1(t, x; T)$:

$$\langle \mathcal{L}_2 \rangle B_1 = \langle \mathcal{L}_1 Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle) \rangle B_0,$$

with a zero terminal condition at $t = T$. Since we are mainly interested in the first correction to B_0 , we rewrite this equation in terms of $\tilde{B}_1(t, x; T) = \sqrt{\varepsilon} B_1(t, x; T)$:

$$\langle \mathcal{L}_2 \rangle \tilde{B}_1 = \mathcal{A} B_0, \quad (2.12)$$

with a zero terminal condition and where the operator $\mathcal{A}$ is defined by

$$\mathcal{A} = \sqrt{\varepsilon} \langle \mathcal{L}_1 Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle) \rangle.$$

Observe that $\tilde{B}_1$ is of the same order $\sqrt{\varepsilon}$ as the source term $\mathcal{A} B_0$ in (2.12).

The operator $\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle$ is given by

$$\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle = \frac{1}{2} (f(y)^2 - \langle f^2 \rangle) \frac{\partial^2}{\partial x^2} - (\lambda_1 f(y) - \langle \lambda_1 f \rangle) \frac{\partial}{\partial x},$$

and we introduce the centered $\phi(y)$ and $\psi(y)$ which are solutions of

$$Q\phi = f(y)^2 - \langle f^2 \rangle \quad \text{and} \quad Q\psi = \lambda_1 f(y) - \langle \lambda_1 f \rangle.$$

The operator $\mathcal{A}$ can now be written

$$\mathcal{A} = V_1 \frac{\partial}{\partial x} + V_2 \frac{\partial^2}{\partial x^2} + V_3 \frac{\partial^3}{\partial x^3}, \quad (2.13)$$

involving the constant quantities

$$\begin{aligned} V_3 &= \frac{\nu}{\sqrt{2\alpha}} \rho \langle f \phi' \rangle \\ V_2 &= -\frac{\nu}{\sqrt{2\alpha}} \left(\rho \langle \lambda_1 \phi' \rangle + \sqrt{1 - \rho^2} \langle \lambda_2 \phi' \rangle \right) - \nu \rho \sqrt{\frac{2}{\alpha}} \langle f \psi' \rangle \\ V_1 &= \nu \sqrt{\frac{2}{\alpha}} \left(\rho \langle \lambda_1 \psi' \rangle + \sqrt{1 - \rho^2} \langle \lambda_2 \psi' \rangle \right), \end{aligned}$$

complicated functions of the model parameters, including market prices of risk, and the volatility function f . Notice that no skew ($\rho = 0$) implies $V_3 = 0$ and a market price of risk such that $\lambda_1 f$ is constant implies $V_1 = 0$. Using the change of variable

$s = T - t$, (2.10) and (2.13), equation (2.12) becomes

$$\begin{aligned} \frac{\partial \tilde{B}_1}{\partial s} = & \frac{1}{2} \tilde{\sigma}^2 \frac{\partial^2 \tilde{B}_1}{\partial x^2} + \kappa(\bar{\theta} - x) \frac{\partial \tilde{B}_1}{\partial x} - x \tilde{B}_1 \\ & + e^{-a_{\text{vas}}(s) - \tau(s; \kappa)x} (V_3 \tau(s; \kappa)^3 - V_2 \tau(s; \kappa)^2 + V_1 \tau(s; \kappa)) , \end{aligned} \quad (2.14)$$

with the initial condition $\tilde{B}_1(T - 0, x; T) = 0$.

We now try to find a solution of the form

$$\tilde{B}_1(T - s, x; T) = D(s) e^{-a_{\text{vas}}(s) - \tau(s; \kappa)x} ,$$

which leads to the following equation for $D(s)$:

$$D'(s) = V_3 \tau^3(s) - V_2 \tau^2(s) + V_1 \tau(s) ,$$

Using $\tau(0) = 0$ and $a_{\text{vas}}(0) = 0$, we get an explicit expression for $D(s)$ written here as a function of $\tau(s; \kappa) = (1 - e^{-\kappa s})/\kappa$:

$$\begin{aligned} D(s) = & \frac{V_3}{\kappa^3} \left(s - \tau(s; \kappa) - \frac{1}{2} \kappa \tau(s; \kappa)^2 - \frac{1}{3} \kappa^2 \tau(s; \kappa)^3 \right) \\ & - \frac{V_2}{\kappa^2} \left(s - \tau(s; \kappa) - \frac{1}{2} \kappa \tau(s; \kappa)^2 \right) + \frac{V_1}{\kappa} (s - \tau(s; \kappa)) , \end{aligned} \quad (2.15)$$

and finally the corrected bond price

$$B(t, x, y; T) \approx B_0(T - s, x; T) + \tilde{B}_1(T - s, x; T) = e^{-a_{\text{vas}}(s)} (1 + D(s)) e^{-\tau(s; \kappa)x} , \quad (2.16)$$

where D is a small factor of order $1/\sqrt{\alpha}$.

In Figure 2.1, we show the effect of the small correction $\tilde{B}_1$ on the bond prices (top) and on the corresponding yield curve (bottom).

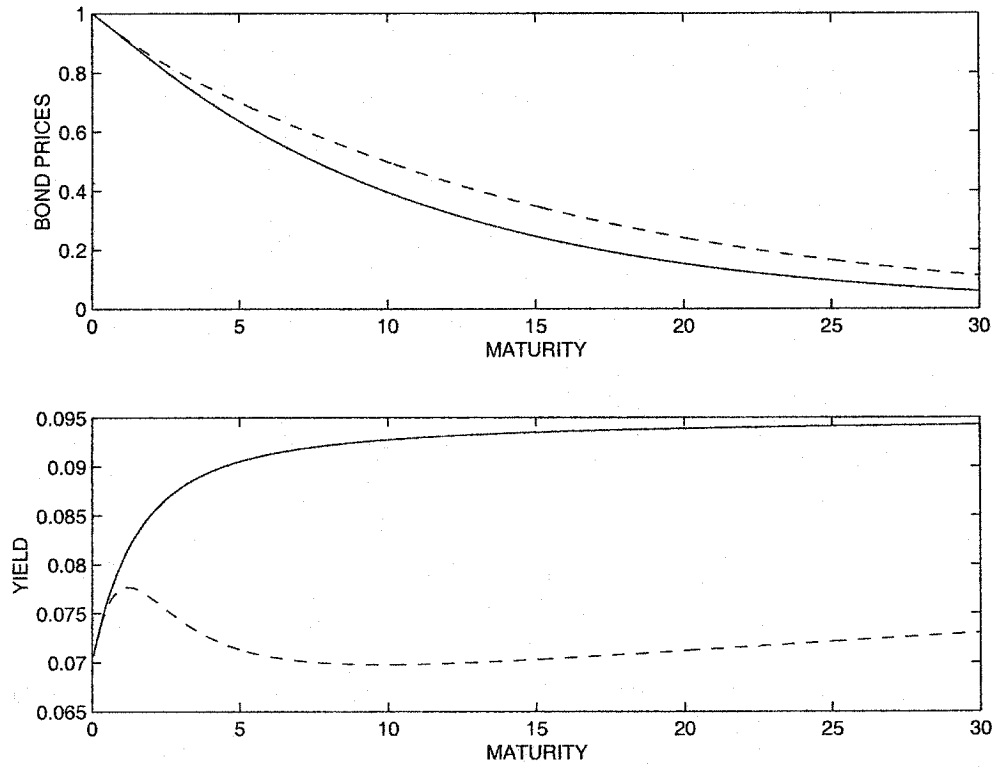


Figure 2.1: Top: bond prices and corrected bond prices (dotted curve). Bottom: yield curve and corrected yield curve (dotted curve) in the simulated Vasicek model (constant and stochastic volatility) with: $\kappa = 1$, $\bar{\theta} = 0.1$ and $\bar{\sigma} = 0.1$ as in Figure A.1. Concerning the correction we have used $V_3 = 1/\sqrt{\alpha}$ assuming a nonzero skew, $\alpha = 10^3$ and $\lambda_1 = \lambda_2 = 0$ implying $V_1 = 0$ and $V_2 = 0$. Maturity u runs from 0 to 30 years and the initial rate is $x = 0.07$.

2.1.7 Corrections for Bond Option Prices

We consider now the bond option pricing problem in the context of the stochastic volatility Vasicek model introduced in section 2.1.1. The short rate process x_t and the OU process y_t driving the volatility are given by equations (2.1) and (2.3). The option on the zero-coupon bond is as in appendix A.0.6, with payoff function $h(B(T_0, T))$. Its premium at time t is denoted by $H(t, x, y; T, T_0)$. It is now a function of the current rate $x_t = x$ and the current volatility level $y_t = y$, given by

$$H(t, x, y; T, T_0) = \mathbb{E}^{(B, \lambda_2)} \left[e^{-\int_t^{T_0} x_s ds} h(B(T_0, T)) \mid x_t = x, y_t = y \right], \quad (2.17)$$

where the bond price (T_0, T) is a function of x_{T_0} and y_{T_0} given by formula (2.3) with t replaced by T_0 :

$$B(T_0, T) = B(T_0, x_{T_0}, y_{T_0}; T) = \mathbb{E}^{(\lambda_1, \lambda_2)} \left[e^{-\int_{T_0}^T x_s ds} | x_{T_0}, y_{T_0} \right].$$

By the Feynman-Kac formula, this premium $H(t, x, y; T_0, T)$ is again the solution of the partial differential equation 2.4:

$$\begin{aligned} \frac{\partial H}{\partial t} + \frac{1}{2} f(y)^2 \frac{\partial^2 H}{\partial x^2} + (\kappa(x_\infty - x) - \lambda_1(y) f(y)) \frac{\partial H}{\partial x} - xH \\ + \beta \rho f(y) \frac{\partial^2 H}{\partial x \partial y} + \frac{1}{2} \beta^2 \frac{\partial^2 H}{\partial y^2} \\ + \left(\alpha(m - y) - \beta \left[\rho \lambda_1(y) - \lambda_2(y) \sqrt{1 - \rho^2} \right] \right) \frac{\partial H}{\partial y} = 0, \end{aligned}$$

but the terminal condition at time T_0 becomes

$$\begin{aligned} H(T_0, x, y; T, T_0) &= h \left(\mathbb{E}^{(\lambda_1, \lambda_2)} \left[e^{-\int_{T_0}^T x_s ds} | x_{T_0} = x, y_{T_0} = y \right] \right), \\ &= h(B(T_0, x, y; T)), \end{aligned} \quad (2.18)$$

where, as in (2.3), $B(T_0, x, y; T)$ denotes the bond price $B(T_0, T)$ at time T_0 as a function of the present short-rate x and level y of the volatility-driving process.

Observe that the terminal condition depends also on the volatility variable y . Fortunately, in the fast mean-reverting case (α large) we have already obtained, in section 2.1.1, an expansion of the bond price $B(T_0, x, y; T)$ and we have seen that, up to order $\mathcal{O}(1/\sqrt{\alpha})$, this expansion is independent of y . The bond option prices $H(t, x, y; T, T_0)$ will be abbreviated $H(t, x, y)$ and expanded as

$$H(t, x, y) = H_0(t, x, y) + \sqrt{\varepsilon} H_1(t, x, y) + \dots \quad (2.19)$$

Assuming first that the payoff function h is smooth, one can expand the terminal condition (2.18) by using the first two terms in the bond price expansion (2.16)

$$H(T_0, x, y) = h(B_0(T_0, x; T)) + \sqrt{\varepsilon} B_1(T_0, x; T) h'(B_0(T_0, x; T)) + \dots, \quad (2.20)$$

and observe that, up to the order $\mathcal{O}(\sqrt{\varepsilon})$, it does not depend on y .

Zero-order term

Expanding the equation $\mathcal{L}^\varepsilon H = 0$, the term of order $1/\varepsilon$ demonstrates that $H_0(t, x, y)$ does not depend on y :

$$H_0(t, x, y) = H_0(t, x).$$

The terms of order $1/\sqrt{\varepsilon}$ give the same conclusion for $H_1(t, x, y)$:

$$H_1(t, x, y) = H_1(t, x).$$

The $\mathcal{O}(1)$ terms lead to a Poisson equation in H_2 whose solvability condition is

$$\langle \mathcal{L}_2 H_0 \rangle = 0 \tag{2.21}$$

where $\langle \cdot \rangle$ denotes the averaging with respect to the invariant distribution of the OU process y_t . Since $H_0(t, x)$ is independent of y , it is the solution of the partial differential equation

$$\langle \mathcal{L}_2 \rangle H_0 = 0,$$

with the terminal condition $H_0(T_0, x) = h(B_0(T_0, x; T))$ at time T_0 . This is exactly the problem solved in the case of a constant volatility $\bar{\sigma}$ and $\bar{\theta} = x_\infty - \langle \lambda_1 f \rangle / \kappa$. Consequently

$$H_0(t, x) = H_{vas}(t, x; T, T_0),$$

where $H_{vas}(t, x; T, T_0)$ is the constant parameter Vasicek option price given in appendix A.0.6. It can be written as a Gaussian integral:

$$H_0(t, x) = \mathbb{E} \left[e^{-\int_t^{T_0} \bar{x}_s ds} h(B_{vas}(T_0, x; T)) \right] \quad (2.22)$$

$$= \mathbb{E} \left[e^{-\int_t^{T_0} \bar{x}_s ds} h \left(e^{-a_{vas}(T-T_0) - \tau(T-T_0; \kappa) \bar{x}_{T_0}} \right) \right], \quad (2.23)$$

where $\bar{x}_t$ is the risk-neutral one-factor process with averaged parameters $\kappa, \bar{\theta}$ and $\bar{\sigma}$. The expectation is taken with respect to the martingal measure $\mathbb{P}$.

First-order correction

The $\mathcal{O}(\sqrt{\varepsilon})$ terms give a Poisson equation in H_3 whose solvability condition, as in section 2.1.1, reduces to

$$\langle \mathcal{L}_2 \rangle \tilde{H}_1 = \mathcal{A} H_0, \quad (2.24)$$

where we have set $\tilde{H}_1(t, x) = \sqrt{\varepsilon} H_1(t, x)$ and the operator $\mathcal{A}$ is given by (2.13). Using (2.20), the terminal condition at time T_0 is now

$$\tilde{H}_1(T_0, x) = \tilde{B}_1(T_0, x; T) h'(B_{vas}(T_0, x; T)).$$

The solution to this problem has the following probabilistic representation

$$\begin{aligned} \tilde{H}_1(t, x) = \mathbb{E} \left[e^{-\int_t^{T_0} \bar{x}_s ds} \tilde{B}_1(T_0, \bar{x}_{T_0}; T) h'(B_0(T_0, \bar{x}_{T_0}; T)) \right. \\ \left. - \int_t^{T_0} e^{-\int_t^u \bar{x}_s ds} \mathcal{A} H_{vas}(u, \bar{x}_u) du \middle| \bar{x}_t = x \right] \end{aligned}$$

which reduces again to Gaussian integrals since under $\mathbb{P}$, conditional on $\bar{x}_t = x$, the pair $(\int_t^u \bar{x}_s ds, \bar{x}_u)$ is normally distributed for every $t \leq u \leq T_0$.

Using the representation (2.22) for the leading term $H_0(t, x)$ we obtain the corrected bond option pricing formula

$$\begin{aligned} H(t, x, y; T, T_0) &\approx H_0(t, x) + \tilde{H}_1(t, x) \\ &= \mathbb{E} \left[e^{-\int_t^{T_0} \bar{x}_s ds} \left[h(B_0(T_0, \bar{x}_{T_0}; T)) + \tilde{B}_1(T_0, \bar{x}_{T_0}; T) h'(B_0(T_0, \bar{x}_{T_0}; T)) \right] \right] \\ &\quad - \mathbb{E} \left[\int_t^{T_0} e^{-\int_t^u \bar{x}_s ds} \mathcal{A} H_0(u, \bar{x}_u) du | \bar{x}_t = x \right]. \end{aligned}$$

Since the quantity

$$\left[h(B_{vas}(T_0, x; T)) + \tilde{B}_1(T_0, x; T) h'(B_{vas}(T_0, x; T)) \right],$$

is independent of y and is the expansion, up to order $1/\sqrt{\alpha}$, of the terminal condition $h(B(T_0, x, y; T))$, it is also equal, up to order $1/\sqrt{\alpha}$, to the average in the variable y of the terminal condition

$$\langle h(B(T_0, x, y; T)) \rangle,$$

with respect to the invariant distribution of OU process y_t . This last remark enables us to treat the case of non smooth payoff functions since the integral with respect to y regularizes the derivatives of h .

For instance, in the case of a European call bond option, the first term $H_0(t, x)$ is equal to $C_{vas}(t, x)$ obtained explicitly in the constant volatility case (A.10-A.11-A.12) and the correction is

$$\begin{aligned} \tilde{H}_1(t, x) &= \mathbb{E} \left[e^{-\int_t^{T_0} \bar{x}_s ds} \tilde{B}_1(T_0, \bar{x}_{T_0}; T) 1_{\{B_0(T_0, \bar{x}_{T_0}; T) > K\}} | \bar{x}_t = x \right] \\ &\quad - \mathbb{E} \left[\int_t^{T_0} e^{-\int_t^u \bar{x}_s ds} \mathcal{A} C_{vas}(u, \bar{x}_u) du | \bar{x}_t = x \right]. \end{aligned}$$

Using the function D defined in (2.15) and the formula (A.10) for $C_{vas}(t, x)$, the

Model Parameters	Correction Parameters
Rate of mean-reversion of short-rate κ	κ
Long-run mean under $\mathbb{P}$ x_∞	θ
Specific volatility distribution $f(\cdot)$	Mean volatility $\bar{\sigma}$
Rate of mean-reversion of volatility α	Group parameter V_1
Mean-level m of y_t	Group parameter V_2
"V-vol" β	Group parameter V_3
Interest rate risk premium $\lambda_1(\cdot)$	
Volatility risk premium $\lambda_2(\cdot)$	

corrected call price becomes

$$C_{vas}(t, x) + D(T - T_0)B_{vas}(t, x; T)N(h_1) - \int_t^{T_0} \mathbb{E} \left[e^{-\int_t^u \bar{x}_s ds} \mathcal{A}C_{vas}(u, \bar{x}_u) | \bar{x}_t = x \right] du,$$

where the operator $\mathcal{A}$ is given by (2.13) and h_1 by equation A.11:

$$h_1 = \frac{\log \left(\frac{B(t, x; T)}{B(t, x; T_0)} \right) - \log K + \frac{1}{2}v^2}{v}$$

with

$$v^2 = \frac{\sigma^2}{2\kappa^3} (1 - e^{-2\kappa(T_0-t)}) (1 - e^{-\kappa(T-t)})^2,$$

as in the constant parameter Vasicek model.

2.1.8 Implications

The main implication of the asymptotic calculation is reduction of the parametric dependence of the pricing formulas on a specific model. In the table below, the left column lists the original model parameters, including the risk premium functions and the right column the group parameters that need to be estimated from the yield curve using formula (2.16).

In particular, the sensitivities to the correlation ρ and the rate of mean-reversion α of the *hidden* process y_t are absorbed in the group parameters V_1 , V_2 and V_3 . There is also the added benefit that the function $f(\cdot)$, $B(\cdot)$ and $\lambda_2(\cdot)$ do not have

to be specified, and that to this level of approximation, the present level y of the unobserved volatility-driving process does not need to be estimated.

The stochastic volatility corrected prices of bond options also depend on these same parameters in the right column to this order. That is, although the group parameters do not give us enough to recover the law of the processes (x_t, y_t) , they do give us enough to price other interest-rate derivatives under fast mean-reverting stochastic volatility.

2.2 The Exponential Ornstein-Uhlenbeck Stochastic Volatility Model

We now specialize further to a stochastic volatility model, of the type treated in the previous section, where the volatility of x_t is an exponential of the y_t process. We call this the Exponential-Ornstein-Uhlenbeck stochastic volatility (**ExpOU SV**) model. By focusing on one particular choice of the function f which relates the volatility of the x process to the process y we aim to explain more features of the asymptotic approach and derive more explicit formulas.

2.2.1 Market Price of Risk

Once again we shall not assume that the model is presented under the risk neutral measure but instead explicitly parameterize the Girsanov change of measure. This is in keeping with the philosophy outlined in the introduction. That is, we assume “market prices of risk” $\lambda = (\lambda_1, \lambda_2)$ such that the dynamics of $X = (x, y)$ may be represented formally as in 2.1 with (W, Z) now Brownian motion with respect to the risk neutral probability and the drift modified by an amount $\sigma\lambda$.

Here, in the single factor setting, λ_1 parameterizes the inability to hedge against changes in the short rate and λ_2 parameterizes the inability to hedge against changes in the short rate volatility. A similar interpretation applies in the multifactor case where λ_1 parameterizes the inability to hedge against changes in a particular factor - perhaps representing the level, slope or curvature of the yield curve - and λ_2

parameterizes the inability to hedge against changes in this factor's volatility.

2.2.2 Short rate process

The two-scale Vasicek model can be specified by the real world parameters

$$\Omega = (\kappa, x_\infty, m, \nu, \rho, \tilde{\lambda}_1, \tilde{\lambda}_2, \tilde{\epsilon}).$$

The risk neutral process is

$$\begin{aligned} dx_t &= \left(\kappa(x_\infty - x_t) - \tilde{\lambda}_1 e^{y_t} \right) dt + e^{y_t} dW_t \\ dy_t &= \left(\frac{1}{\tilde{\epsilon}}(m - y_t) - \tilde{\lambda}_1 \frac{\nu}{2} \rho - \tilde{\lambda}_2 \frac{\nu}{2} \sqrt{1 - \rho^2} \right) dt + \frac{\nu}{\sqrt{2\tilde{\epsilon}}} \left(\rho dW_t + \sqrt{1 - \rho^2} dZ_t \right) \end{aligned} \quad (2.25)$$

where $(\tilde{\lambda}_1, \tilde{\lambda}_2)$ are market prices of risk (associated with the short rate and short rate volatility), $\tilde{\epsilon} \ll 1$ is a scale factor and ρ allows for correlation between the short rate and short rate volatility.

2.2.3 Corrections to Bond Prices

From the results of the previous section and elementary moment calculations we are able to write down explicitly the correction to bond prices arising from introduction of correlated stochastic volatility. We calculate averaged risk adjusted parameters

$$\begin{aligned} \bar{\sigma} &= e^{m+\nu^2} \\ \bar{\theta} &= x_\infty + \frac{\bar{\sigma} \tilde{\lambda}_1}{\kappa} e^{-\nu^2/2} \end{aligned}$$

and group parameters

$$\begin{aligned}\tilde{V}_1 &= -\frac{\tilde{\lambda}_1}{\sqrt{2}} \left(\rho \tilde{\lambda}_1 + \sqrt{1 - \rho^2} \tilde{\lambda}_2 \right) e^{m + \frac{1}{2} \nu^2} \\ \tilde{V}_2 &= -\frac{1}{\sqrt{2}} \left(\rho \tilde{\lambda}_1 + \sqrt{1 - \rho^2} \tilde{\lambda}_2 \right) \bar{\sigma}^2 - \frac{e^{3m} \rho \tilde{\lambda}_1 \sqrt{2}}{\nu} \left(e^{\nu^2} - e^{\frac{1}{2} \nu^2} \right) \\ \tilde{V}_3 &= -\frac{e^{3m} \rho}{\sqrt{2}} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right).\end{aligned}$$

Then, defining the function

$$\tilde{V}(t) = \tilde{V}_1 \int_0^t \tau(s; \kappa) ds - \tilde{V}_2 \int_0^t \tau^2(s; \kappa) ds + \tilde{V}_3 \int_0^t \tau^3(s; \kappa) ds$$

where

$$\tau(s; \kappa) = \frac{1 - e^{-\kappa t}}{\kappa},$$

the bond price is given by

$$B^{\bar{\epsilon}}(t, x, T) = \left(1 - \sqrt{\bar{\epsilon}} \nu \tilde{V}(T - t) \right) B_{vas}(x, t, T; \kappa, \theta, \sigma),$$

where $B_{vas}(x, t, T; \kappa, \theta, \sigma)$ is the standard Vasicek bond price

$$B_{vas}^-(t, x; T) = e^{-a_{vas}(T-t, x) - \tau(T-t; \kappa)x}$$

with

$$a_{vas}(s) = \tilde{\theta} (s - \tau(s; \kappa)) + \frac{\sigma^2}{4\kappa} \tau(s; \kappa)^2$$

reviewed in appendix A.

2.2.4 Corrections to the Moment Generating Function

As explained in appendix B it is useful to have at our disposal the moment generating function B.8 for the joint negative short rate and negative integrated short rate process

$(-x_s, -\int x_s):$

$$g(t, x, T; \alpha, \beta) := \mathbb{E}_t \left[\exp \left(-\alpha \cdot X(T) - \beta \cdot \int_t^T X(u) du \right) \right].$$

Here α and β are constants unrelated to those appearing in chapter 2. The bond price is a special case obtained by setting $\beta = 1$ and $\alpha = 0$. The moment generating function is no harder to derive than the bond price and we omit the details as they are similar to the calculations already performed. Using the same averaged parameters and group parameters as for the bond price the correction to the moment generating function is proportional to

$$\begin{aligned} \tilde{V}(t) = & \tilde{V}_1 \int_0^t (\alpha \tau'(s) + \beta \tau(s)) ds - \tilde{V}_2 \int_0^t (\alpha \tau'(s) + \beta \tau(s))^2 ds \\ & + \tilde{V}_3 \int_0^t (\alpha \tau'(s) + \beta \tau(s))^3 ds. \end{aligned}$$

The generating function is given by

$$\begin{aligned} g_{\text{expousv}}(t, T; \alpha, \beta) &= \left(1 - \sqrt{\tilde{\epsilon}} \nu \tilde{V}(T-t) \right) \exp(-a_{\text{vas}}(T-t; \alpha, \beta) \\ &\quad - b_{\text{vas}}(T-t; \alpha, \beta) x) + O(\tilde{\epsilon}) \\ &= \exp(-a_{\text{expousv}}(T-t; \alpha, \beta) - b_{\text{vas}}(T-t; \alpha, \beta) x) + O(\tilde{\epsilon}) \end{aligned}$$

where the coefficient $b_{\text{vas}}(t; \alpha, \beta)$ of x is the same as in the constant coefficient Vasicek model:

$$b_{\text{vas}}(t; \alpha, \beta) = \alpha \tau'(t) + \beta \tau(t)$$

The coefficient

$$a_{\text{vas}}(t; \alpha, \beta) = \kappa \theta \int_0^t (\alpha \tau'(s) + \beta \tau(s)) ds - \frac{\sigma^2}{2} \int_0^t (\alpha \tau'(s) + \beta \tau(s))^2 ds$$

is not the same as in the constant coefficient Vasicek model. It is modified by a small correction term $\sqrt{\tilde{\epsilon}}\nu V$. We have

$$\begin{aligned} b_{\text{expousv}}(t; \alpha, \beta) &= b_{\text{vas}}(t; \alpha, \beta) = (\alpha\tau'(t) + \beta\tau(t)) \\ a_{\text{expousv}}(t; \alpha, \beta) &= a_{\text{expousv}}(t; \alpha, \beta) + \sqrt{\tilde{\epsilon}}\nu\tilde{V} \\ &= \kappa\theta \int_0^t (\alpha\tau'(s) + \beta\tau(s)) ds - \frac{\sigma^2}{2} \int_0^t (\alpha\tau'(s) + \tau(s))^2 ds + \sqrt{\tilde{\epsilon}}\nu\tilde{V} \end{aligned}$$

where again $\tau(s) = \tau(s; \kappa) = \frac{1-e^{-\kappa s}}{\kappa}$.

2.2.5 A Clean Reparameterization

The volatility process has root mean square level $\sigma = \exp(m + \nu^2)$ and its standard deviation is $\xi = \exp(\nu^2/2) \approx 1$. A convenient scale parameter is $\epsilon := \rho\xi\sqrt{\tilde{\epsilon}/2}$, and to facilitate comparison with the standard Vasicek model we write $\lambda_1 := \tilde{\lambda}_1/\xi \approx \tilde{\lambda}_1$. It is also convenient to absorb another factor into the market price of volatility risk by letting $\lambda_2 := \tilde{\lambda}_2\sqrt{1-\rho^2}/(\rho\xi)$. In this manner we arrive at a new parameterization

$$\Omega = \{\kappa, \theta, \sigma, \xi, \lambda_1, \lambda_2, \epsilon\}.$$

The stochastic correction term $\sqrt{\tilde{\epsilon}}\nu\tilde{V}$ becomes simply ϵV where

$$\begin{aligned} V_1 &= -\lambda_1(\lambda_1 + \lambda_2)\sigma \\ V_2 &= -(\lambda_1 + \lambda_2)\sigma^2 - 2\lambda_1\sigma^3(\xi^{-4} - \xi^{-1}) \\ V_3 &= -\sigma^3\sqrt{2}(\xi^3 - \xi^{-1}) \\ V(t) &= V_1 \int_0^t \tau(s; \kappa) ds - V_2 \int_0^t \tau^2(s; \kappa) ds + V_3 \int_0^t \tau^3(s; \kappa) ds \end{aligned}$$

Calculating integrals of polynomials in $\tau(s; \kappa) = \frac{1-e^{-\kappa s}}{\kappa}$:

$$\int_0^t \tau^n(s) ds = \frac{t}{\kappa^n} - \sum_{j=1}^n \frac{\tau^j(t)}{j\kappa^{n-j+1}}$$

presents no difficulty.

2.2.6 An Analytic Comparison of Small Jumps and Volatility Fluctuations

The exponential Ornstein-Uhlenbeck stochastic volatility model allows for bursts of volatility which may allow for large changes in the short rate. Excursions by the y_t process are magnified by the exponential function, resulting in high volatility. It is plausible then that the effect on yields may be similar to a Vasicek model to which we add jumps. In this section we show that this is indeed the case and that to first order in the parameter determining jump magnitude the correction to bond yields brought about by jumps can be replicated by the ExpOU SV model.

To the Vasicek short rate model we add memoryless exponentially distributed jumps with Poisson arrival times. The model is specified by

$$\Omega = (\kappa, x_\infty, \lambda, \sigma, \eta, \mu),$$

although using $\theta := x_\infty - \frac{\lambda\sigma}{\kappa}$ is more convenient. The short rate obeys the SDE

$$dx_t = \kappa \left(x_\infty - \frac{\lambda\sigma}{\kappa} - x_t \right) dt + \Delta J + \sigma dW_t \quad (2.27)$$

or simply

$$dx_t = \kappa (\theta - x_t) dt + \Delta J + \sigma dW_t$$

where J is a jump process with arrival intensity η . Without loss of generality we assume the exponentially distributed jumps have positive mean jump size $\mu > 0$. Negative jumps are added in a similar fashion.

The generator for the new process is the same as the vanilla Vasicek with an additional term

$$\mathcal{L}_J = \eta \left[\frac{1}{1 + \mu b_{\text{vas}}(t; \alpha, \beta)} - 1 \right]. \quad (2.28)$$

The moment generating function B.8 remains exponential affine, with only the deterministic coefficient “ a ” modified by the integral of $\mathcal{L}_J$. We have

$$\begin{aligned}
 g(t, x, T; \alpha, \beta) &= \exp(-a_{\text{vas+j}}(T-t; \alpha, \beta) - b_{\text{vas+j}}(T-t; \alpha, \beta)x) \\
 b_{\text{vas+j}}(t; \alpha, \beta) &= b_{\text{vas}}(t; \alpha, \beta) = \alpha\tau'(t) + \beta\tau(t) \\
 a_{\text{vas+j}}(t; \alpha, \beta) &= a_{\text{vas}}(t; \alpha, \beta) - \int_0^t \mathcal{L}_J(s) ds \\
 &= \kappa\theta \int_0^t (\alpha\tau'(s) + \beta\tau(s)) ds - \frac{\sigma^2}{2} \int_0^t (\alpha\tau'(s) + \beta\tau(s))^2 ds \\
 &\quad + \eta t + \eta I_{\text{vas+j}}(t)
 \end{aligned}$$

where

$$\begin{aligned}
 I_{\text{vas+j}} &= \int_0^t \frac{1}{1 + \mu(\alpha\tau'(s) + \beta\tau(s))} ds \\
 &= \frac{\kappa}{\kappa + \beta\mu} \left[t + \log \left(1 + \mu \frac{\beta - \kappa\alpha}{1 + \mu\alpha} \tau(t) \right) \right]
 \end{aligned}$$

The last integral is left to appendix C. The addition of jumps to the ExpOU SV model is achieved in the same manner, the stochastic volatility correction failing to interact with the correction to the generating function caused by the addition of jumps. The functional form of the jump and stochastic volatility corrections to the generating function are quite different, but we can attempt to understand them by expanding them in a non-orthogonal basis of monotonic functions of maturity. We recall the useful quantity τ and define also a “jump maturity” $\tilde{t}$ as follows:

$$\begin{aligned}
 \tau(t; \kappa) &:= \frac{1 - e^{-\kappa t}}{\kappa} \\
 \tilde{\mu} &:= \mu \frac{\beta - \alpha\kappa}{1 + \alpha\mu}, \\
 \tilde{t}(\tau; \mu) &:= \frac{\log(1 + \tilde{\mu}\tau(t))}{\tilde{\mu}}
 \end{aligned}$$

The jump maturity is related to τ in similar manner to that in which t is related to

τ :

$$\tau = \frac{e^{\tilde{\mu}\tilde{t}-1}}{\tilde{\mu}}.$$

The jump maturity allows us to express the generating function coefficient “a” for the respective Vasicek models as a linear combination of members of the bases

$$\begin{aligned}\mathcal{A}_{\text{expousv+j}} &:= \{1, t, \tilde{t}(t), \tau(t), \tau^2(t), \tau^3(t)\} \\ \mathcal{A}_{\text{expousv}} &:= \{1, t, \tau(t), \tau^2(t), \tau^3(t)\} \\ \mathcal{A}_{\text{vas+j}} &:= \{1, t, \tilde{t}(t), \tau(t), \tau^2(t)\} \\ \mathcal{A}_{\text{vas}} &:= \{1, t, \tau(t), \tau^2(t)\}.\end{aligned}$$

This can be seen from the integrals carried out in the appendix. The b term is affine in $\tau(t)$ for all models and therefore spanned by

$$\mathcal{B}_{\text{vas}} := \{1, \tau(t)\}$$

For concreteness we consider the one factor setting and suppose that we wish to fit today's yield curve. The bond prices are obtained by setting $\alpha = 0$ and $\beta = 1$ which implies $q = 1$, and $\tilde{\mu} = \mu$. The yields are spanned by

$$\begin{aligned}\mathcal{Y}_{\text{expousv+j}} &:= \left\{1, \frac{\tilde{t}}{t}, \frac{\tau(t)}{t}, \frac{\tau^2(t)}{t}, \frac{\tau^3(t)}{t}\right\} \\ \mathcal{Y}_{\text{expousv}} &:= \left\{1, \frac{\tau(t)}{t}, \frac{\tau^2(t)}{t}, \frac{\tau^3(t)}{t}\right\} \\ \mathcal{Y}_{\text{vas+j}} &:= \left\{1, \frac{\tilde{t}}{t}, \frac{\tau(t)}{t}, \frac{\tau^2(t)}{t}\right\} \\ \mathcal{Y}_{\text{vas}} &:= \left\{1, \frac{\tau(t)}{t}, \frac{\tau^2(t)}{t}\right\}\end{aligned}$$

and one can already get an idea of the relative flexibility of the models. For example, to compare vas+j and expousv we can approximate $\tilde{t}$ by a power series in $\tau(t)$

$$\tilde{t} = \tau - \frac{\mu}{2}\tau^2 + \frac{1}{3}\mu^2\tau^3 - \frac{1}{4}\mu^3\tau^4 + O(\mu^4).$$

It is apparent that up to order μ^2 the yield curve for the Vasicek model with jumps can be replicated by the ExpOU SV model. By contrast, the Vasicek model without the stochastic correction term can replicate at best terms up to order μ . (Put another way, the addition of jumps to the two scale model when μ is small adds little to the ability to fit bond prices, the difference being of order μ^3). This potential for replication opens up the possibility of using the ExpOU SV doubly stochastic process and Vasicek jump processes as substitutes in calculation. It is not difficult to arrive at explicit formulas for the first few terms in the yield basis. For the yields for the ExpOU SV model with jumps are approximately

$$\begin{aligned}
y(t) &\approx y_0 + y_1 \frac{\tau(t)}{t} + y_2 \frac{\tau^2(t)}{t} + y_3 \frac{\tau^3(t)}{t} + y_4 \frac{\tau^4(t)}{t} \\
y_0 &= \frac{\kappa \theta + \epsilon V^{(1)}}{\kappa} - \frac{1/2 \sigma^2 + \epsilon V^{(1)}}{\kappa^2} + \frac{\epsilon V^{(3)}}{\kappa^3} + \frac{\eta}{1 + \kappa} \\
y_1 &= -\frac{\kappa \theta + \epsilon V^{(1)}}{\kappa} + \frac{1/2 \sigma^2 + \epsilon V^{(2)}}{\kappa^2} - \frac{\epsilon V^{(3)}}{\kappa^3} - \frac{\eta \kappa \mu}{\kappa + \mu} \\
y_2 &= 2 \frac{1/2 \sigma^2 + \epsilon V^{(2)}}{\kappa} - 1/2 \frac{\epsilon V^{(3)}}{\kappa^2} + \frac{\eta \kappa \mu^2}{2 \kappa + 2 \mu} \\
y_3 &= -1/3 \frac{\epsilon V^{(3)}}{\kappa} - 1/3 \frac{\eta \kappa \mu^3}{\kappa + \mu} \\
y_4 &= 1/4 \frac{\eta \kappa \mu^4}{\kappa + \mu}
\end{aligned}$$

or even more explicitly

$$\begin{aligned}
y_0 &= \frac{\kappa \theta - \epsilon \lambda_1 (\lambda_1 + \lambda_2) \sigma}{\kappa} - (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-2} \\
&\quad - \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-3} + \frac{\eta}{1 + \kappa} \\
y_1 &= -\frac{\kappa \theta - \epsilon \lambda_1 (\lambda_1 + \lambda_2) \sigma}{\kappa} + (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-2} \\
&\quad + \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-3} - \frac{\eta \kappa \mu}{\kappa + \mu} \\
y_2 &= 2 (1/2 \sigma^2 + \epsilon (-(\lambda_1 + \lambda_2) \sigma^2 - 2 \lambda_1 \sigma^3 (\xi^{-4} - \xi^{-1}))) \kappa^{-1} \\
&\quad + 1/2 \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-2} + \frac{\eta \kappa \mu^2}{2 \kappa + 2 \mu} \\
y_3 &= 1/3 \epsilon \sigma^3 \sqrt{2} (\xi^3 - \xi^{-1}) \kappa^{-1} - 1/3 \frac{\eta \kappa \mu^3}{\kappa + \mu} \\
y_4 &= 1/4 \frac{\eta \kappa \mu^4}{\kappa + \mu}
\end{aligned}$$

We have not pushed this idea further in a pricing or estimation setting.

2.2.7 Correction to the Conditional Density

The reader unfamiliar with fixed income literature can consider this a set of calculations for the approximate conditional probability density of a two dimensional (two factor) diffusion process. The interesting thing about the calculation is that the conditional density is (to first order approximation) a function of the first factor (the x_t process) alone. This arises because the second variable (the y_t process) is assumed to fluctuate rapidly in comparison with the first. The difference in time scales between the x_t and y_t processes - loosely the ratio between the mean reversion rates of - is parameterized by a scale parameter $\epsilon \ll 1$.

The conditional density or transition probabilities we calculate are a perturbation of those for the constant parameter Ornstein-Uhlenbeck process with which one might estimate a traditional Vasicek model. We briefly review this, making a slight change of notation.

Conditional Density for the Ornstein-Uhlenbeck Process

The primary process, representing the short rate, is denoted

$$dx_t = \kappa(x_\infty - x)dt + \sigma dW^*$$

under the real world measure. It transforms to

$$dx_t = [\kappa(x_\infty - x) - \lambda\sigma]dt + \sigma dW$$

under the risk neutral measure when a simple market price of risk λ is assumed. This simplifies to

$$d\tilde{x} = -\kappa\tilde{x}dt + dW$$

under the change of variable $\tilde{x} = (x - \theta + \frac{\sigma\lambda}{\kappa})/\sigma$. This Ornstein-Uhlenbeck process is gaussian with conditional mean at time t_2 equal to $e^{-\kappa(t_2-t_1)}$ times the initial value of $\tilde{x}$ at time t_1 . The conditional density of the original process x_t is therefore

$$p(x_1, t_1, x_2, t_2) = \frac{1}{\sigma\nu(t_1, t_2)\sqrt{2\pi}} \exp\left(-\frac{(x_2 - x_1(1 - e^{-\kappa(t_2-t_1)}))^2}{2\sigma\nu(t_1, t_2)}\right)$$

where

$$\nu^2(t_1, t_2) = (1 - e^{-2\kappa(t_2-t_1)})/2\kappa$$

is the conditional variance.

Conditional Density for the Ornstein-Uhlenbeck Process with Fluctuating Volatility

To avoid being swamped in notation we consider the joint process

$$dx_t^\varepsilon = \kappa(\theta - x_t^\varepsilon)dt + e^{y_t^\varepsilon} dZ_t \quad (2.29)$$

$$dy_t^\varepsilon = \frac{\alpha}{\varepsilon}(m - y_t^\varepsilon)dt + \frac{\beta}{\sqrt{\varepsilon}}\left(\rho dZ_t + \sqrt{1 - \rho^2}dB_t\right) \quad (2.30)$$

with scale parameter $\epsilon \ll 1$. We do not explicitly accomodate market price of risk parameters here, and without loss of generality it suffices to treat the slightly cleaned up version:

$$dx_t^\epsilon = -\kappa x_t^\epsilon dt + \sigma e^{y_t^\epsilon} dZ_t \quad (2.31)$$

$$dy_t^\epsilon = -\frac{1}{\epsilon} y_t^\epsilon dt + \sqrt{2} \frac{\nu}{\sqrt{\epsilon}} \left(\rho dZ_t + \sqrt{1 - \rho^2} dB_t \right) \quad (2.32)$$

which is convenient because the invariant density for the Ornstein-Uhlenbeck process y is Gaussian with mean zero and variance ν^2 . Furthermore, the reversion of the x factor is also a distraction and we make the change of variable $z_t^\epsilon := e^{kt} x_t^\epsilon$ so that

$$dz_t^\epsilon = -\kappa z_t^\epsilon dt + \sigma e^{y_t^\epsilon} dZ_t \quad (2.33)$$

$$dy_t^\epsilon = -\frac{1}{\epsilon} y_t^\epsilon dt + \sqrt{2} \frac{\nu}{\sqrt{\epsilon}} \left(\rho dZ_t + \sqrt{1 - \rho^2} dB_t \right) \quad (2.34)$$

Now let $\tilde{p}(z_1, y_1, t_1; z_2, y_2, t_2)$ be the transition probability for this process and let $\bar{p}(z_1, y_1, t_1; z_2, t_2)$ be the marginal with y_2 integrated out. It will turn out that in fact

$$\bar{p}(z_1, y_1, t_1; z_2, t_2) = p_0(z_1, t_1; z_2, t_2) + \sqrt{\epsilon} p_1(z_1, t_1; z_2, t_2) + O(\epsilon)$$

for some functions p_0 and p_1 . Here p_0 is a simple gaussian (fundamental solution to the heat equation). The variance entering p_0 is that which we would expect from sampling the stationary (equilibrium) distribution of y . The term p_1 is a more complicated correction term which takes into account the fact that the y process has not yet reached equilibrium. Thus it makes sense up to order ϵ to speak about a conditional density for the process x_t alone. This is what we wish to calculate.

Asymptotic Approach to the Backward Kolmogorov Equation

The probability $\tilde{p}(z_1, y_1, t_1; z_2, y_2, t_2)$ satisfies a backward equation in the variables z_1, y_1 and a forward equation in the variables z_2, y_2 . We choose to fix z_2, y_2, t_2 and work with the backward equation. By a translation argument it is sufficient to fix

a particularly simple z_2 and t_2 . Namely, $z_2 = 0$ and $t_2 = 0$. For notational convenience we then replace t by $-t$ so that the backward equation for $\tilde{p}(z_1, y_1, t_1; z_2, y_2, t_2)$ translates to a PDE for $p(z, y, t) := \tilde{p}(z, y, -t; 0, y_2, 0)$, namely

$$\left(\frac{\partial}{\partial t} + \mathcal{L}^\varepsilon\right) p(z, y, t) = \left\{ \frac{\partial}{\partial t} + \frac{1}{\varepsilon} Q + \frac{1}{\sqrt{\varepsilon}} \mathcal{L}_1 + \mathcal{L}_2 \right\} p(z, y, t) = 0 \quad (2.35)$$

where

$$\begin{aligned} Q &= \frac{1}{2} 2\nu^2 \frac{\partial^2}{\partial y^2} - y \frac{\partial}{\partial y}, \\ \mathcal{L}_1 &= \sqrt{2\nu\rho} e^{kt+y} \frac{\partial^2}{\partial y \partial z}, \\ \mathcal{L}_2 &= \frac{1}{2} e^{2kt+2y} \frac{\partial^2}{\partial z^2} \end{aligned}$$

The boundary condition on $p(z, y, t)$ is somewhat beside the point as we will not be solving for the higher order terms in y_2 . It is easier to consider the boundary behavior of $\bar{p}(z, \cdot, t)$ which is the Dirac function $\delta_0(z)$ at time $t = 0$ supported at $z = 0$. The PDE has an asymptotic solution which we can find by defining p_0 and p_1 to be solutions of

$$\left\{ \frac{\partial}{\partial t} + \langle \mathcal{L}_2 \rangle_{OU} \right\} p_0(z, t) = 0 \quad (2.36)$$

$$\left\{ \frac{\partial}{\partial t} + \langle \mathcal{L}_2 \rangle_{OU} \right\} p_1(z, t) = \mathcal{A} p_0(z, t) \quad (2.37)$$

where the left hand side operator $\langle \mathcal{L}_2 \rangle_{OU}$ is the averaged version of $\mathcal{L}_2$

$$\langle \mathcal{L}_2 \rangle_{OU} = \frac{1}{2} e^{2kt} \langle e^{2y} \rangle_{OU} \frac{\partial^2}{\partial z^2}$$

and the operator $\mathcal{A}$ is

$$\mathcal{A} = \langle \mathcal{L}_1 Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle_{OU}) \rangle_{OU}.$$

Guided by earlier calculations we have designed $\mathcal{A}$ here so that $p_0(z, t) + \sqrt{\epsilon}p_1(z, t)$ as defined by 2.36 and 2.37 formally solves the PDE (2.35) up to order ϵ . We omit these manipulations and turn to the problem of unraveling the action of the operator $\mathcal{A}$.

Determining the Forcing Term

The action of the operator $\mathcal{A}$ can be deduced by assuming a solution $\phi(y)$ to

$$Q\phi(y) = e^{2y} - \langle e^{2y} \rangle_{OU}.$$

Then for any “nice” $\Psi(z, y, t)$,

$$\begin{aligned} \langle \mathcal{L}_1 Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle_{OU}) \rangle_{OU} \Psi(z, y, t) &= \langle \mathcal{L}_1 Q^{-1} (\mathcal{L}_2 - \langle \mathcal{L}_2 \rangle_{OU}) \Psi \rangle_{OU} \\ &= \left\langle \mathcal{L}_1 Q^{-1} \left(\frac{1}{2} e^{2\kappa t + 2y} - \left\langle \frac{1}{2} e^{2\kappa t + 2y} \right\rangle \right) \frac{\partial^2}{\partial z^2} \Psi(z, t) \right\rangle \\ &= \left\langle \mathcal{L}_1 Q^{-1} \left(Q \frac{1}{2} e^{2\kappa t} \phi(y) \frac{\partial^2}{\partial z^2} \Psi(z, t) \right) \right\rangle \\ &= \left\langle \mathcal{L}_1 \frac{1}{2} e^{2\kappa t} \phi(y) \frac{\partial^2}{\partial z^2} \Psi(z, t) \right\rangle \\ &= \left\langle \sqrt{2\nu} \rho e^{\kappa t + y} \frac{\partial^2}{\partial y \partial z} \frac{1}{2} e^{2\kappa t} \phi(y) \frac{\partial^2}{\partial z^2} \Psi(z, t) \right\rangle \\ &= \sqrt{2\nu} \rho e^{2\kappa t} \frac{1}{2} e^{\kappa t} \langle \phi'(y) e^y \rangle \frac{\partial^3}{\partial z^3} \Psi(z, t) \\ &= -\sqrt{2\nu} \rho e^{3\kappa t} \frac{1}{2} \frac{1}{\nu^2} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right) \frac{\partial^3}{\partial z^3} \Psi(z, t) \end{aligned}$$

The last equality is obtained by integration by parts. Using $Q\phi(y) = e^{2y} - e^{2\nu^2}$ one can calculate

$$\begin{aligned} \langle \phi'(y) e^y \rangle_{OU} &= \int_{-\infty}^{\infty} \phi'(y) e^y \frac{1}{\sqrt{2\pi\nu^2}} e^{-\frac{y^2}{2\nu^2}} dy \\ &= -\frac{1}{\nu^2} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right). \end{aligned}$$

Thus

$$\mathcal{A} = -ae^{3kt} \frac{\partial^3}{\partial z^3}$$

where

$$a = \frac{\rho}{\nu\sqrt{2}} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right).$$

We exploit the relative simplicity of this operator.

The Fast Mixing Limit

Substituting the explicit representation of the operator $\mathcal{A}$ into the partial differential equation 2.37 we obtain:

$$\left\{ \frac{\partial}{\partial t} + \langle \mathcal{L}_2 \rangle_{OU} \right\} p_0(z, t) = 0 \quad (2.38)$$

$$\left\{ \frac{\partial}{\partial t} + \langle \mathcal{L}_2 \rangle_{OU} \right\} p_1(z, t) = -ae^{3kt} \frac{\partial^3}{\partial z^3} p_0(z, t). \quad (2.39)$$

Solution of this PDE is facilitated by introducing a new time coordinate

$$\tau_{2\kappa}(t) := \tau(t; 2\kappa) = \frac{1 - e^{-2kt}}{2k}.$$

We also exploit the monotonicity of $t \rightarrow \tau(t; 2\kappa)$ and define scaled functions $q_0(z, \tau)$ and $q_1(z, \tau)$ implicitly:

$$q_0(z, \tau_{2\kappa}(t)) := p_0(z, t),$$

$$q_1(z, \tau_{2\kappa}(t)) := p_1(z, t).$$

With these definitions equations 2.38 transform to

$$\left\{ \frac{\partial}{\partial \tau} + \frac{\bar{\sigma}^2}{2} \frac{\partial^2}{\partial z^2} \right\} q_0(z, \tau_{2\kappa}) = 0$$

where $\bar{\sigma} = \sigma \sqrt{\langle e^{2y} \rangle_{OU}} = \sigma e^{\nu^2}$. The initial value is the Dirac distribution $\delta_0(z)$ and the solution is just the fundamental solution of the heat equation

$$q_0(z, \tau_{2\kappa}) = \frac{1}{\sqrt{2\pi\tau_{2\kappa}\bar{\sigma}^2}} \exp\left(\frac{-z^2}{2\tau_{2\kappa}\bar{\sigma}^2}\right)$$

This demonstrates that once again, a few changes of variables notwithstanding, the fast mixing limit is given by the standard result for the constant parameter process.

Stochastic Volatility Correction

The second equation (2.39) becomes

$$\left\{ \frac{\partial}{\partial \tau} + \frac{\bar{\sigma}^2}{2} \frac{\partial^2}{\partial z^2} \right\} q_1(z, \tau_{2\kappa}) = -a(1 - 2k\tau_{2\kappa})^{5/2} \frac{\partial^3}{\partial z^3} q_0(z, \tau_{2\kappa})$$

which one can solve using Duhamel's principle [26]. The nonhomogeneous component of the solution is the part we want:

$$q_1(z, \tau_{2\kappa}) = -a \int_0^{\tau_{2\kappa}} \int_{-\infty}^{\infty} q_0(z - z', \tau_{2\kappa} - \tau'_{2\kappa}) (1 - 2k\tau_{2\kappa})^{5/2} \frac{\partial^3}{\partial z'^3} q_0(z', \tau'_{2\kappa}) dz' d\tau'_{2\kappa}.$$

given our boundary conditions. We now demonstrate a solution, up to a one dimensional numerical quadrature, for $q_1(z, \tau_{2\kappa})$. It is convenient to rescale variables again.

The function $r_1(z, \tau_{2\kappa}) := \bar{\sigma}^4 q_1(z\bar{\sigma}, \tau_{2\kappa})$ can be computed

$$\begin{aligned} r_1(z, \tau_{2\kappa}) &:= \bar{\sigma}^4 q_1(z\bar{\sigma}, \tau_{2\kappa}) \\ &= -a\bar{\sigma}^4 \int_0^{\tau_{2\kappa}} \int_{-\infty}^{\infty} q_0(\bar{\sigma}z - z', \tau_{2\kappa} - \tau'_{2\kappa}) (1 - 2k\tau_{2\kappa})^{5/2} \frac{\partial^3}{\partial z'^3} q_0(z', \tau'_{2\kappa}) dz' d\tau'_{2\kappa} \\ &= -\frac{a}{\sqrt{2\pi}} \int_0^{\tau_{2\kappa}} \sqrt{\tau_{2\kappa}/\tau'_{2\kappa} - 1} (1 - 2k\tau'_{2\kappa})^{3/2} \left(\frac{3z}{\tau'_{2\kappa}{}^2} - \frac{3z}{\tau'_{2\kappa}{}^3} - \frac{\tau_{2\kappa}^2 z^3}{\tau'_{2\kappa}{}^5} \right) \dots \\ &\quad \exp\left(\frac{1}{2} \left(\frac{\tau_{2\kappa}^3}{\tau'_{2\kappa}{}^3} - 1 \right) \frac{z^2}{\tau_{2\kappa} - \tau'_{2\kappa}}\right) d\tau'_{2\kappa}. \end{aligned} \tag{2.40}$$

where the last equality is deferred momentarily and we recall

$$a = \frac{\rho}{\nu\sqrt{2}} \left(e^{9\nu^2/2} - e^{5\nu^2/2} \right).$$

Tracing back through our numerous substitutions one may thereby compute the first order correction to the likelihood function arising from fluctuations in the y_t process. We hope to use this result in future estimation work, but caution the reader that there are subtleties in its application due to the assumed time scale separation. The equality 2.40 is elementary but not obvious. Using another change of variable

$$\tilde{z} \rightarrow \tilde{z}/\bar{\sigma},$$

the algebraic observation

$$\frac{(z - \tilde{z})^2}{\tau - \tilde{\tau}} + \frac{\tilde{z}^2}{\tilde{\tau}} = \frac{\left(\tilde{z} - \frac{\tau}{\tilde{\tau}}z\right)^2}{\frac{\tilde{\tau}}{\tau}(\tau - \tilde{\tau})} - \left(\frac{\tau^3}{\tilde{\tau}^3} - 1\right) \frac{z^2}{\tau - \tilde{\tau}}$$

and the elementary facts

$$\begin{aligned} \int_{-\infty}^{\infty} x e^{-(x-a)^2/2\sigma^2} dx &= a\sqrt{2\pi\sigma^2} \\ \int_{-\infty}^{\infty} x^3 e^{-(x-a)^2/2\sigma^2} dx &= a^3\sqrt{2\pi}\sigma + 3a\sqrt{2\pi}\sigma^3 \end{aligned}$$

we verify this (dropping the subscripts on τ_{2k} and τ'_{2k}),

$$\begin{aligned}
r_1(z, \tau) &= \bar{\sigma}^4 q_1(\bar{\sigma}z, \tau) \\
&= -a\bar{\sigma}^4 \int_0^\tau \int_{-\infty}^\infty q_0(z - \tilde{z}, \tau - \tilde{\tau}) (1 - 2k\tau)^{5/2} \frac{\partial^3}{\partial z'^3} q_0(\tilde{z}, \tilde{\tau}) d\tilde{z} d\tilde{\tau} \\
&= -a\bar{\sigma}^4 \int_0^\tau \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})\bar{\sigma}^2}} \exp\left(\frac{-(z\bar{\sigma} - \tilde{z})^2}{2(\tau - \tilde{\tau})\bar{\sigma}^2}\right) (1 - 2k\tau)^{5/2} \dots \\
&\quad \frac{\partial^3}{\partial z'^3} \frac{1}{\sqrt{2\pi\tilde{\tau}\bar{\sigma}^2}} \exp\left(\frac{-\tilde{z}^2}{2\tilde{\tau}\bar{\sigma}^2}\right) d\tilde{z} d\tilde{\tau} \\
&= -a\bar{\sigma}^4 \int_0^\tau (1 - 2k\tau)^{5/2} \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})\bar{\sigma}^2}} \frac{1}{\sqrt{2\pi\tilde{\tau}\bar{\sigma}^2}} \dots \\
&\quad \int_{-\infty}^\infty \exp\left(\frac{-(z\bar{\sigma} - \tilde{z})^2}{2(\tau - \tilde{\tau})\bar{\sigma}^2}\right) \frac{\partial^3}{\partial z'^3} \exp\left(\frac{-\tilde{z}^2}{2\tilde{\tau}\bar{\sigma}^2}\right) d\tilde{z} d\tilde{\tau} \\
&= -a \int_0^\tau (1 - 2k\tau)^{5/2} \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})}} \frac{1}{\sqrt{2\pi\tilde{\tau}}} \dots \\
&\quad \int_{-\infty}^\infty \exp\left(\frac{-(z - \tilde{z})^2}{2(\tau - \tilde{\tau})}\right) \frac{\partial^3}{\partial z'^3} \exp\left(\frac{-\tilde{z}^2}{2\tilde{\tau}}\right) d\tilde{z} d\tilde{\tau} \\
&= -a \int_0^\tau (1 - 2k\tau)^{5/2} \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})}} \frac{1}{\sqrt{2\pi\tilde{\tau}}} \dots \\
&\quad \int_{-\infty}^\infty \exp\left(\frac{-(z - \tilde{z})^2}{2(\tau - \tilde{\tau})}\right) \left(\frac{3\tilde{z}}{\tilde{\tau}^2} - \frac{\tilde{z}^3}{\tilde{\tau}^3}\right) \exp\left(\frac{-\tilde{z}^2}{2\tilde{\tau}}\right) d\tilde{z} d\tilde{\tau} \\
&= -a \int_0^\tau (1 - 2k\tau)^{5/2} \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})}} \frac{1}{\sqrt{2\pi\tilde{\tau}}} \dots \\
&\quad \int_{-\infty}^\infty \left(\frac{3\tilde{z}}{\tilde{\tau}^2} - \frac{\tilde{z}^3}{\tilde{\tau}^3}\right) \exp\left(\frac{-(z - \tilde{z})^2}{2(\tau - \tilde{\tau})} - \frac{\tilde{z}^2}{2\tilde{\tau}}\right) d\tilde{z} d\tilde{\tau} \\
&= -a \int_0^\tau (1 - 2k\tau)^{5/2} \frac{1}{\sqrt{2\pi(\tau - \tilde{\tau})}} \frac{1}{\sqrt{2\pi\tilde{\tau}}} \dots \\
&\quad \int_{-\infty}^\infty \left(\frac{3\tilde{z}}{\tilde{\tau}^2} - \frac{\tilde{z}^3}{\tilde{\tau}^3}\right) \exp\left(-\frac{1}{2} \frac{(\tilde{z} - \frac{\tau}{\tilde{\tau}}z)^2}{\frac{\tilde{\tau}}{\tau}(\tau - \tilde{\tau})} - \frac{1}{2} \left(1 - \frac{\tau^3}{2\tilde{\tau}^3}\right) \frac{z^2}{\tau - \tilde{\tau}}\right) d\tilde{z} d\tilde{\tau}
\end{aligned}$$

$$\begin{aligned}
&= -\frac{a}{2\pi} \int_0^\tau (1-2k\tau)^{5/2} \frac{1}{\sqrt{\tau-\tilde{\tau}}} \frac{1}{\sqrt{\tilde{\tau}}} \exp\left(-\frac{1}{2}\left(1-\frac{\tau^3}{2\tilde{\tau}^3}\right) \frac{z^2}{\tau-\tilde{\tau}}\right) \dots \\
&\quad \int_{-\infty}^{\infty} \left(\frac{3\tilde{z}}{\tilde{\tau}^2} - \frac{\tilde{z}^3}{\tilde{\tau}^3}\right) \exp\left(-\frac{1}{2} \frac{(\tilde{z} - \frac{\tau}{\tilde{\tau}}z)^2}{\frac{\tilde{\tau}}{\tau}(\tau-\tilde{\tau})}\right) d\tilde{z} d\tilde{\tau} \\
&= -\frac{a}{\sqrt{2\pi}} \int_0^\tau \sqrt{\tau/\tilde{\tau}-1} (1-2k\tilde{\tau})^{3/2} \left(\frac{3z}{\tilde{\tau}^2} - \frac{3z}{\tilde{\tau}^3} - \frac{\tau^2 z^3}{\tilde{\tau}^5}\right) \exp\left(\frac{1}{2}\left(\frac{\tau^3}{\tilde{\tau}^3}-1\right) \frac{z^2}{\tau-\tilde{\tau}}\right) d\tilde{\tau}
\end{aligned}$$

The final simplification here uses

$$\int_{-\infty}^{\infty} \left(\frac{3\tilde{z}}{\tilde{\tau}^2} - \frac{\tilde{z}^3}{\tilde{\tau}^3}\right) \exp\left(-\frac{1}{2} \frac{(\tilde{z} - \frac{\tau}{\tilde{\tau}}z)^2}{\frac{\tilde{\tau}}{\tau}(\tau-\tilde{\tau})}\right) d\tilde{z} = \sqrt{2\pi}(\tilde{\tau}-\tau) \left(\frac{3z}{\tilde{\tau}^2} - \frac{3z}{\tilde{\tau}^3} - \frac{z^3\tau^2}{\tilde{\tau}^5}\right).$$

2.3 A Separation of Time Scales in US Bond Yields

The analytics of the previous sections assume that the time scale on which volatility or rather the volatility driving process y_t reverts to a mean level is shorter than the time scale on which the x_t process reverts. This supposition is supported by direct evidence. In section 2.3.2 we examine the dynamics of US short rates (or rather, an estimate of this quantity based on swap yields) and demonstrate fast mean reversion of short rate volatility over a period of a few weeks - much shorter than the typical mean reversion time of the short rate itself which is measured in years.

Having established a separation of time scales for the real world dynamics of the short rate we turn to the question of whether a separation of time scales exists under the risk neutral measure. The analysis here is necessarily more model dependent and open to interpretation, but the preliminary findings are striking. For each yield curve in an 1111 day sample we fit several parsimonious models drawn from the popular Vasicek and CIR jump family and compare the ability of model predicted yield curves to match the data. Similar tests are performed for the exponential Ornstein-Uhlenbeck model with stochastic volatility corrections, with restrictions placed on the parameter space to facilitate a fair comparison.

The results are dramatic. Prior to July 1998 both the corrected Vasicek and CIR models fit all sixty points on the yield curve quite well, but after this time qualitative changes in the yield curve occur and there is no longer any comparison. The CIR

model with jumps and four free parameters fails hopelessly to fit the “dip” in the yield curve and cannot avoid mispricing bonds by as much as fifty basis points. By contrast the corrected Vasicek model with four free parameters fits even “wild” yield curves like a glove. A typical fit is shown in figure 2.8 where the pricing errors are less than five basis points. We are tempted to conclude that *yield curve shapes, particularly during the period July-September 1998, present indirect evidence of a separation of scales under the risk neutral measure*. This scale separation may be explained by market segmentation. It appears that models which are able to separate out short term and long term effects are less likely to exhibit specification errors during crisis periods such as this.

2.3.1 Data Set: Implied US Zero Coupon Bond Prices

Yield curve data stretching over many maturities free of institutional anomalies arising from treasury specials and other effects is difficult to come by. As it is not our intention to micro model bond market quirks we take as our starting point a time series of synthetic zero coupon bond yields which was preprocessed by Shinghoi Lee¹ from USD-LIBOR swap rates and cash deposit rates. We consider that the regularization method used to back out the zero curve is largely model independent and appropriate for our purposes.

The dataset covers a 1111 calendar day period from August 8 1996 to August 20 1999, including the dramatic events of the summer of 1998. For most of the period prior to August 27 1998 the yield curve is upward sloping concave downwards. At the end of this month a negative hump in the short end of the curve is evident. Relative to the very short and very long maturity bonds the two year yields are substantially discounted. This discrepancy peaks at around a hundred basis points in October and persists until the third week in November. The crossover is evident in figure 2.2

¹Morgan Stanley Dean Witter

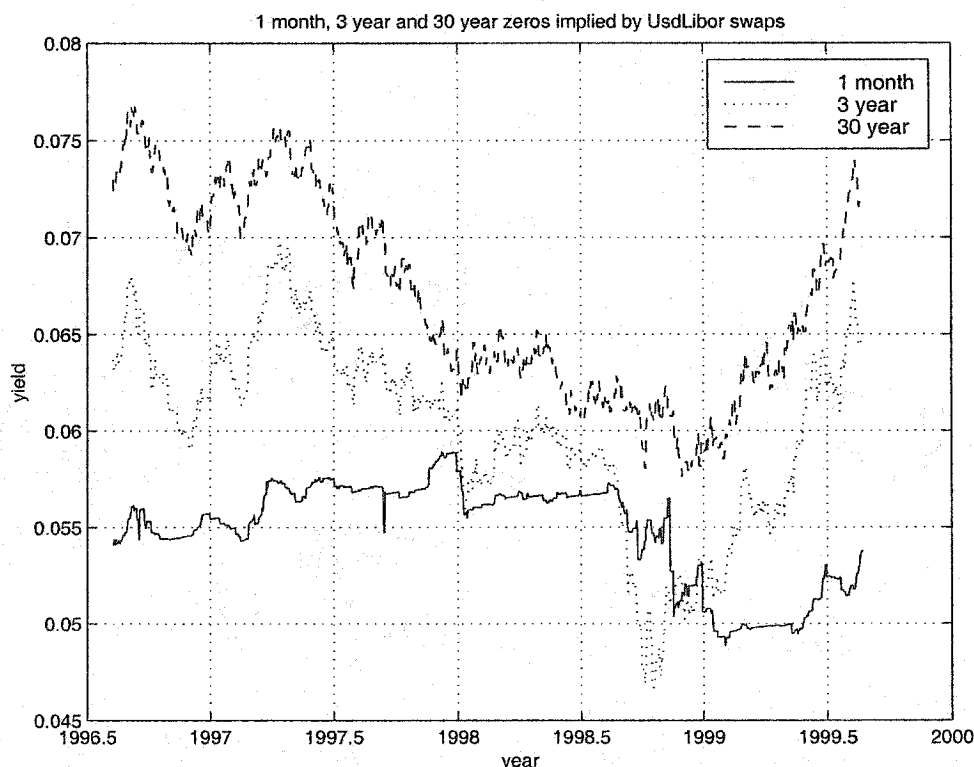


Figure 2.2: Approximate short (one month), medium term (three years) and long (thirty years) maturity synthetic zero coupon yields for the period August 8 1996 to August 20 1999. The synthetic yields were “cooked” using USD-LIBOR swap rates and cash deposit rates. Note the relative surge in three year bond prices around August-November 1998.

2.3.2 Analysis of Interest Rate Dynamics

In this section we examine the dynamics of empirical short term and long term yields. Using a test described in the following section we present evidence for rapid oscillations in short rate volatility on a time scale much shorter than the typical time scale of fluctuations in the short rate itself. The latter is measured in years, the former in weeks. Though the test is somewhat model dependent (designed around the ExpOU SV process) the qualitative results suggest generality.

We find no evidence for rapid fluctuations in the volatility of thirty year yields.

The Method of Variogram Analysis

Variogram analysis can be used in the analysis of any time series. We are particularly interested in US short rates as the ExpOU SV model is ostensibly a short rate model. Here we use techniques and notation analogous to those in [33] we model the normalized short rate fluctuations

$$D_j = \frac{x_j - x_{j-1}}{\sqrt{\Delta t}}$$

as

$$D_j = e_j^y \epsilon_j$$

where ϵ_j are independent identically distributed Gaussian random variables with variance one. Here x_j is the proxied short rate (overnight cash rate) for the j 'th trading day and Δt is one day. This is consistent with the exponential Ornstein-Uhlenbeck volatility model up to a small short rate drift term. We discard this term for simplicity as it makes negligible difference to the results which follow.

We examine the **empirical structure function** or **variogram** of the log absolute value of the normalized fluctuations. That is, we examine

$$V_j := \frac{1}{N} \sum_n (X_{n+j} - X_n)^2$$

where

$$X_j := \log|D_j| = y_j + \log|\epsilon_j|$$

To avoid the singularities from the noise $\log|\epsilon_j|$ we pass the X_j through a median filter. The graph of V_j as a function of j is fitted to the theoretical curve

$$\tilde{V}_j = 2\psi^2 + 2\frac{\beta^2}{2\alpha}(1 - \exp(-j/l)) \quad (2.41)$$

The constants ψ^2 , $\frac{\beta^2}{2\alpha}$ and l are the variance of the noise $\log|\epsilon_j|$, the variance of the volatility process y_j , and the decorrelation length of the volatility process y_j respectively. We are most interested in the latter, since $l = 1/(\alpha\Delta t)$. A short

decorrelation length (relative to the mean reversion time of the short rate which) will validate the chapter's main assumption that $\epsilon = 1/\alpha$ is small.

In fitting the parametric function 2.41 to points (j, V_j) we used the matlab LeastSq routine, making some minor modifications which guaranteed stability of the search.

Variograms for US Interest Rates

We first fitted structure functions for time series of US short rates. A decision was made to exclude non trading days following examination of the variograms when weekends were included. For example the lack of trading on weekends showed up prominently in the variogram 2.4 when Saturdays and Sundays were included in the data. As predicted by a simple calculation this adds a periodic component to the theoretical variogram which obscures the mean reversion estimate somewhat. The periodic component is substantially reduced by excluding Saturdays and almost eliminated by excluding both Saturdays and Sundays. We therefore treat Friday close to Monday close as a single trading day.

Price quantization creates further problems. The zero probability event (under ExpOU SV or any similar model) that interest rates are precisely the same from one day to the next is empirically anything but zero probability. Zero changes present quite a distraction to the structure function 2.41 and to get around this problem we used a simple median filter. Thus avoiding the logarithm of zero, we reveal the structure function for the short rate shown in figure 2.3. The fitted curve implies a decorrelation length of around eleven and a half days. Simulation was used to test the effect of the median filter and bootstrap errors in the decorrelation time.

The standard deviation of decorrelation time estimates in bootstrapped samples is around four days. This rough sampling estimate overstates our confidence in the precise time scale of mean reversion and as we have emphasized before, is unimportant. The fact that volatility mean reverts on the order of weeks rather than years (as illustrated by the qualitative observation that the empirical structure function is curved in accordance with 2.41 rather than flat) is sufficient justification for our asymptotic approach.

Interestingly, when a similar structure function analysis was performed on time

series of long maturity yields we did *not* find evidence of a separation of scales. The structure function shown in figure 2.5 gives no indication of a short decay time and provides no evidence for rapid mean reversion of long yield volatility.

Variograms of Simulated Data

The qualitative difference between the observed structure function for the short rate and thirty year yields bolsters the claim that rapid mean reversion of short rate volatility is an empirical phenomenon and not an artifact of the variogram methodology. This finding is consistent with joint work with Knut Solna, in which more model independent spectral tests were used. We hope to expand on this in future work.

It is natural to ask whether short rate models, if necessary incorporating multiple factors and jumps, can replicate the qualitative features of the variogram 2.3. We simulate short rates for a number of simple models from the affine jump diffusion family and for the ExpOU SV model. These are reviewed:

Vasicek and CIR Jump Diffusions

We recall that the ExpOU SV model is a member of the stochastic volatility family with volatility specifying function $f(y) = e^y$. For generality we also add exponentially distributed Poisson jumps (up and/or down) whose arrival intensity and mean are denoted $\eta^{+/-}$ and $\mu^{+/-}$ respectively. The jumps are added using calculations in appendix C. The risk neutral process for our Vasicek type model may be written

$$dx_t = (\kappa(x_\infty - x_t) - \lambda_1 e^{y_t}) dt + e^{y_t} dW_t + \Delta J_t^+ - \Delta J_t^- \quad (2.42)$$

$$dy_t = \left(\alpha(m - y_t) - \lambda_1 \beta \rho - \lambda_2 \beta \sqrt{1 - \rho^2} \right) dt + \beta \left(\rho dW_t + \sqrt{1 - \rho^2} dZ_t \right) \quad (2.43)$$

where $J^{+/-}$ are jump processes with arrival intensity $\eta^{+/-}$. Jumps are exponentially distributed with mean jump size $\mu^{+/-}$. A family of Vasicek type jump diffusions is obtained.

A family of multifactor CIR type jump diffusion models is obtained by specifying the “real world ” CIR parameters $(\tilde{\kappa}, x_\infty, \lambda_1, \sigma)$ for each independent factor. Each

factor evolves as

$$dx = (\tilde{\kappa}x_{\infty} - (\tilde{\kappa} + \lambda_1)x) dt + \sigma\sqrt{x}dW$$

where dW is Brownian motion with respect to the risk neutral measure. This has the advantage that $(\kappa, x_{\infty}, \sigma)$ parameterize the real world process

$$dx = \tilde{\kappa}(x_{\infty} - x) dt + \sigma\sqrt{x}dB$$

but the representation is for our purposes somewhat redundant. It is simplest to model the risk neutral process as

$$dx = \kappa(\theta - x) dt + \sigma\sqrt{x}dW$$

where

$$\begin{aligned}\kappa &:= \tilde{\kappa} + \lambda_1 \\ \theta &:= \frac{\tilde{\kappa} + \lambda_1}{\tilde{\kappa}} x_{\infty}\end{aligned}$$

As in the case of the Vasicek type models we allow Poisson exponentially distributed jumps to be added to each factor. We thank Dmitri Kachin² for pointing out the closed form solutions for bond prices with these jumps. The details are left to appendix C.

Variogram Results for Simulated Dynamics

With the family of multifactor jump diffusion models described in the previous section we performed exploratory tests in an attempt to determine which processes and, in a rough sense, parameters were required to match the qualitative structure function 2.3.

For example, we performed the variogram analysis on data simulated using two and three factor square root processes. The precise parameter values were not considered crucial for any given example. Examples are given in tables 2.1 and 2.2. Our goal was to try different parameter choices and try to replicate the shape in figure

²Morgan Stanley Dean Witter

2.3. For example, the parameters in table 2.1 for the three factor model are close to estimates from a Kalman filter supplied by Alois Geyer in unpublished work. However we have modified greatly the mean reversion of the third factor. The third factor is often correlated with convexity of the yield curve and volatility, so we have shrunk its typical oscillation time down to a couple of weeks. However in this example we were, despite our efforts to tweak parameters and introduce a short time scale, unable to match the empirical variogram.

Those processes which were able to reproduce curvature in the structure function consistent with the data were either close to our model 2.25 or represented a serious departure from common sense. An example of an unrealistic model is a single factor CIR model with fairly large jumps and parameters κ and η set to around thirty. This is unrealistic because the short rate process itself certainly does not display rapid oscillations on the order of weeks. Similar comments apply to the single factor Vasicek models with Poisson exponential jumps. It was not enough to add jumps with a suggestive frequency of around two weeks (this non-result is fairly obvious given the independence of the diffusion and jump components on the short rate).

Model	Factor	κ	x_∞	σ
CIR (3 factor)	1	0.0671	0.0397	0.216
	2	0.588	0.0153	1.14
	3	26	0.00218	6.23
CIR (2 factor)	1	0.0710	0.043	0.216
	2	0.0774	0.0157	0.0132

Table 2.1: Parameters used in simulations for the purpose of variogram analysis. The CIR parameters are approximate estimates from a Kalman filter, although for the three factor model the mean reversion rate of the third factor has been greatly exaggerated. This sets the mean reversion time of the third factor at around two weeks and was done in attempt to replicate the fast mean reverting component we find in the data.

The variogram for a simulated three factor CIR model with parameters given in table 2.1 is shown in figure 2.3.2. Noise in the structure function data points V_j notwithstanding there is a clear contrast with the variogram for a simulated ExpOU SV model shown in figure 2.6. Four years of data were used in creating 2.6 to facilitate

Model	κ	x_∞	σ	α	m	β	ρ	η^+	μ^+	η^-	μ^-
ExpOU SV	0.7	0.06		26	-5	2	0.6				
Vasicek+J	0.7	0.05	0.006					26	0.01		
Vasicek+J	0.7	0.05	0.006					13	0.01	13	0.01
CIR+J	26	0.05	0.0854					26	0.01		

Table 2.2: Parameters for some Vasicek and CIR models used to simulate data sets for the purpose of variogram analysis. The parameters $\nu^\pm, \mu^\pm$ govern the arrival intensity and mean size of jumps in the positive and/or negative directions.

comparison with figure 2.3. The agreement between the ExpOU SV model and data is good.

2.3.3 Contemporaneous Yield Curve Fitting

For each day in our dataset period we fit a selection of models to the contemporaneous yield curve, minimizing the sum of squared errors of yields sampled at sixty maturities ranging from one month to thirty years. The optimization was performed using the matlab LeastSq routine. For a given day, up to sixty searches were performed beginning at different starting guesses. For all models we found that the parameter values were quite unstable. Fitting σ proved to be particularly unreliable, and for some calibration runs this was fixed.

Contemporaneous Yield Curve Fits for One Factor Jumps Models

Prior to August 1998 the single factor Vasicek and CIR models perform quite well, but at this time the synthetic bonds in the two to five year range become more expensive and a dip in the yield curve appears. The CIR model fitted by least squares to the bond yields of all maturities misprices the (normally) liquid midrange bonds by dozens of basis points. As demonstrated in figure 2.3.3, the three parameters κ, r_∞ and σ provide surprisingly little flexibility. In an attempt to improve the fit without introducing unobservable factors we also fitted the single factor models with jumps in one or both directions. A snapshot and corresponding parameter values are reported for a particular day, Sep 6 1998, in figure 2.8. The addition of jumps did little to

improve the fit on this and many other days.

Contemporaneous Fits with Stochastic Volatility Corrections

Unimpressed by the ability of the single factor jump models to fit the initial term structure we used the bond pricing approximations developed in this paper to repeated this least squares fitting procedure. Two approaches were tried here. In the first instance we fit directly the correction terms parameterized by V_1 , V_2 and V_3 appearing in equation 2.13. In the second instance we adopt the particular model 2.25 and place restrictions on the parameters $\{\kappa, r_\infty, \sigma, \alpha, m, \beta, \rho, \lambda_1, \lambda_2\}$ in order to reduce the number of degrees of freedom to something commensurate with the single factor CIR model. After some experimentation we found that very good fits were achieved when all parameters but κ, r_∞ and the market prices of risk λ_1, λ_2 were fixed. A snapshot from one such calibration run is shown in figure 2.8. It is typical of the fits achieved by the model and represents a striking improvement over CIR and Vasicek models without stochastic volatility corrections. The short end peculiarities are reproduced beautifully and the fitted curve stays within a five basis point band of the data for almost every day in the calibration run. Similarly results are obtained when we fit V_1 , V_2 and V_3 directly, although in these searches we generally had at least five degrees of freedom and a good yield curve fit less surprising.

Table 2.3 and figure 2.11 summarize the 1111 yield curve fits using the stochastic corrections. For the most part the correction terms are very small. However, as is made evident by figure 2.10, dramatic changes took place during 1998.

2.3.4 Tentative Conclusions

If nothing else, the ExpOU SV Vasicek model provides a parsimonious parameterization of possible yield curve shapes which is significantly more flexible than competing econometric single factor models with constant coefficients. It appears that the separation of time scales enable us to capture unusual short end behavior whilst maintaining the match with long term bond yields. We have shown that consistent

Parameter	Mean	Std
κ	0.3	0.12
r_∞	0.049	0.015
σ	0.0059*	(fixed)
V_1	0.006	0.0079
V_2	-0.000069	0.028
V_3	0.00001	0.000043

Table 2.3: Moments of contemporaneously fitted parameters for a Vasicek model with stochastic volatility corrections. The parameters were calibrated by least squares for each day in a four year period. This table reports the group parameters V_1 , V_2 , V_3 which parameterize the stochastic volatility correction, as well as the usual parameters κ , r_∞ , σ from the Vasicek model. The stochastic correction terms include market prices of risk for the underlying short rate and short rate volatility. Most of the instability in the parameters V_2 and V_3 can be attributed to a period in the summer of 1998 when the bond market was in a state of crisis. At this time, as can be seen from 2.10, the parameters V_2 and V_3 move far from their means in an attempt to match the unusual behaviour in the short end of the yield curve. This “dip” or “lip up” is evident in figure 2.8.

snug fits such as that in figure 2.8 can be achieved without the introduction of deterministic “fudge factors” or time dependent coefficients as in the Hull-White approach [42]. By keeping to a low dimensional parameter space we can aim for more reliable risk analysis and hedging, avoiding the overfitting problems treated by Lee [47], Frey [6] and others. Further work is needed in this regard.

The analysis of structure functions in section 2.3.2 suggests new directions in affine modeling. A possible shortcoming of affine models is that, after estimation of parameters has been performed, a binding of the typical fluctuation time of the factors to the typical fluctuation time of the factor volatility emerges. This implicit identification of time scales does not appear to be supported by a careful examination of US short rates and as we have shown is not crucial to the tractability of the model. By contrast the exponential Ornstein-Uhlenbeck stochastic volatility model, an example from a class amenable to our methods, allows the natural volatility time scale to be much shorter than that of the short rate. Consequently the variogram for the two scale model in figure 2.6 is an excellent match to the data.

In principle, multifactor estimation by likelihood, moments or otherwise may result in one rapidly mean reverting factor and a structure similar to 2.25. In practice however the factors have other work to do and whilst one factor may mean revert faster than others it is unlikely to represent a real separation of time scale volatility. Moreover if such a rapidly mean reverting factor was to emerge from a multifactor estimation scheme then its very existence is problematic because as our analytics have demonstrated, the effect of the additional factor is largely independent of the exact specification of its dynamics. The likelihood function will be very flat.

We suggest that short time scale fluctuations can be directly accounted for using our analytics (and future developments of the same), thereby removing a possible distraction, and allowing multifactor estimation to proceed more smoothly.

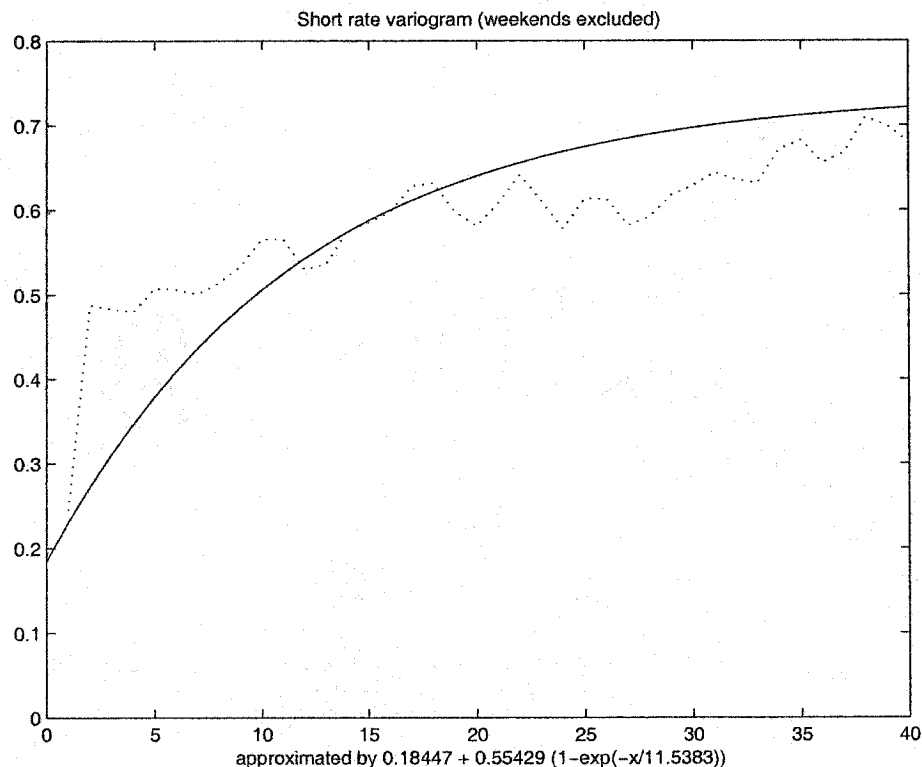


Figure 2.3: Variogram for the logarithm of the absolute value of the normalized differenced short rate, illustrating evidence for a fast mean reverting component in short rate volatility consistent with the exponential Ornstein-Uhlenbeck stochastic volatility model. The x-axis represents the lag or window length j (in days). The y-axis represents the structure function as defined in the text. The dotted line represents the numerical results for V_j . The solid line is a least squares fitted theoretical prediction based on the ExpOU SV model and given by equation 2.41. The fitted curve implies a decorrelation length of around eleven and a half days. To avoid taking logarithms of zero a median filter was applied. Simulation was used to test the effect of the median filter and bootstrap errors in the decorrelation time. The standard deviation of decorrelation time estimates in bootstrapped samples is around four days.

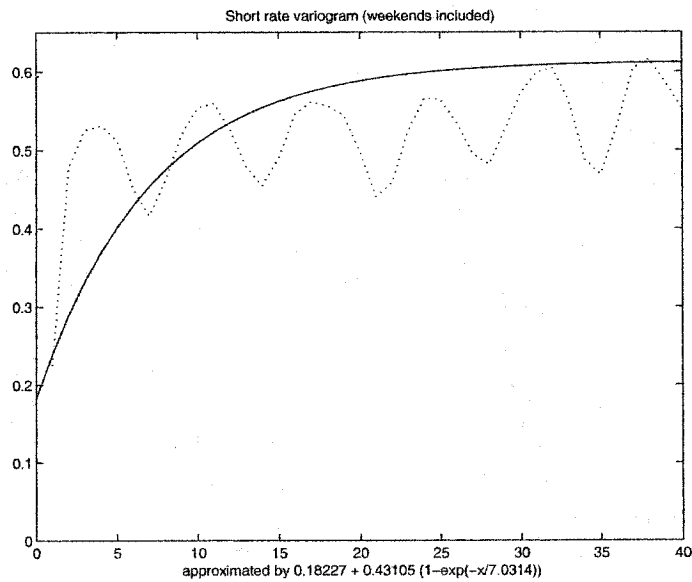


Figure 2.4: Variogram for the logarithm of the absolute value of the differenced short rate when weekends are included. The non trading days add a periodic component to the structure function and present a distraction.

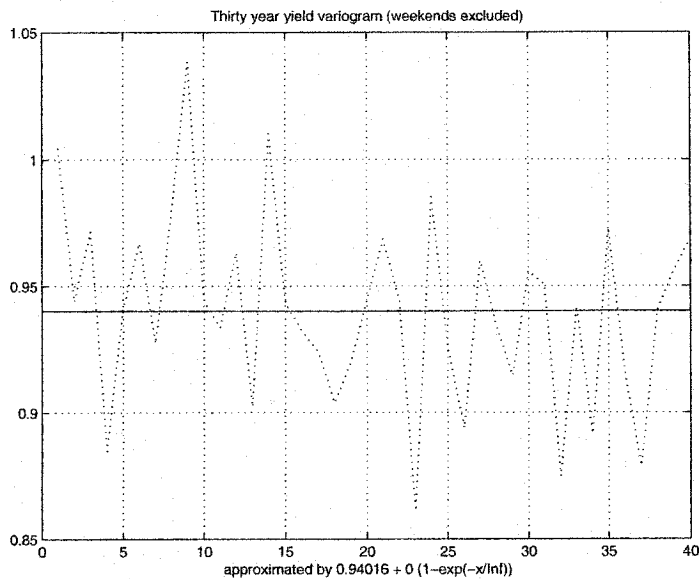


Figure 2.5: Variogram for the logarithm of the absolute value of the differenced thirty year yield when weekends are excluded. In contrast to the structure function corresponding to the short rate in figure 2.3, no curvature is evident above the noise. We find no evidence for rapid mean reversion in the volatility of long maturity yields.

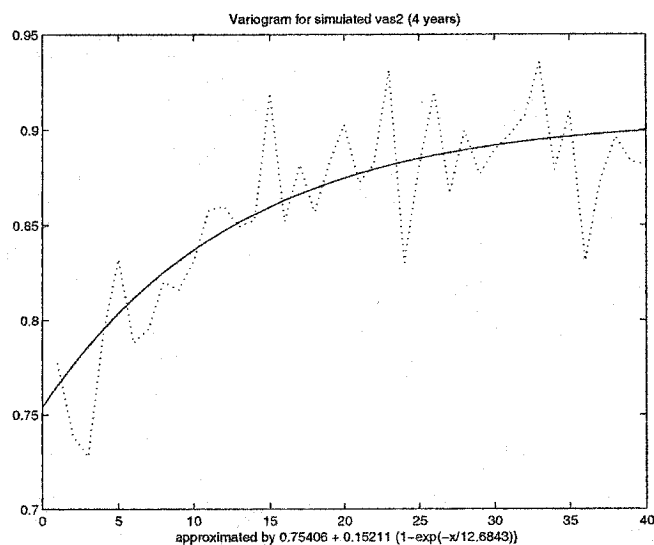


Figure 2.6: Variogram for simulations from an ExpOU SV Vasicek model. The parameters used in this simulation were $a = 0.7$, $x_\infty = 0.06$, $\alpha = 26$, $m = -5$, $\beta = 2$, $\rho = 0.6$, $\lambda_1 = 0$ and $\lambda_2 = 0$. Numerical results are dotted. The solid line is a least squares fit as explained in the text. The agreement with the data (compare figure 2.3) is excellent.

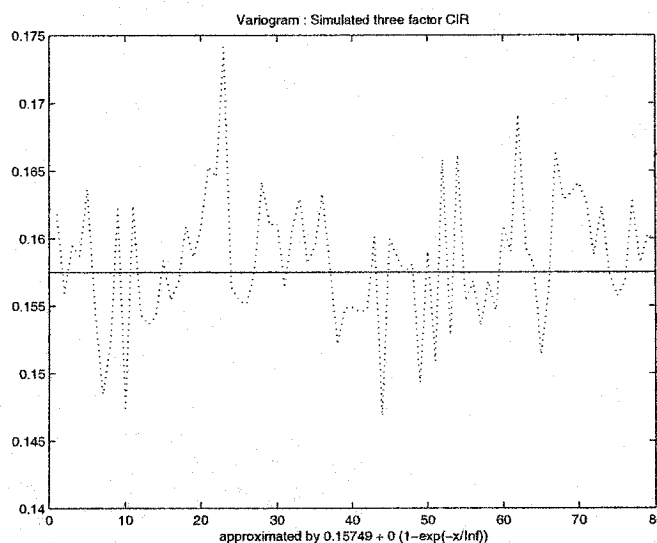


Figure 2.7: Variogram for simulated three factor CIR data. Note that the reversion rate of the third factor (often associated with convexity of the yield curve and volatility) has been increased ten fold to give the model a fighting chance of replicating the two week mean reversion in volatility characterized by the shape of the structure function in figure 2.3. It is not able to do so.

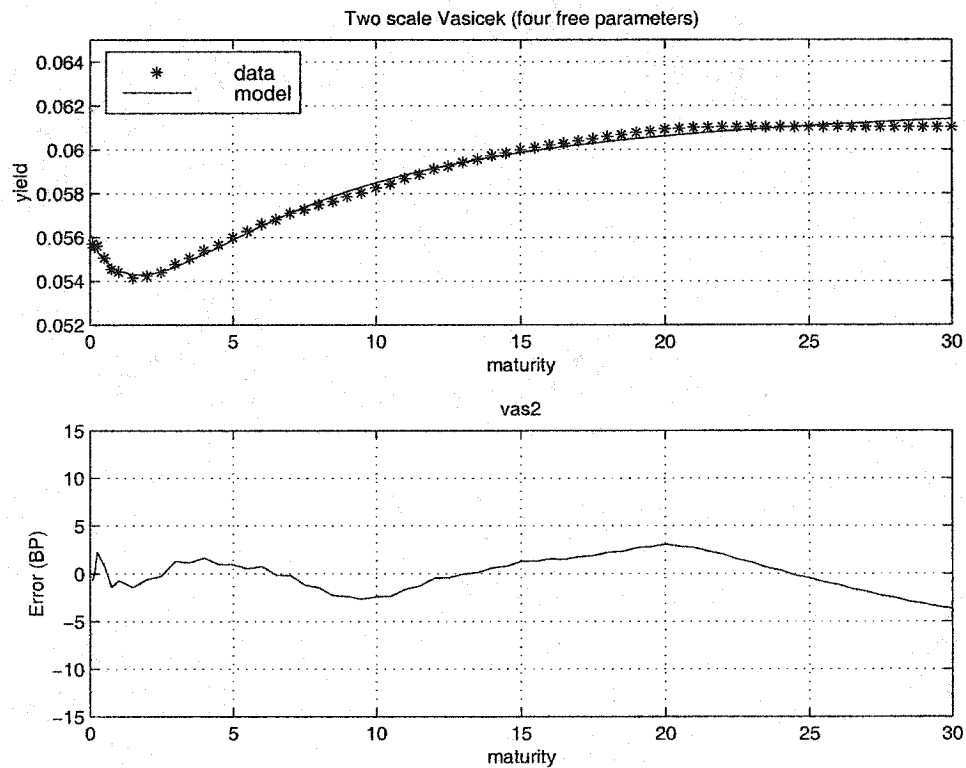


Figure 2.8: Yield curve fit with a single factor corrected Vasicek model (four free parameters) for September 6 1998. The dotted line is a synthetic yield curve cooked from swap rates. The solid line is a least squares fit of the exponential Ornstein-Uhlenbeck stochastic volatility model. This fit was typical of over a thousand contemporaneously fitted yield curves and should be compared with the fit for a single factor CIR model with jumps shown in figure 2.3.3. The separation of time scales in the ExpOU SV model helps the model fit the dip in the short end of the yield curve as well as the overall upward sloping shape. Errors expressed in basis points (1/10,000) are shown below.

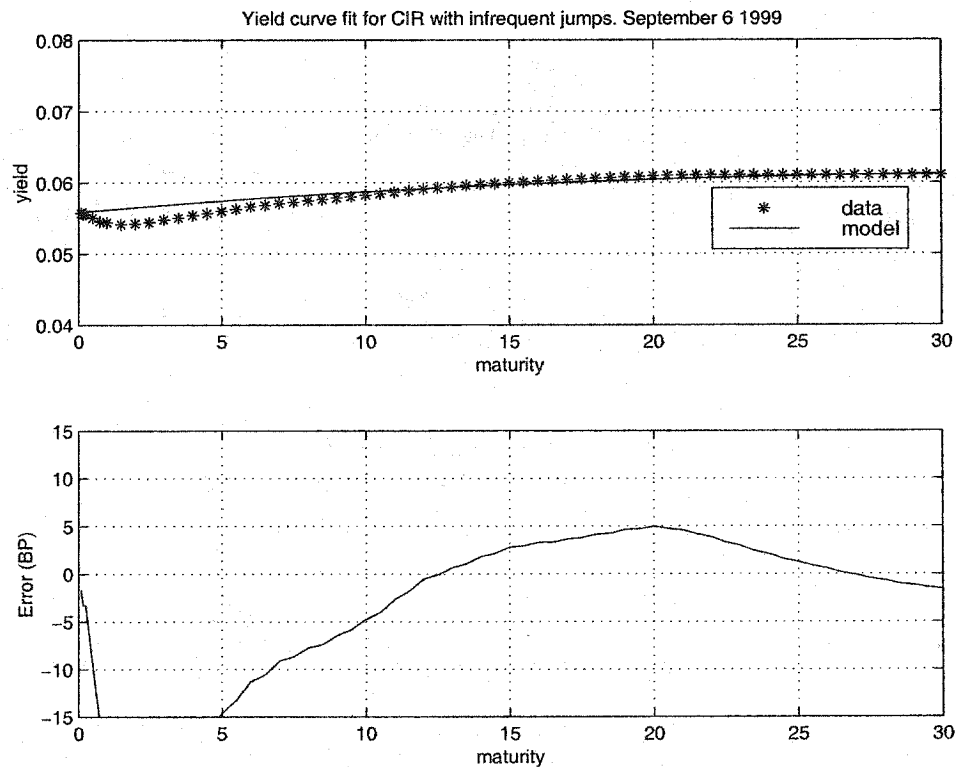


Figure 2.9: Yield curve fit with a single factor CIR model with jumps, September 6 1998. The dotted line is a synthetic yield curve cooked from swap rates. The solid line is a least squares fit using a jump model as described in the text. The model cannot accomodate the somewhat unusual shape of the short end of the yield curve. Errors, expressed in basis points on the bottom plot, are significantly larger than those the the ExpOU SV model in figure 2.8.

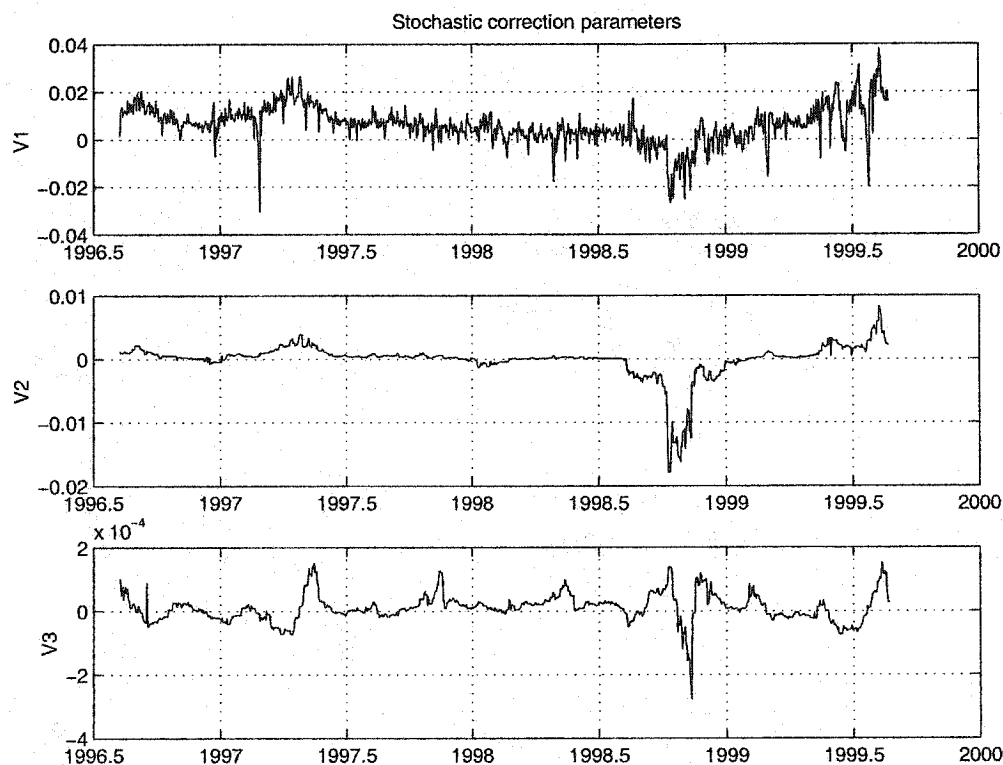


Figure 2.10: Time series of group parameters V_1 , V_2 , V_3 entering the stochastic volatility corrections used to assist the contemporaneous fitting of yield curves. Most of the instability in the parameters V_2 and V_3 can be attributed to a period in the summer of 1998 when the bond market was in a state of crisis. As the stochastic correction terms include market prices of risk for the underlying short rate and short rate volatility one is tempted to interpret this as evidence of time changing market risk premia. Other explanations however, such as market segmentation and liquidity premia, are possible.

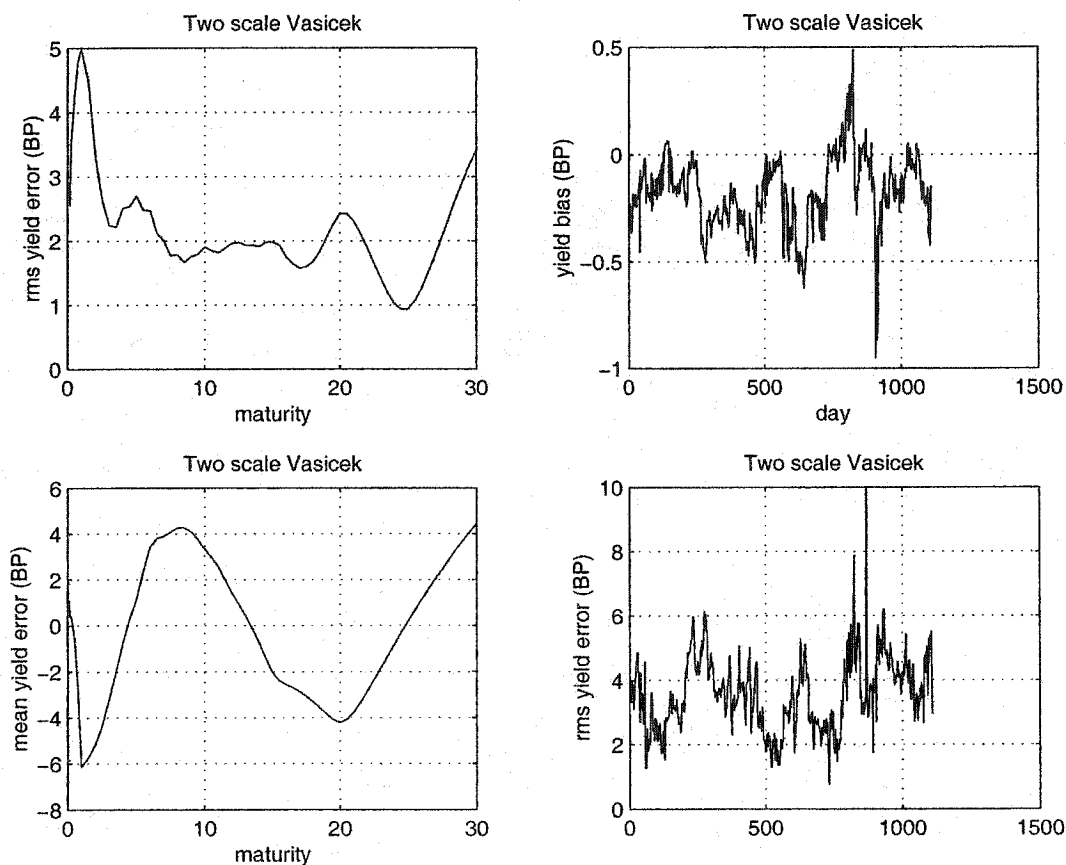


Figure 2.11: Contemporaneous yield curve fitting errors for the ExpOU SV Vasicek model, restricted to a four dimensional parameter space. Errors are expressed in basis points. Sixty starting points were used in searching for the best fit at each day in a four year period. The top left plot reports, as a function of maturity, the sample standard deviation of the root mean square yield fitting error across the 1111 day data set. The bottom left plot reports the mean of the same quantity, also as a function of maturity. The top right time series shows the small average yield error across maturities for each day in the sample. Most importantly, the bottom right plot shows the root mean square yield fitting error averaged across maturities for each day in the sample. The typical fitting error is on the order of three to four basis points, even during the summer of 1998 when other candidate models are off by dozens of basis points.

Chapter 3

A regime switching Vasicek model

In this chapter we consider a process x_t obeying

$$dx_t = \kappa(\theta(y_t) - x_t) dt + \sigma(y_t) dW_t, \quad (3.1)$$

where the secondary process y_t switches between two states with intensity λ . The secondary process is assumed for simplicity to be independent of the Brownian motion driving x_t . We propose a new approach to calculating the expectation

$$u(t, x) = \mathbb{E}_0 \left[\exp \left(- \int_0^t x_s ds \right) \right]$$

of the exponentiated negative integrated regime switching x_s process under the law of the same. This amounts to solving the PDE,

$$-\frac{\partial}{\partial t} u + \kappa(\theta(y) - x) \frac{\partial}{\partial x} u + \frac{1}{2} \sigma^2(y) \frac{\partial^2}{\partial x^2} u - xu + \lambda Qu = 0$$

with initial values $u(0, x) = 1$. Here λ parameterizes the arrival intensity of regime switches through the generator

$$\lambda Q = -\lambda \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

of the y_t process. We derive an asymptotic solution in powers of $1/\lambda$ and demonstrate that terms in this expansion can be explicitly calculated.

We thereby construct formulas which may be used for approximating the price of a discount bond. The same results carry over to the modeling of spreads and the pricing of credit derivatives. The formulas capture effects which are independent of the present regime as well as those which depend on knowledge of the present value of y_t . We discuss qualitative features of the regime switching process, notable bimodality and predictability, demonstrating the existence of the same amongst hundreds of time series of US corporate spreads.

3.1 A Not So Simple Regime Switching Process

We consider the process

$$dx_t = \kappa (\theta(y_t) - x_t) dt + \sigma(y_t) dW \quad (3.2)$$

where the process y is an independent Poisson Markov chain taking two discrete values (as for example, in Pinsky [52]). The volatility and mean reversion level θ are allowed to fluctuate, but for tractability we fix the mean reversion rate κ . The process y is a Markov chain taking values in one of two states which we label 1 and 2, with (2×2) generator:

$$\lambda Q(p) = -\lambda \begin{pmatrix} q & -p \\ -q & p \end{pmatrix}.$$

Here we allow an arbitrary invariant distribution, $\mu = (p, q)'$. However, the main pricing results presented in this chapter apply to the special case $p = q = 1/2$, a somewhat unnatural restriction shared by Hansen and Poulsen ("A Simple Regime Switching Model", 2000 [37]) who also assume regime independent σ . We refer to the regime switching process (3.2), in which both $\theta(y_t)$ and $\sigma(y_t)$ are stochastic, as "not so simple", explaining in the following sections why regime dependent volatility is such a useful modeling tool.

A sample path and histogram for a regime switching process are shown in figure 3.1.3. The mean reversion rate is $\kappa = 2$. The level to which the process reverts switches between $\theta_{high} = 0.15$ and $\theta_{low} = 0.05$ with intensity $\lambda = 0.5$. The diffusion parameter is also regime dependent. In the high θ regime $\sigma_{high} = \sqrt{0.0035}$ whereas in the low regime $\sigma_{low} = \sqrt{0.0005}$. The simulation extends for around twenty five years with a time step corresponding to a single business day.

3.1.1 Bimodal Distributions

The histogram of x_t values is shown below the x_t time series in figure 3.1.3. The distribution is clearly bimodal, our parameters having been chosen to illustrate this possibility. As first suggested to the author by Zhifeng Zhang¹, bimodality in the histograms of corporate spreads is not uncommon although hard to define UN controversially. We take up the task of placing this gut feeling on a firmer footing in section 3.5 where many spread time series are examined and compared to simulations of simple diffusions.

3.1.2 Gentle Correlation Between Level and Volatility

The relationship between level x_t and instantaneous variance σ of interest rates is a strong implication of most short rate models, and has attracted considerable attention. For example, in a family of models

$$dx_t = \kappa(\theta - x_t)dt + \sigma x_t^\gamma dW_t$$

one might wish to estimate the coefficient γ most consistent with observed data (the Vasicek model predicts $\gamma = 0$, the CIR model predicts $\gamma = 1/2$). Techniques applicable in this instance (and in other instances with more complex drift specifications) have been developed by Hansen and Scheikman [38], Ait-Sahalia [1], Lo [2] and others.

For the regime switching process (3.2), the relationship between level and volatility is a little more complex. If we consider x_t (but not the regime y_t) to be observable

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then, in principle, all past values of x_t contain information about the current regime y_t and therefore contain information about the diffusion parameter $\sigma(y_t)$. Though we do not in this thesis treat directly this estimation problem, (a promising approach is the EM algorithm going back to Dempster et al [16]), in accordance with common sense, we can predict the sign of observed correlation between level and volatility in long time series.

3.1.3 Jumping Safely and Landing Softly

An attractive feature of the process (3.2), somewhat easier to appreciate with the naked eye than the correlation between x_t and y_t , is the appearance of what we might loosely term “soft jumps”. These jumps are changes in the level of x_t over a relatively short period (evident in figure 3.2), which are possible but highly unlikely in a diffusion process with constant parameters. “Soft jump” behavior is more common in nature than pure jumps which serve a similar modeling role (and also occur in practice).

We desire soft jumps in modeling for some of the same reasons that we might desire jumps. Jumps enable us to drastically increase the possibility of rare events and fatten the tails of short horizon conditional distributions. However, jump behaviour can be problematic in some applications. In particular, it is difficult to add downward jumps to models without either increasing greatly the possibility of negative x_t values or reducing the tractability of pricing calculations. By contrast, the regime switching process described here accommodates soft jumps in both directions with minimal chance of negative x_t values.

What the regime switching process lacks is the potential for repeated jumps in the same direction. Put another way, the process cannot accomodate jumps whose direction comes as a surprise. In order to model repeated jumps in the same direction we return to the existing literature. In particular, important work on pricing with a class of affine jump diffusions was carried out by Duffie, Pan, Singleton [23].

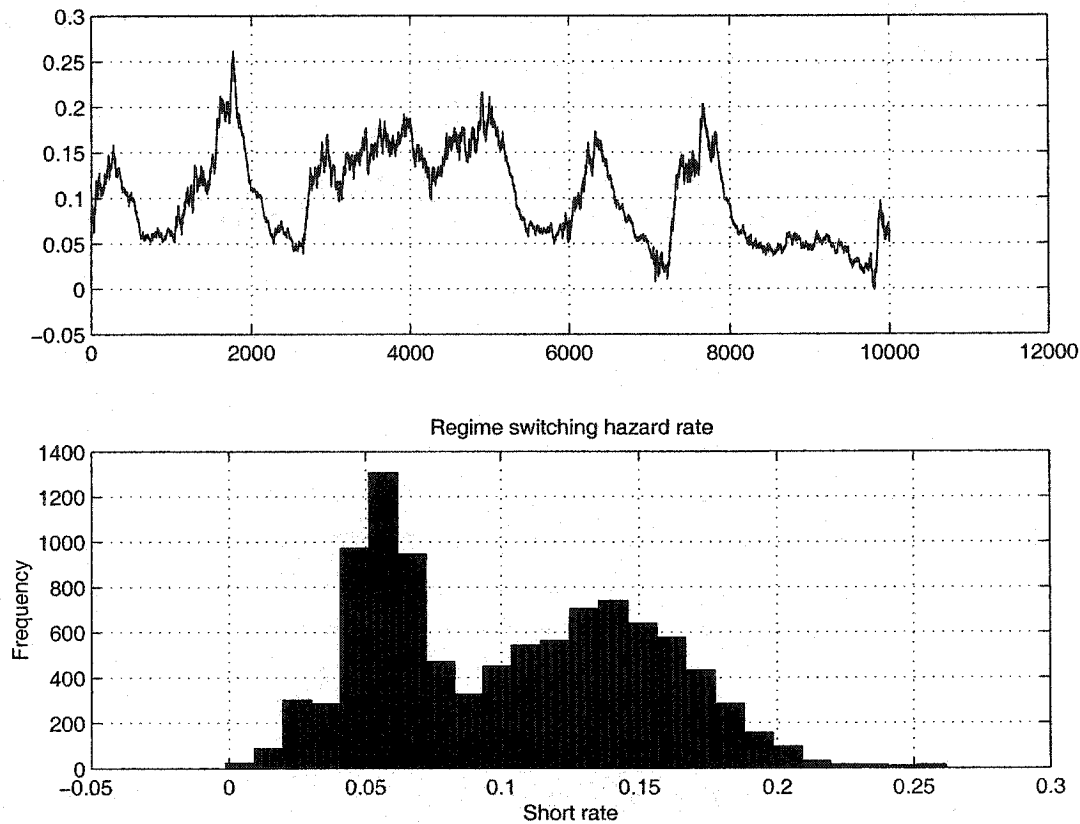


Figure 3.1: Sample path and histogram for a regime switching process. Parameters $\kappa = 2, \theta_1 = 15\%, \theta_2 = 5\%, \sigma_1 = \sqrt{0.0035}, \sigma_2 = \sqrt{0.0005}, \lambda = 0.5$. Time step is around one business day. Sample is twenty five years. The unconditional distribution is strongly bimodal due to our choice of parameters. It need not be. High volatility is associated with high levels and smaller fluctuations accompany low levels.

3.1.4 Minimizing the Probability of Negative x_t Values

Figure 3.2 compares a second simulated time series using a regime switching model (with parameters $\theta_{low} = 0.025, \theta_{high} = 0.065, \kappa = 3, \sigma_{low} = 0.01, \sigma_{high} = 0.04, \lambda = 1$) with a constant parameter Ornstein-Uhlenbeck process (parameters $\theta = 0.0451, \kappa = 1.321, \sigma = 0.0327$). The parameters have been chosen so that the unconditional means and variances of the two time series are very similar. However, the regime switching model is designed so that low volatility is associated with low x_t values, thus greatly reducing the probability of x_t being negative. Simulating fifty thousand

years of monthly x_t values for both processes, we found only 112 occasions when the regime switching process was negative (approximately 1 in 5000) compared with 8573 occasions for the Ornstein-Uhlenbeck process (approximately 1 in 70). The histograms for these long simulations are shown in figure 3.3.

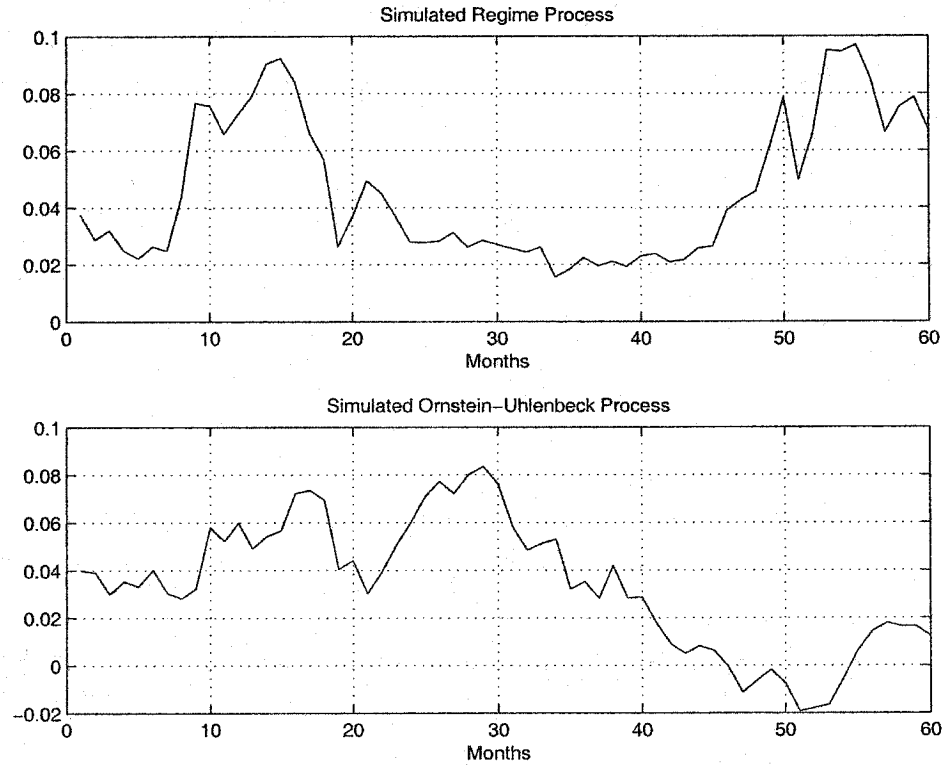


Figure 3.2: Simulated regime switching process (top) and Ornstein-Uhlenbeck process (bottom). Parameters for the regime switching model: $\theta_{low} = 0.025$, $\theta_{high} = 0.065$, $\kappa = 3$, $\sigma_{low} = 0.01$, $\sigma_{high} = 0.04$, $\lambda = 1$. Parameters for the Ornstein-Uhlenbeck process: $\theta = 0.0451$, $\kappa = 1.321$, $\sigma = 0.0327$. Both models have very similar unconditional means and variances

3.1.5 Trends and Predictability

When a regime switch occurs, the x_t process is drawn toward the new mean $\theta(y_t)$, and in doing so displays predictability and a tendency to trend. We demonstrate in section 3.5 that this behavior is consistent with observed spread dynamics.

3.2 Not So Simple Regime Switching Models

So far we have considered the joint process (x_t, y_t) , but said nothing about the intended application. In fact, we have two potential uses in mind. In the first, most obvious, application, x_t represents the short rate and y_t the interest rate regime. In the second application, which has provided most of the motivation for our regime switching work, the process x_t represents the instantaneous probability of default on obligations by an issuer of a bond. In bond cases, the bond price calculation is crucial. We discuss the interest rate application first, before turning to a more detailed explanation of the credit side.

3.2.1 Application to Interest Rate Modeling

As described in [37], one application of our work is as a regime switching short rate model. In order to frame our results in an econometric context, we should specify, as in the case of the stochastic volatility models of chapter 2, a Girsanov transformation relating the risk neutral and real world measures². As the regime switching process introduces only one additional source of risk, this approach, detailed by Hansen and Poulsen [37] is feasible.

For brevity, however, and with an acknowledged bias toward credit rather than interest rate applications, we will follow the prevailing market practice of modeling the process x_t directly under the risk neutral measure. Since we do not address estimation issues in depth we are willing to thereby forego the privilege of making strong claims concerning the real world measure. Alternatively one can, after the fact as it were, consider those simple changes of measure which leave the SDE (3.2) intact, mindful of the dangers of simple risk premia specifications (as critiqued by Duarte

²We remind the reader that, to unfairly compress a very important issue, under technical conditions the price of any claim on future cashflows may be priced by calculating the expectation of these cashflows with respect to a risk neutral measure under which such prices are martingales. Whilst there are special idealized situations where financial products “span” market uncertainty and a unique Girsanov transformation relating the risk neutral and real world measures can be calculated, typically markets are *incomplete* which is to say that not all risk may be hedged and not all financial products may be replicated with combinations of other products (in particular, exposure to the short rate itself cannot truly be hedged). The market price of risk parameterizes the *choice* of risk neutral measure.

[17] for example).

We also remark that the nature of the mathematics in this chapter is quite different from that of the last chapter in respect to the roles of the market price of risk parameters. Correlation between the short rate process x_t and the volatility process y_t previously played a crucial role, ensuring an interesting interplay between volatility risk premia and short rate premia. In this chapter, the processes x_t and y_t are uncorrelated and these effects are not accounted for. The interest arises from “higher terms”.

3.2.2 Application to Credit Derivative Modeling

The Generic Interest Rate Swap

An **interest rate swap** involves an agreement by two parties to exchange cashflows. Typically one cashflow is derived from fixed coupon payments owed to the holder of a bond and the other cashflow is dependent on a floating interest rate such as LIBOR. For a trade to occur there must exist a level of fixed coupon payments, usually expressed as a yield on a notional principle amount, such that both parties are willing to literally swap cashflows with no further compensation paid to either party. These “fair” yields or **swap rates** are very precisely traded. The party receiving payments based on the bond coupons is, in effect, financing the purchase of the bond in a streamlined manner (because for him or her the deal is similar to borrowing money at LIBOR, buying the bond, and thereafter paying back the loan at LIBOR while receiving the bond coupons). For many reasons [15] these contracts are very popular. In the US the volume of swap deals, as measured by the notional principle amount of the deal, is currently around five times the volume of the underlying bonds on which the deals are based.

The Generic Default Swap

An introduction to **credit swaps** is provided by Duffie [21]. A credit swap, also known as a **default swap**, is similar in structure to an interest rate swap but based upon cashflows paid to the holder of a bond which is *not* considered risk free.

One side (“**long** the bond”) of the credit swap is the bond, effectively financing the purchase of a defaultable bond or **reference asset** which might be issued by a corporation or government agency. A default swap between two parties proceeds as in an interest rate swap with an agreed notional P . In the absence of a **credit event** pertaining to the reference asset (such as failure of the issuers to pay coupon payments, bankruptcy and so forth) one party is, in effect, paying an insurance premium to the other and hoping for default of the reference asset. When such a **default** occurs, the exchange of cashflows between credit swap counterparties ceases. The defaulted bond now has a substantially reduced market value (on average half the face value but varying greatly) and the party who was long this bond loses out. The losing party (in effect) must pay the counterparty an amount equal to the difference between the principle P and the same amount written down to the market value of the defaulted asset (thus replicating the financial losses incurred by actual owners of the asset).

The “insurance premia” (**swap spread**) is expressed as a differential between risk-free benchmark yields (such as LIBOR) and the swap rate. Spreads are a reasonably liquid market indication of the credit-worthiness of a company - the volume of US corporate credit swaps is the US being around one fifth of the volume on the corporate bonds themselves. For this reason, spreads are used to calibrate models for the default of assets which are in turn used to price and hedge complex **credit derivative** products involving multiple cashflows based on multiple reference assets. Credit derivatives possess analytic pricing solutions only in very simple cases and typically require intensive simulation, though approximate methods such as those used by Moody’s rating agency exist. Therefore, importance is placed on models where the default probabilities are modeled exogenously with fairly simple dynamics.

Reduced Form Models for Default

Also known as **intensity** models, **reduced form** models represent the instantaneous probability of default by the issuer of a bond directly, usually as a CIR process (as in Lando [46] 1994) or similar diffusion. The evolution of the process is usually assumed to occur under a martingale (risk-neutral) measure. Modulo stochastic recovery rates and accounting quirks, the default intensity is proportional to market traded spreads.

Recently Duffie and Lando [22] help bridge the divide between intensity models and “structural” models - going back to Altman[3] 1968 - in which default probabilities arise endogenously.

We argue in section 3.5 that the regime switching process (x_t, y_t) (together with the analytics we provide and numerical schemes developed by others) is a valuable tool for modeling simple and complex financial products which are sensitive to credit risk. In this application, the x_t process represents the *risk neutral* instantaneous intensity of **default** on obligations by the issuer of a bond (an event with immediate ramifications for credit swaps, basket deals and other credit derivatives sensitive to the bond in question). The intensity of default is (*ceterus paribus*) proportional to the credit spread.

Jarrow and Lando[54] discuss the connection between risk neutral and real world measures in the context of reduced form models for credit. However, one must be particularly wary about relating risk neutral and real world measures in this context, since historical default rates are (at the time of writing) much lower than those implied by spreads. We have no desire to speculate whether there are sound economic reasons for this phenomenon, nor do we wish to specify a change of measure predicated on these reasons or their absence.

Instead, we take the view that the analytic formulas derived in this thesis are useful for pricing complex credit sensitive products involving many assets in relation to prices of simpler products (such as default swaps) with cashflows sensitive to a single asset. We focus on the bond price calculation, which may have several interpretations, rather than the specifics of various credit derivative products. Our empirical work focuses on qualitative aspects common to many spread series and, as such, is relatively robust to risk premia considerations.

Multifactor Spread Models

A comprehensive model for default intensities might consist of numerous common and idiosyncratic factors, modeled as diffusions or jump diffusions. Whilst identification and estimation of common factors is a highly nontrivial undertaking (outside the scope of this thesis), the mathematics of pricing multifactor models for spreads is relatively

simple provide the factors are independent processes. If the single factor “bond price” calculation can be performed instantly, quite complicated products involving dozens or hundreds of reference assets can be modeled.

Our empirical work suggests that spread dynamics are rather complicated and for this reason any enriching of the class of processes (for which closed form bond prices are available) ought to be welcome. We supply the mathematics which brings at least one, albeit simple, regime switching diffusion literally up to speed, thus enriching the library of diffusions which can feasibly be used as building blocks for realistic models of defaults. Though we shall not delving into the details of credit derivative pricing (see for example [45]), we reassure the reader that the “bond price” formulas obtained carry over directly.

The Need for Speed

When modeling the term structure of interest rates the reduction in computing time achieved by our formulas (as compared with, for example, the numerical solution of ordinary differential equations for a general affine model) is of limited practical significance. Even the most impatient of traders will wait a tenth of a second or so for a price.

The same can not be said for modeling credit derivatives. Cashflows of basket default swaps critically depend on the timing and ordering of defaults. Defaults must be simulated many times and valuations made for each path. The calculation of cashflows for each path typically involves state-dependent bond prices and leads to a serious algorithmic bottleneck (even a straightforward default swap valuation can involve hundreds of calculations similar to the evaluation of a discount bond). Numerical solution of ordinary differential equations is undesirable. Numerical solution of ordinary differential equations in the context of hedging and risk management might be better described as “infeasible”.

We discuss the performance of simulation, numerical and analytic approaches in section 3.3. A rough comparison provided in table 3.2, which is intended to give the reader a basic idea as to those applications in which simulation, numerical methods and analytic methods can feasibly be used.

3.2.3 Interpreting Regimes

When modeling the short rate the two values taken by the secondary process y_t are interpreted as states or regimes and, depending on the application, allotted further meaning in relation to business cycles, monetary policy, or market sentiment. Economic motivation for modeling regime switches in short term interest rates is discussed in [36],[5],[4], [48] and [8].

The macroeconomist's desire for regime switches in short rate models transfers over to credit derivatives. In the credit world, changes in regimes arise from the same and additional sources, sometimes with devastating effects. An example of a regime switch in sentiment, suggested by Joseph Langsam, is the move from a regime where very few Russian banks were willing to default to one in which the practice was noticeably more common (and we dare say acceptable).

At a more industry or firm specific level regime switches are equally important. A firm may be modeled as moving between periods where it's credit quality is generally perceived to be good and periods where its credit quality is significantly less certain. In the "good credit" regime, the company balance sheets are look good, the spread to LIBOR of a particular bond issued by the company is low, and spread volatility is also low due to the comparatively slow arrival of new information or perceived risks.

In the "bad credit" regime, the firm balance sheet is shaky, spreads are much larger, and spreads are also subject to increased fluctuation as every new release of information (from earnings reports to court proceedings) has the potential to greatly increase the perceived likelihood of default. Actions by the company or pure good fortune may quickly move the firm from a seemingly precarious situation to one where the risks are removed.

The lurches in perceived credit quality are helped by ratings agencies who publish quantized assessments. Intensities of default are far from linear in the rating categories and movement from one rating level to another can trigger regime shifts in spreads. In addition, firms release information sporadically and the degree of imperfect information held by market participants' as to the issuer credit-worthiness is likely to undergo rapid changes.

Further interpretation may involve models where regime shifts arise endogenously.

Regime shifts in intensity models arising from regime shifts in the productive capabilities of the firm can no doubt be engineered, perhaps using the imperfect information framework of Duffie and Lando [22]. Alternately, regime switches may be interpreted as an approximation to more complex default intensity dynamics arising endogenously in structural models.

3.2.4 Special Case i) Stochastic Volatility

Two special cases of our model are worth mentioning. If the mean reversion level, θ , is fixed (independent of the regime y_t) and only the volatility changes, we have a stochastic volatility model where the volatility takes one of two discrete values. Though this situation may seem to be similar in spirit to the stochastic volatility analysis in the previous chapter, it is actually rather different due to the absence of directly modeled correlation between the x_t and y_t processes. The calculations in the previous chapter capture directly the first order effect of correlation between the short rate x_t and volatility driving process y_t and, we recall, this correction is heavily dependent on the correlation parameter ρ . On the other hand in this chapter the y_t process is independent of the Brownian motion driving x_t . When the formulas in this chapter are specialized to the case where θ is independent of y_t we are left with terms “higher” than the correlation effect.

We hasten to add that although there is no direct correlation effect in this special case, there is an indirect correlation effect implicit in the fully fledged model where both θ and σ are regime dependent. The process x_t will be correlated with $\theta(y_t)$ in a complicated manner, depending on initial conditions and the rate of mean reversion κ .

3.2.5 Special Case ii) Stochastic Mean

The second special case is obtained by assuming that volatility is independent of the regime, leaving us with a “stochastic mean” model. This special case is precisely the model treated by Hansen and Poulsen (“A simple regime switching term structure model” 2000 [37]). Fixing the diffusion parameter σ to be regime independent makes

possible a decomposition of the bond pricing formula which simplifies pricing. This arises because the short rate itself can be broken down into three components:

$$\begin{aligned} x_s &= a_s + b_s + c_s, \\ a_s &= x_0 e^{-\kappa s} + \kappa \int_0^s e^{-\kappa(s-u)} \theta(y_{t=0}) du + \sigma \int_0^s e^{-\kappa(s-u)} dW_u \\ b_s &= 2\kappa\theta \int_0^s e^{-\kappa(s-u)} \chi(y_s) ds \\ c_s &= \sigma \int_0^s e^{-\kappa(s-u)} \lambda ds \end{aligned}$$

where $\chi(y_s)$ is the indicator function equal to one if the current regime y_s is the same as the initial regime $y_{t=0}$ and zero otherwise. As pointed out in [37], this approach fails if the diffusion coefficient σ is stochastic.

In a simple regime switching model five parameters in this model as compared with six in a standard two factor model so the special case is interesting as (we claim) a rich behavior is obtained. From the full model with fluctuations in both θ and σ we determine that the first order effects in the computation of bond prices are captured by introducing only a stochastic mean (regime dependent $\theta(y_t)$) rather than both stochastic mean and stochastic volatility (regime dependent $\theta(y_t)$ and regime dependent $\sigma(y_t)$). The sentiment is shared by Schlögl and Sommer [55], Hansen and Poulsen[37] and others. The simpler stochastic mean model performs well on the critical smoothing tests of section 3.5.3 and for some applications may be every bit as adequate as the full regime switching model.

Regime independent volatility (or more precisely, regime independent σ) does have its limitations however. In particular it is less useful as a model for processes which should remain positive with high probability because, as previously explained, the regime dependent $\sigma(y_t)$ can be used to minimize the frequency of negative x_t values. A second empirically desirable feature, correlation between level and volatility, is also lost.

3.2.6 Recent Work on Regime Switching Models

Ang and Bekaert [4] discuss the difficulties associated with estimation of a more general class of regime switching models, pointing out that even when these are used to generate the data the regime switching models may not perform well on statistical tests. They show, however, that forecasting ability is good even when difficulties in small sample estimation of regime switching processes prevents the models from matching moments as well as simpler models.

Bekaert, Hodrick and Marshall [8] propose a regime switching model for short-term interest rates and discuss the difficulties of estimation. They argue that differences between ex post frequencies of regimes within a given data sample and ex ante probabilities of these regimes (a “Peso” problem) need to be carefully addressed. By looking at small sample distributions generated by a regime-switching model they offer a reassessment of the famous “expectations paradox”.

Bansal and Zhou[5] add to the empirical support for regime switching processes. They conclude that their model with discrete regime shifts outperforms the multifactor CIR model on all tests. They build a discrete time model and provide approximate recursive formulas for bond prices.

Hansen and Poulsen 2000 [37] is the paper closest to the work in this chapter. They discuss a two state regime switching Vasicek model with short rate

$$dx_t = \kappa (\theta(y_t) - x_t) dt + \sigma dW_t.$$

identical a special case of (3.1) where only the mean reversion level parameter $\theta(y_t)$ is regime dependent. With an emphasis on econometric modeling of the term structure the authors address the existence of a unique martingale measure and discuss a nice approximate backward scheme for numerically estimating bond prices.

3.2.7 Limitations, Assumptions and Future Directions

In this exploratory work we make some simplifying assumptions driven by mathematical convenience rather than empirical study. For most of the analytics here we

assume that the y_t process has a long run unconditional distribution which does not favor one state over another. Things turn out to be particularly nice if we are able to impose the restriction that both regimes occur, in the long run, equally often. We are not able to say with certainty whether this key symmetry assumption is necessary, or whether a model with regime dependent rates of mean reversion (that is, κ depending on y_t) is equally tractable.

Comparing this chapter with the last, we remark that while the addition of stochasticity to the long run mean θ complicates the mathematics, the removal of correlation between the W_t and y_t processes considerably simplifies our task. Furthermore we have simplified the y_t process from a continuous valued Ornstein-Uhlenbeck process to a simple two state Markov chain.

We would like to think that the theoretical difficulties associated with pricing in regime switching models have prevented them from gaining wider acceptance in spite of their excellent empirical performance in modeling the short rate. We would furthermore like to think that the analytic advances presented here will change all that. However, as speculated by Ang and Bekaert[4], the difficulties associated with estimation may be a more important reason, given that the existing techniques for pricing are adequate for many purposes. In regard to estimation, we hope, in future work, to carry through the asymptotics of the conditional density function for regime switching models (such as the one analyzed in this chapter).

However, in regard to applications in the modeling of spreads, we strongly suspect that any estimation technique which fails to treat more than one time series at a time may be suboptimal. Common sense would suggest that amongst the thousands of time series of spreads of issues from various companies and agencies there would exist common or frequently occurring characteristics and that these features should be exploited in the estimation of models for individual spreads.

In keeping with this philosophy we have provided evidence of qualitative features (trending and bimodality) common across many spreads, and illustrated the use of quantitative criteria applied to model simulations. Of course more work needs to be done before a convincing procedure for jointly estimating thousands of spread dynamics is implemented.

3.3 Some Approaches to Pricing with Regime Switches

We recall the process

$$dx_t = \kappa(\theta(y_t) - x_t)dt + \sigma(y_t)dW. \quad (3.3)$$

with λ , the arrival intensity of regime switches, entering the generator

$$\lambda Q = -\lambda \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

of the y_t process.

3.3.1 The Task

The bond price

$$u(t, x) = \mathbb{E}_0 \left[\exp \left(- \int_0^t x_s ds \right) \right]$$

solves the pricing PDE

$$-\frac{\partial}{\partial t}u + F(x, y)\frac{\partial}{\partial x}u + \frac{1}{2}\sigma^2(y)\frac{\partial^2}{\partial x^2}u - xu + \lambda Qu = 0$$

with boundary condition $u(0, x) = 1$. We let $F(x, y) = \kappa(\theta(y) - x)$ and use t to represent maturity, not time.

Since y takes discrete values one can think of the PDE as a pair of PDE's, each of which represents the solution conditional on a particular choice of $y \in \{1, 2\}$. Specializing to the case of bonds ($\alpha = 0, \beta = 1$) we have

$$\begin{aligned} -\frac{\partial}{\partial t}u_1 + F(x, 1)\frac{\partial}{\partial x}u_1 + \frac{1}{2}\sigma_1^2\frac{\partial^2}{\partial x^2}u_1 - xu_1 + \lambda q_{11}u_1 &= -\lambda q_{12}u_2 \\ -\frac{\partial}{\partial t}u_2 + F(x, 2)\frac{\partial}{\partial x}u_2 + \frac{1}{2}\sigma_2^2\frac{\partial^2}{\partial x^2}u_2 - xu_2 + \lambda q_{22}u_2 &= -\lambda q_{21}u_1 \end{aligned} \quad (3.4)$$

Here $q_{i,j}$ is the (i, j) element in the matrix representation of Q . We've arranged the coupling terms on the right hand side. These terms prevent the PDE from being

solved in a straightforward manner.

3.3.2 Notation

Notation in this chapter varies somewhat from the notation in chapter 2. We typically use t to denote time to maturity (rather than the present time) and $u(t, x)$ in place of $B(t, x; T)$ for the bond price (suggesting uses other than the bond price for the same mathematical quantity). Subscripts other than t usually indicate regimes (either 1 or 2) or index terms in an asymptotics series.

3.3.3 Pricing by Simulation

Simulation of Arrival Times

Conditional on a sequence of regime switch times $t_1, \dots, t_k < t$ one can calculate the generating function by recursion, utilizing the fact that the moment generating function for a Vasicek model where the parameters are piecewise constant functions of maturity is known. The “forward” calculation in appendix B.3.6 demonstrates how to simplify the the piecewise constant Vasicek moment generating function.

An exercise in iterated expectations shows that the regime switching model moment generating function and bond price may be estimated by simulating default times and averaging the bond price for the piecewise constant Vasicek model with parameters determined by the regimes. For practical applications control variates may be calculated from well known probability properties of the y_t process, though in testing we avoided this possible source of bugs.

Brute Force Simulation

As a precaution we checked the arrival time simulation against brute force simulation of the short rate process. This method is noisy and there is little that can be said for it.

3.3.4 Recursive Schemes Involving Temporal Discretization

In a slightly different setting Bansal and Zhou [5] provide a recursive scheme for pricing bonds using simple approximations at the level of the pricing PDE. Other authors including Piazzesi [51] have detailed similar approaches involving a temporal discretization and approximations at each time step. The details vary from model to model but the basic idea is simple and common to lattice methods and recombining trees.

One first discretizes time into intervals smaller than the typical time between regime switches so that the probability of more than one regime switch occurring within any given interval is small. We next exploit our ability to compute the relevant expectation *conditional on knowledge of the regime*. The regime choice and time to maturity parametrize nodes on a lattice. The transition probabilities for the finite time step are estimated (or calculated precisely) from the arrival intensity of regime switches. The law of iterated expectations is used to produce recursive formulas allowing us to move backwards along the lattice from the known terminal value to the present day.

Efficiency of Simulation Techniques

Hansen and Poulsen design a backward scheme for a special case of our model where volatility is regime dependent, after first separating out the regime effects as cleanly as possible. We reproduce their comparison of numerical schemes in table 3.1. In this table “naive simulation” refers to brute force Monte Carlo and “smart simulation” is what we refer to as arrival time simulation.

As can be seen from the table the recursive method is relatively less noisy than simulation of default times for an example with regime switches occurring around once a year. The errors introduced by the numerical schemes such as this can be traced to the failure to account for multiple regime switches occurring within a single time interval. When regime switches are rare the pricing can proceed very quickly (and the bulk of our work in this chapter is of little use). Conversely, when regime switches are frequent the pricing is slow and noisy. The number of floating point

Method	10-yr yield (BP)	Uncertainty (BP)	Runtime (sec)
Naive simulation	732.8	1.0	60
Smart simulation	732.9	1.0	1.7
Backward induction	733.1	1.0	0.1

Table 3.1: Reproduction of a comparison of numerical methods for pricing a ten year bond from Hansen and Poulsen[37]. Parameters $\theta = \{0.05, 0.10\}$, $\sigma = 0.02$, $x_0 = 0.05$, $\kappa = 1$, $\lambda = 1$. The backward induction technique as applied to a regime switching model with regime independent σ is described in [37] and summarized in section 3.3. It exploits a decomposition of the short rate not available when σ is stochastic. Naive simulation is referred to in the text as “brute force” simulation. It uses simulation of the short rate process x_t , and is noisier than “smart” simulation (referred to in the text as “arrival time simulation”) in which we calculate piecewise constant parameter bond prices averaged over sequences of regime arrival times.

Bond calculations	1	100	1,000	100,000	10,000,000
Application	Single bond	Default swap	Credit derivative	Complex synthetic	Hedging and risk mgmt
Naive simulation	1 min	1½ hrs	16 hrs	2 months	20 years
Smart simulation	1.7 sec	3 min	30 min	2 days	6 months
Backward induction	0.1 sec	10 sec	1½ min	2½ hrs	1½ weeks
Analytic formulas	0.001 sec	0.1 sec	1 sec	1½ min	2½ hours

Table 3.2: Rough, indicative relative comparison of computation times for numerical and analytic methods, extrapolating the performance of backward induction as devised by Hansen and Poulsen [37]. The times for a particular application are not to be taken to literally and are intended only to provide an approximate indication of the relative scope and practicality of numerical and analytic methods. This table assumes that pricing is carried out to within a basis point on a single machine with specifications given in [37]. We have attempted to be conservative in relating the computational complexity of applications to the complexity of pricing a individual bond.

operations required to price within specified error bounds is roughly proportional to the arrival intensity of regime switches.

For this reason our main result for the regime switching model (the rapid fluctuation asymptotic formulas) complements recursive numerical schemes. In future work we hope to provide slow fluctuation asymptotics designed for situations where regime switches are rare. The benefit of slow fluctuation approximations is that they can be tailored for use in the recursive step of a backward numerical scheme.

In consultation with Shinghoi Lee³, and using the relative speed of the backward induction method compared to “smart” simulation as reproduced in 3.1, we constructed a rough table 3.2 of computation times for simulation, numerical and analytic techniques. We hope that this provides the reader not familiar with the day to day operations of a credit derivative operation an approximate indication of the computational challenges faced. Clearly the scope of applications as indicated by the column headings are subject to debate, though we have tried to be conservative in estimating the computational complexity of these products and relating the computational burden to the computation of a single bond price.

3.3.5 Analytic Pricing Approaches

We will present analytic formulas for bond prices developed using a fast mixing approach to the pricing PDE. Before doing so, we mention some obvious and reasonable approximations. The first is a fast mixing approximation. The second is a slow mixing approximation. These estimates are very intuitive, and will enable us to demonstrate the advantages of our analytics to follow.

Unconditional Averaging: The “Mean Vasicek” Approximation

A simple and intuitive estimate for bond prices is obtained by replacing the parameters $\theta(y_t)$ and $\sigma^2(y_t)$ with their unconditional means $\bar{\theta} = p\theta_1 + q\theta_2$ and $\bar{\sigma}^2 = \sqrt{p\sigma_1^2 + q\sigma_2^2}$. One then prices with the constant parameter Vasicek formula detailed in the introduction, substituting these averaged parameters.

³Morgan Stanley Dean Witter

This approximation is accurate if the Markov process y_t switches regimes very rapidly. An obvious defect of this approximation is that the bond price is independent of the initial regime, an obvious contradiction. However if the regime switches occur very rapidly then the initial regime is quickly forgotten and the estimate is reasonable. We shall see that it is possible, building on this averaging estimate, to obtain very accurate analytic formulas.

Conditional Averaging, The “Dynamic Mean Vasicek” Approximation

A similar approximation requiring only a little more work is achieved by replacing the parameters $\theta(y_t)$ and $\sigma^2(y_t)$ with their means *conditional on the initial value of y_t* . That is to say that we replace the θ in the constant parameter Vasicek PDE with a deterministic function of maturity $\theta(t)$ equal to the expected value $\mathbb{E}[\theta(y_t)|y_0]$ and similarly replace $\sigma^2(t)$ with a deterministic function $\sigma^2(t)$ of equal to the expected value $\mathbb{E}[\sigma^2(y_t)|y_0]$. These functions $\theta(t)$ and $\sigma^2(t)$ of maturity are not complicated. They begin at the initial regime and decay exponentially toward the long run averages $\bar{\theta}$ and $\bar{\sigma}^2$ respectively. The partial differential equation can still be solved and keep the simple exponential affine form. For example if the y_t process spends equal amounts of time in each of the two states then the bond price $u_{\text{dmv}}(t, x)$ is given by

$$u_{\text{dmv}}(t, x) = \exp(-a_{\text{dmv}}(t) - \tau(t)x) \quad (3.5)$$

$$\tau(t) = \frac{1 - e^{-\kappa t}}{\kappa} \quad (3.6)$$

$$a_{\text{dmv}}(t) = \kappa \int_0^t (\bar{\theta} - e^{-\lambda s} \underline{\theta}) \tau(s) ds - \frac{1}{2} \int_0^t (\bar{\sigma}^2 - e^{-\lambda s} \underline{\sigma}^2) \tau^2(s) ds \quad (3.7)$$

where integrals of $\tau^n(s; \kappa)$ present no difficulty.

3.4 Rapid Fluctuation Asymptotics

We describe a method of calculating bond prices in an approximate manner under the assumption that the y_t process reverts to it's mean level quickly as compared to the speed at which the x_t process mean reverts. Put another way, we assume in

this “fast mixing” approximation that the time scale on which regime switches take place is short as compared to a longer time scale of order $1/\kappa$ over which the short rate mean reverts. A numerical analysis and discussion placing these time scales on a firmer footing will follow.

For clarity of exposition we describe the calculation of bond prices. The calculation of the more general and useful moment generating function is similar.

3.4.1 Introduction to the rapid fluctuation asymptotics

For clarity we calculate the first few terms in the rapid fluctuation approximation “by hand” and outline the main ideas.

Separation of scales

We assume $\lambda = 1/\epsilon$ is “large”, although a posteriori we cannot be overly precise about what large means. There are two important time scales with which to compare the average time one spends in a particular regime. The first is the maturity T of the bond we wish to price. The second is the mean reversion time or half life determined by the rate of mean reversion κ of the short rate. We anticipate best results when $\kappa \ll \lambda$ and $1/\lambda \gg T$, although as is clear from the numerical tests, acceptable accuracy is obtained even when the regime switches occur relatively slowly.

We assume $\epsilon \ll 1$ and look for an asymptotic solution

$$u^\epsilon = u_0 + \epsilon u_1 + \epsilon^2 u_2 + \dots \quad (3.8)$$

Substituting this into the bond pricing PDE (3.4) and matching coefficients in increasing powers of ϵ we have

$$0 = -Qu_0 \quad (3.9)$$

$$\begin{aligned} \left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_0 &= -Qu_1 \\ \left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_1 &= -Qu_2 \\ \left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_2 &= -Qu_3 \end{aligned} \quad (3.10)$$

and so on, where

$$\mathcal{D} = \kappa(\theta(y) - x) \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2(y) \frac{\partial^2}{\partial x^2}.$$

Solubility and balancing conditions

The first equation is satisfied if and only if u_0 is independent of y . The remaining equations are pairs of coupled PDE's. Our approach to solving them involves first partially decoupling each pair into what we shall call solubility and balancing conditions. We observe that the right hand sides of the remaining equations are antisymmetric (mean zero with respect to the invariant measure $\mu = [1/2, 1/2]$). Indeed everything in the image of Q must be mean zero since

$$\langle Qf, \mu \rangle = \langle f, Q^* \mu \rangle = \langle f, 0 \rangle = 0.$$

Thus the left hand sides of each PDE in our chain (3.9 – 3.10) must also have mean zero:

$$\overline{\left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_k} = 0. \quad (3.11)$$

We call (3.11) the solubility condition. Subtracting the left hand side of the solubility equation from the PDE gives

$$\left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_k - \overline{\left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right) u_k} = -Q u_{k+1}. \quad (3.12)$$

We call (3.12) the balancing condition for u_{k+1} . Note that both sides of the equation are antisymmetric so indeed have one equation and not two. We shall calculate by hand the first terms in the asymptotic solution (3.8) by alternating between the solubility and balancing equations. It will be useful to introduce bar/underbar notation all for vector valued functions. For instance, $\mathcal{D}_1 = \overline{\mathcal{D}} + \underline{\mathcal{D}}$, $\mathcal{D}_2 = \overline{\mathcal{D}} - \underline{\mathcal{D}}$, and if u_{ki} is the i 'th component of u_k , $u_{k1} = \overline{u_k} + \underline{u_k}$, $u_{k2} = \overline{u_k} - \underline{u_k}$ and so on. These substitutions and a little algebra reduce the balancing and solubility equations to a simple form

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \overline{u_k}(t, x) = -\underline{\mathcal{D}u_k}(t, x) \quad (3.13)$$

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \underline{u_k}(t, x) = -\underline{\mathcal{D}\overline{u_k}}(t, x) - \underline{u_{k+1}}(t, x) \quad (3.14)$$

We remark that the solubility condition (3.13) is an observation regarding the range of the operator Q whereas the balancing condition (3.14) is a statement concerning the domain of the same. The solubility condition says simply that the operator Q is “antisymmetrizing” and furthermore, according to the balancing equation, only antisymmetric parts survive the action of Q . In other words, Q takes the antisymmetric part of its argument and produces a purely antisymmetric function. This feature will enable us to untangle the coupled PDE's by hopping between antisymmetric and symmetric components as we proceed from one time scale to the next.

First term in the expansion

Since u_0 is independent of y a little algebra reduces the first solubility condition to

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \overline{u_0}(t, x) = 0.$$

where $\overline{\mathcal{D}} = \kappa(\overline{\theta} - x) \frac{\partial}{\partial x} + \frac{1}{2}\overline{\sigma}^2 \frac{\partial^2}{\partial x^2}$. The unsurprising solution to (3.4.1) is

$$\overline{u}_0(t, x) = \exp(-a_0(t) - \tau(t)x) \quad (3.15)$$

where

$$\tau(t) = \frac{1 - e^{-\kappa t}}{\kappa} \quad (3.16)$$

$$a_0(t) = \kappa \overline{\theta} \int_0^t \tau(s) ds - \frac{1}{2} \overline{\sigma}^2 \int_0^t \tau^2(s) ds \quad (3.17)$$

$$= \overline{\theta}(t - \tau) - \frac{1}{2} \overline{\sigma}^2 \left(\frac{t - \tau}{\kappa^2} + \frac{\tau^2}{2\kappa} \right) \quad (3.18)$$

That is, we take as the first term in the rapid fluctuation asymptotics the usual Vasicek bond price with averaged parameters $\overline{\theta}$ and $\overline{\sigma}^2$. In the limit $\epsilon \rightarrow 0$ this is the only term in the solution. The process behaves as if at every instant the mean reversion level and instantaneous variance are chosen by a coin flip.

Second term in the expansion

The balancing condition (3.12) for $\underline{u}_1$ can be simplified by hand. We write out both components of the equation and substitute $u_{0,1} = \overline{u}_0 + \underline{u}_0$, $u_{0,2} = \overline{u}_0 - \underline{u}_0$, $\mathcal{D}_1 = \overline{\mathcal{D}} + \underline{\mathcal{D}}$, $\mathcal{D}_2 = \overline{\mathcal{D}} - \underline{\mathcal{D}}$, noting that $\underline{u}_0 = 0$. We also observe that

$$Qu_k = \begin{pmatrix} \underline{u}_k \\ -\underline{u}_k \end{pmatrix}$$

in the case $p = q = 1/2$ for all k and in particular for $k = 1$. As previously noted both components of the balancing equation are equivalent. After a little simplification one has a solution for $\underline{u}_1$ in terms of the previously derived $\overline{u}_0(t, x)$:

$$\underline{u}_1 = \underline{\mathcal{D}} \overline{u}_0$$

Thus

$$\begin{aligned}\underline{u}_1(t, x) &= \left(-\kappa \underline{\theta} \frac{\partial}{\partial x} + \frac{1}{2} \underline{\sigma}^2 \frac{\partial^2}{\partial x^2} \right) \underline{u}_0(t, x) \\ &= \left(\kappa \underline{\theta} \tau(t) + \frac{1}{2} \underline{\sigma}^2 \tau^2(t) \right) \exp(-a_0(t) - \tau(t)x)\end{aligned}\quad (3.19)$$

We now turn to the solubility condition (3.11) in order to find $\overline{u}_1$. After some similar algebra one has

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x \right) \overline{u}_1(t, x) = -\underline{\mathcal{D}} \underline{u}_1(t, x).$$

The boundary condition is $\overline{u}_1(0, x) = 0$. Due to the particular form of the right hand side we can solve this equation exactly. Details are deferred to the following section. The solution is approximated by

$$\overline{u}_1(t, x) \approx -\frac{1}{4} \kappa \underline{\theta} \underline{\sigma}^2 u_0(t, x) \quad (3.20)$$

Combining this with (3.19) and (3.15) we have solved for the bond price up to an error of order ϵ^2 .

$$u(t, x, y) = \overline{u}_0(t, x) \left(1 + \epsilon \overline{u}_1(t, x) \pm \epsilon \underline{u}_1(t, x) \right) + O(\epsilon^2)$$

The plus sign is taken in the first state $y = 1$, the minus sign in the state $y = -1$.

3.4.2 Rapid fluctuation asymptotics in detail

Solubility and balancing equations viewed as a basis change

Whilst the solubility and balancing equations can be simplified by hand in the case $p = q = 1/2$, the task is essentially a change of basis which we now describe. The idea is to decompose $\mathbb{R}^2$ into a “symmetric” part and an “antisymmetric” part, though for p not equal to $1/2$ the latter terminology is a little misleading.

Let $\mathbb{F}$ be the space of functions which depend on both x and y . Let

$$\mathcal{W} = \{f \in \mathbb{F} : \mathbb{E}^\mu f = 0\},$$

where $\mathbb{E}^\mu$ indicates the expectation with respect to a measure $\mu = (p, q)$ and $p + q = 1$. Let $\mathcal{V}$ denote the orthogonal complement of $\mathcal{W}$ in $\mathbb{F}$ (with respect to μ_k). Then $\mathbb{F} = \mathcal{V} + \mathcal{W}$. We can easily calculate the unique v in $\mathcal{V}$ and w in $\mathcal{W}$ such that $u = v + w$. The projections onto these subspaces and their adjoints in the canonical basis are

$$P_v = \begin{bmatrix} p & q \\ p & q \end{bmatrix} \quad P_w = \begin{bmatrix} q & -q \\ -p & p \end{bmatrix}$$

$$P_v^* = \begin{bmatrix} p & p^2/q \\ q^2/p & q \end{bmatrix} \quad P_w^* = \begin{bmatrix} q & -p^2/q \\ -q^2/p & p \end{bmatrix}$$

The spaces $\mathcal{W}$ and $\mathcal{V}$ are spanned by

$$e_w = \begin{bmatrix} \sqrt{q/p} \\ -\sqrt{p/q} \end{bmatrix} \quad e_v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The transformation of coordinates when changing bases from the canonical basis to $\{e_v, e_w\}$ is given by

$$C = \begin{bmatrix} p & q \\ \sqrt{pq} & -\sqrt{pq} \end{bmatrix}, \quad C^{-1} = \begin{bmatrix} 1 & \sqrt{q/p} \\ 1 & -\sqrt{p/q} \end{bmatrix}$$

An operator represented by $\mathcal{A}$ in the canonical basis has representation $\tilde{\mathcal{A}} = C^{-1}\mathcal{A}C$ in the $\{e_v, e_w\}$ basis. For example in the symmetric case $p = q$ we have $\tilde{Q} = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$. If p is close to q this is “almost” the case. In fact if $p = \frac{1+\xi}{2}$ then

$$\tilde{Q} = \lambda \begin{bmatrix} (\frac{1}{4}\xi^2 - \frac{1}{16}\xi^4 + O(\xi^6)) & (\frac{1}{2}\xi - \frac{1}{2}\xi^2 + O(\xi^6)) \\ (-\frac{1}{2}\xi - \frac{1}{2}\xi^2 - \frac{1}{2}\xi^3 - \frac{1}{2}\xi^4 - \frac{3}{8}\xi^5 + O(\xi^6)) & (-1 - \frac{1}{4}\xi^2 + \frac{1}{16}\xi^4 + O(\xi^6)) \end{bmatrix}$$

Multiples of the identity matrix which are independent of y are left unchanged by the basis change. For our purposes it suffices to calculate the transformation of arbitrary

multiplication operators taking the form

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

in the canonical basis. These transform to (exactly)

$$\tilde{A} = \begin{bmatrix} \frac{a+b}{2} + \frac{a-b}{2}\xi & \frac{a-b}{2}(1+\xi) \\ \frac{a-b}{2}(1+\xi) & \frac{a+b}{2} + \frac{a-b}{2}\xi \end{bmatrix}$$

in the new basis. If a is close to b the operator is left largely unchanged and is almost diagonal.

We now apply the transformation to the PDEs (3.9) through (3.10) in the case $\xi = 0$ where the invariant measure is symmetric. The case $p \neq q$ is not treated here. Recall that after matching coefficients the k 'th pair of PDE's are

$$-\frac{\partial u_{k,1}}{\partial t} + F_1(x) \frac{\partial u_{k,1}}{\partial x} + \frac{1}{2} \sigma_1^2 \frac{\partial^2}{\partial x^2} u_{k,1} - x u_{k,1} + q_{1,1} u_{k+1,1} = -q_{1,2} u_{k+1,2} \quad (3.21)$$

$$-\frac{\partial u_{k,2}}{\partial t} + F_2(x) \frac{\partial}{\partial x} u_{k,2} + \frac{1}{2} \sigma_2^2 \frac{\partial^2}{\partial x^2} u_{k,2} - x u_{k,2} + q_{2,1} u_{k+1,1} = -q_{2,2} u_{k+1,2} \quad (3.22)$$

where $q_{i,j}$ is the i, j 'th coefficient of Q (without the scaling term $\epsilon = 1/\lambda$), $u_{k,i}$ is the i 'th component of u_k and $F_y(x) = \kappa(\theta(y) - x)$. Alternatively,

$$-\frac{\partial}{\partial t} u_k + \mathcal{D} u_k - x u_k = -Q u_{k+1}$$

where

$$\begin{aligned} Q &= -\frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ D &= \kappa(\Theta - x\mathbf{1}) \frac{\partial}{\partial x} + \frac{1}{2} \Sigma^2 \frac{\partial^2}{\partial x^2} \\ \Theta &= \begin{bmatrix} \theta_1 & 0 \\ 0 & \theta_2 \end{bmatrix} \quad \Sigma^2 = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \end{aligned}$$

After the change of basis with $p = q = 1/2$ we have

$$-\frac{\partial}{\partial t}\tilde{u}_k + \tilde{\mathcal{D}}\tilde{u}_k - x\tilde{u}_k = -\tilde{Q}u_{k+1} \quad (3.23)$$

where the new components of $\tilde{u}_k$ are precisely $\overline{u}_k = \frac{u_{k1}+u_{k2}}{2}$ and $\underline{u}_k = \frac{u_{k1}-u_{k2}}{2}$. Also

$$\begin{aligned} \tilde{Q} &= \begin{bmatrix} 0 & 0 \\ 0 & -\lambda \end{bmatrix} \\ \tilde{\mathcal{D}} &= \kappa(\tilde{\Theta} - x\mathbf{1}) \frac{\partial}{\partial x} + \frac{1}{2}\tilde{\Sigma}^2 \frac{\partial^2}{\partial x^2} = \begin{pmatrix} \overline{\mathcal{D}} & \underline{\mathcal{D}} \\ \underline{\mathcal{D}} & \overline{\mathcal{D}} \end{pmatrix} \\ \overline{\mathcal{D}} &= \kappa(\overline{\theta} - x) \frac{\partial}{\partial x} + \frac{1}{2}\overline{\sigma}^2 \frac{\partial^2}{\partial x^2} \\ \underline{\mathcal{D}} &= \kappa\underline{\theta} \frac{\partial}{\partial x} + \frac{1}{2}\underline{\sigma}^2 \frac{\partial^2}{\partial x^2} \\ \tilde{\Theta} &= \begin{bmatrix} \overline{\theta} & \underline{\theta} \\ \underline{\theta} & \overline{\theta} \end{bmatrix} \quad \tilde{\Sigma}^2 = \begin{bmatrix} \overline{\sigma}^2 & \underline{\sigma}^2 \\ \underline{\sigma}^2 & \overline{\sigma}^2 \end{bmatrix} \end{aligned}$$

so one sees that in the new basis the components of (3.23) are the solubility and balancing conditions

$$\left(-\frac{\partial}{\partial t} + \overline{\mathcal{D}} - x\right) \overline{u}_k(t, x) = -\underline{\mathcal{D}}u_{k+1}(t, x) \quad (3.24)$$

$$\left(-\frac{\partial}{\partial t} + \underline{\mathcal{D}} - x\right) \underline{u}_k(t, x) = -\overline{\mathcal{D}}u_{k+1}(t, x) - \underline{u}_{k+1}(t, x) \quad (3.25)$$

respectively. The boundary conditions are $\overline{u}_k(0, x) = 1$ and $\underline{u}_k(0, x) = 0$ for all k with the sole exception of $\overline{u}_0(0, x) = 1$. The key to our approach is that the solubility condition can be solved exactly because the right hand side takes a special form.

Algorithm for computing all terms in the rapid fluctuation approximation exactly

The expansion

$$u^\epsilon = \sum_{k=0}^{\infty} \epsilon^k (\overline{u}_k(t, x) \pm \underline{u}_k(t, x))$$

can be determined exactly. Let $\bar{\mathcal{L}} = (-\frac{\partial}{\partial t} + \bar{\mathcal{D}} - x)$ and perform the following steps :

1. Let $k = 0$ and $\underline{u}_0 = 0$.

2. Solve

$$\bar{\mathcal{L}}\bar{u}_k = -\underline{\mathcal{D}}\underline{u}_k$$

with initial condition $\bar{u}_k(0, x) = 1$ for $k = 0$ and $\bar{u}_k(0, x) = 0$ for $k > 0$.

3. Let

$$\underline{u}_{k+1} = \bar{\mathcal{L}}\underline{u}_k + \underline{\mathcal{D}}\bar{u}_k.$$

4. Increment k and return to step 2.

Assuming this task can actually be followed through, we have, formally, met all balancing and solubility conditions. This is indeed the case, although it is perhaps fortuitous that the second step of the algorithm (which involves solving an inhomogeneous partial differential equation) can be carried out explicitly. The ability to solve the inhomogeneous PDE can be traced to the fact that every term in the expansion is a deterministic multiple of $\bar{u}_0$ and such multiples, when appearing as driving terms on the right hand side of the PDE in step two, can be dealt with mechanically. We explain why this is the case in the following section (which the reader may wish to skip).

Aside on probabilistic solutions to certain nonhomogeneous PDE's

In these notes we solve exactly a particular nonhomogeneous PDE of the form (3.26) driven on the right hand side by a deterministic multiple of the homogeneous solution. One can obtain the rather simple solution by guessing or postulating an solution which is a deterministic multiple of an exponential affine function of the state variable. For this reason the reader may wish to skip this explanation.

Alternatively, one can treat the operator on the left hand side as the generator of the graph of a Vasicek process⁴ which may be killed at any time. Then the

⁴Strictly speaking we mean the graph of the Vasicek process with a sign change on the x_t process

probabilistic solution is straightforward and might be thought of as an “integrated” Feynman-Kac formula. We are really just solving a Poisson problem on an infinite strip in $\mathbb{R}^2$ where only one boundary is absorbing, as follows:

To arrive at a solution, we consider the following problem for each fixed t . We define the following processes. Let x_s be a process with generator $\mathcal{D}$ and initial value x at time zero (we are interested in the particular case $\mathcal{D} = \kappa(\theta - x_s)\frac{\partial}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2}{\partial x^2}$). Let $\tilde{x}_s$ be the process x_s killed with an intensity x_s . That is, let the process x_s have a continuous transition probability x_s of falling into a coffin state⁵. The probability of the x_t process dying happens to be equal in magnitude to the process x_t itself.

We let z_s be the process $z_s = (t - s, x_s)$ in $\mathbb{R}^2$ and let $\tilde{z}_s$ be the process z_s killed with intensity x_s . The generator of the process z_s is $-\frac{\partial}{\partial s} + \mathcal{D}$. The generator for $\tilde{x}_s$ is $\mathcal{D} - x$ and the generator for $\tilde{Z}_t$ is $-\frac{\partial}{\partial s} + \mathcal{D} - x$. The nonhomogeneous equation

$$\left(-\frac{\partial}{\partial t} + \mathcal{D} - x\right)u(t, x) = -G(t, x) \quad (3.26)$$

with boundary condition $u(0, x) = 0$ has probabilistic solution

$$\begin{aligned} u(t, x) &= \mathbb{E}^x \left[\int_0^t G(\tilde{z}_s) ds \right] \\ &= \mathbb{E}^x \left[\int_0^t \exp \left(- \int_0^s x_u du \right) G(z_s) ds \right] \\ &= \mathbb{E}^x \left[\int_0^t \exp \left(- \int_0^s x_u du \right) G(t - s, x_s) ds \right] \end{aligned}$$

The first equality results from the well known solution to the Poisson problem (with zero boundary condition). Strictly speaking we’ve extended the definition of $G(t, x)$ so that it is zero in the coffin state. If $G(t, x)$ takes the particular form $g(t)B(t, x)$,

⁵That is to say, being killed. It is convenient to enlarge the space in which x_t lives (and dies) by including another state.

where $B(t, x) = \mathbb{E}^x \left[\exp \left(- \int_0^t x_s ds \right) \right]$ then we have the nice simplification:

$$\begin{aligned} u(t, x) &= \mathbb{E}^x \left[\int_0^t \exp \left(- \int_0^s x_u du \right) g(t-s) B(t-s, x_s) ds \right] \\ &= \int_0^t \mathbb{E}^x \left[\exp \left(- \int_0^s x_u du \right) B(t-s, x_s) \right] g(t-s) ds \\ &= \int_0^t B(t, x) g(t-s) ds \\ &= B(t, x) \int_0^t g(s) ds \end{aligned}$$

This equality is readily verified by substituting $u(t, x) = B(t, x) w(t)$ into the PDE, with $w(t) = \int_0^t g(s) ds$. We observe that

$$\begin{aligned} \left(-\frac{\partial}{\partial t} + \mathcal{D} - x \right) u(t, x) &= w(t) \left(-\frac{\partial}{\partial t} + \mathcal{D} - x \right) B(t, x) - B(t, x) \frac{\partial}{\partial t} w(t) \\ &= 0 - B(t, x) g(t) \\ &= -G(t, x) \end{aligned}$$

by definition of $B(t, x)$.

3.4.3 A simplified algorithm

As mentioned above we can assume that the solutions for the u 's appearing in the expansion (3.8) are multiples of the first symmetric term $u_0(t, x)$. We write

$$\overline{u}_k(t, x) = \overline{v}_k(t) \overline{u}_0(x, t)$$

and

$$\underline{u}_k(t, x) = \underline{v}_k(t) \underline{u}_0(x, t)$$

where the v 's are some functions lying in the space $P(t, \tau)$ of polynomials in t and $\tau(t) = \frac{1-e^{-\kappa t}}{\kappa}$. Viewed as manipulations of the v functions, the algorithm for computing all terms exactly requires evaluating the action of three operators: differentiation

with respect to t , integration with respect to t and multiplication by a polynomial in $\tau(t)$. To see this we conjugate the operator $\bar{\mathcal{L}}$ with the isometry

$$g(t) \mapsto g(t)\bar{u}_0(t, x)$$

and note that $\bar{\mathcal{L}} = -\frac{\partial}{\partial t} + \bar{\mathcal{D}} - x$ transforms to simply $-\frac{\partial}{\partial t}$. Similarly the operator $\underline{\mathcal{D}}$ transforms to multiplication by

$$\kappa \underline{\theta} \tau(t) + \frac{1}{2} \underline{\sigma}^2 \tau^2(t).$$

The range of this multiplication operator is clearly within $P(t, \tau)$. The same is true of the operator $-\frac{\partial}{\partial t}$ since $\frac{\partial}{\partial t} \tau(t) = 1 - \kappa \tau(t)$. and also for the integration operator $\int$ which is calculated cleanly using

$$\int_0^t \tau^n(s) ds = \frac{t}{\kappa^n} - \sum_{j=1}^n \frac{\tau^j(t)}{j \kappa^{n-j+1}}$$

Therefore we restate the algorithm as a calculation of the v 's in the space $P(t, \tau)$,

1. Let $\underline{v}_0 = 0$, $\bar{v}_0 = 1$, and $k = 1$.
2. Let $\underline{v}_k(t) = -\frac{\partial}{\partial t} \underline{v}_{k-1}(t) + \underline{\phi}(\tau) \bar{v}_{k-1}$.
3. Let $\bar{v}_k(t) = \int_0^t \underline{\phi}(\tau(s)) \underline{v}_k(s) ds$,
4. Increment k and return to step 2.

and solve the recursive algorithm to find explicit formulas for the v 's:

$$\underline{v}_k = \left(-\frac{\partial}{\partial t} + \underline{\phi} \cdot \int \cdot \underline{\phi} \right)^{k-1} \underline{\phi} \cdot 1 \quad (3.27)$$

$$\bar{v}_k = \int \cdot \underline{\phi} \left(-\frac{\partial}{\partial t} + \underline{\phi} \cdot \int \cdot \underline{\phi} \right)^{k-1} \underline{\phi} \cdot 1 \quad (3.28)$$

which can be calculated using matrix multiplication only. For concreteness we now describe this representation, using the canonical basis for $P(t, \tau)$ given by $\{p_{m,n} = \tau^m t^n\}_{m,n=0}^\infty$

Integrating by parts we let

$$\begin{aligned} I_{m,n} &:= \int_0^t s^m \tau^n(s) ds \\ &= \frac{1}{m+1} t^{m+1} \tau^n - \frac{1}{1+\kappa} \int_0^t s^{m+1} n (1 - \kappa \tau(s)) \tau^{n-1}(s) ds \\ &= \frac{1}{m+1} t^{m+1} \tau^n + \frac{\kappa n}{m+1} I_{m+1,n} - \frac{n}{m+1} I_{m+1,n-1}. \end{aligned}$$

Rearranging this we have

$$I_{m+1,n} = \frac{1+m}{\kappa n} I_{m,n} - \frac{1}{n\kappa} p_{m+1,n} + \frac{1}{\kappa} I_{m+1,n-1}.$$

A matrix representation of the integration operator is therefore

$$\int \cdot \left(\sum_{m,n} c_{m,n} p_{m,n} \right) = \sum_{m,n} \sum_{r,s} \iota_{m,n;r,s} c_{r,s} p_{m,n}$$

where ι satisfies the relation

$$\iota_{m+1,n;r,s} = \frac{1+m}{\kappa n} \iota_{m,n;r,s} + \frac{1}{\kappa} \iota_{m+1,n-1;r,s} - \frac{1}{\kappa n} \delta(m+1, r) \delta(n, s)$$

All coefficients for particular r, s may be calculated using a diagonal ordering of the m, n tuples. The matrix representation of the differentiation operator $\mathcal{T} = -\frac{\partial}{\partial t}$ is clear. Since

$$-\frac{\partial}{\partial t} p_{m,n} = -m p_{m-1,n} - n p_{m,n-1} + n \kappa p_{m,n}$$

we have

$$\mathcal{T} \sum_{m,n} c_{m,n} p_{m,n} = \sum_{m,n} \sum_{r,s} \mathcal{T}_{m,n;r,s} c_{r,s} p_{m,n}$$

where the matrix coefficients satisfy

$$\mathcal{T}_{m,n;r,s} = -m \delta(m-1, r) \delta(n, s) - n \delta(m, r) \delta(n-1, s) + n \kappa \delta(m, r) \delta(n, s)$$

Finally the matrix representation of the multiplication operator $\underline{\phi}$ is obtained:

$$\underline{\phi} p_{m,n} = -\kappa \underline{\theta} p_{m,n+1} + \frac{1}{2} \underline{\sigma}^2 p_{m,n+2}$$

or in other words

$$\underline{\phi} \sum_{m,n} c_{m,n} p_{m,n} = \sum_{m,n} \sum_{r,s} \phi_{m,n;r,s} c_{r,s} p_{m,n}$$

where

$$\phi_{m,n;r,s} = -\kappa \underline{\theta}(m,r) \delta(n+1,s) + \frac{1}{2} \underline{\sigma}^2 \delta(m,r) \delta(n+2,s)$$

Given the matrix representation for T , $\underline{\phi}$ and $\int$ we can mechanically compute (3.27) and (3.28), although in practice it is simpler to take a more direct approach using a symbolic package. In numerical tests to follow we also examined the accuracy of formulas obtained with a faster approximate matrix representation obtained using the less basis $P(\tau)$ of polynomials in $\tau(t)$. We use the series approximation

$$t = \tau + \frac{1}{2} \kappa \tau^2 + \frac{1}{3} \kappa^2 \tau^3 + \dots$$

and notice that

$$\int_0^t \tau^n(s) ds = \sum_{j=n+1}^{\infty} \frac{\tau^j}{j \kappa^{n-j+1}}$$

so to an arbitrary degree of accuracy T maps $P(\tau)$ into itself. The other operators $-\frac{\partial}{\partial t}$ and $\underline{\phi}$ manifestly map $P(\tau)$ into itself and the approximate matrix representation is given by

$$\begin{aligned} \int \cdot \sum_m c_m \tau^m &= \sum_m \sum_r c_r \ell_{m,r} \tau^m \\ \mathcal{T} \cdot \sum_m c_m \tau^m &= \sum_m \sum_r c_r \mathcal{T}_{m,r} \tau^m \\ \underline{\phi} \cdot \sum_m c_m \tau^m &= \sum_m \sum_r c_r \phi_{m,r} \tau^m \end{aligned}$$

where

$$\iota_{m,r} = r^{-1} \kappa^{-m+r-1} \delta(r > m) \quad (3.29)$$

$$\mathcal{T}_{m,r} = -m\delta(m-1, r) + n\kappa\delta(m, r) \quad (3.30)$$

$$\phi_{m,r} = -\kappa\theta\delta(m+1, r) + \frac{1}{2}\sigma^2\delta(m+2, r) \quad (3.31)$$

We find that this approximate technique works well when regime switches occur rapidly. However it is generally less robust than use of the exact series expansion and we feel there would be few situations where the speed improvement was warranted.

3.4.4 Analytic Formulas for Bond Prices

We present our results in terms of the v_k 's, reminding the reader that the $\bar{v}_k(t, x)$ terms represent corrections to the bond price which are independent of the initial regime and $\underline{v}_k(t, x)$ terms represent corrections which are dependent on the initial regime. The bond prices are perturbations on the Mean Vasicek price $\bar{u}_0(t, x)$. Corrections are multiples of this price also, and are scaled by the factor ϵ . The $\bar{u}_k(t, x)$'s are referred to as **symmetric corrections** while the $\underline{u}_k(t, x)$'s are termed **antisymmetric corrections**. The first antisymmetric correction term is

$$\begin{aligned} \underline{v}_1 &= \frac{1}{4\kappa^2}(1 - \exp(-\kappa t))(2\theta\kappa^2 + \sigma^2 - \sigma^2 \exp(-\kappa t)) \\ &= \frac{1}{2}\kappa\tau(t)\theta + \frac{1}{4}\sigma^2\tau^2(t) \end{aligned}$$

For small maturities we notice that

$$\bar{v}_1 \rightarrow \frac{1}{2}\kappa\theta t \text{ as } t \rightarrow 0$$

where we recall that $\underline{\theta}$ is half the difference between the mean reversion levels for $\theta(y_t)$ in the two regimes. The first symmetric correction term is

$$\begin{aligned} \overline{v_1} = & \frac{1}{96\kappa^5} \{ -3 \exp(-4\kappa t) \underline{\sigma}^4 + 16 \exp(-3\kappa t) \underline{\theta} \underline{\sigma}^2 \kappa^2 \\ & + 16 \exp(-3\kappa t) \underline{\sigma}^4 - 24 \exp(-2\kappa t) \underline{\theta}^2 \kappa^4 - 72 \exp(-2\kappa t) \underline{\theta} \underline{\sigma}^2 \kappa^2 \\ & - 36 \underline{\sigma}^4 \exp(-2\kappa t) + 96 \exp(-\kappa t) \underline{\theta}^2 \kappa^4 \\ & + 144 \underline{\theta} \underline{\sigma}^2 \exp(-\kappa t) \kappa^2 + 48 \underline{\sigma}^4 \exp(-\kappa t) + 48 \kappa^5 t \underline{\theta}^2 + 48 \kappa^3 t \underline{\theta} \underline{\sigma}^2 \\ & + 12 \kappa t \underline{\sigma}^4 - 25 \underline{\sigma}^4 - 88 \underline{\theta} \kappa^2 \underline{\sigma}^2 - 72 \underline{\theta}^2 \kappa^4 \}. \end{aligned}$$

The short maturity limit

$$\overline{v_1} \rightarrow \frac{1}{6} \kappa^2 \underline{\theta}^2 t^3 \text{ as } t \rightarrow 0$$

is not obvious but can also be derived from first principles and is consistent with elementary dimensionality considerations. Higher order corrections are also omitted.

Bond Prices Corrections for a Stochastic Mean Model

We obtain a special case, the stochastic mean model discussed earlier, by setting $\underline{\sigma} = 0$. Only the mean reversion level $\theta(y)$ is driven by the regime switching process. The hazard rate is

$$dx_t = \kappa (\theta(y_t) - x_t) dt + \sigma dW_t$$

The first two antisymmetric and symmetric correction terms are

$$\begin{aligned} \underline{v_1} &= \frac{1}{2} (1 - \exp(-\kappa t)) \underline{\theta} \\ \underline{v_2} &= \frac{1}{4\kappa} \underline{\theta} \{ 5 \exp(-2\kappa t) \underline{\theta}^2 - \exp(-3\kappa t) \underline{\theta}^2 - 7 \exp(-\kappa t) \underline{\theta}^2 + 3 \underline{\theta}^2 \\ &\quad - 2 \kappa t \underline{\theta}^2 + 2 \kappa t \underline{\theta}^2 \exp(-\kappa t) + 2 \kappa^2 \exp(-\kappa t) \} \\ \overline{v_1} &= \frac{1}{4\kappa} \underline{\theta}^2 \{ -\exp(-2\kappa t) + 4 \exp(-\kappa t) + 2 \kappa t - 3 \} \\ \overline{v_2} &= \frac{1}{16\kappa^2} \underline{\theta}^2 \{ -4 \kappa^2 \exp(-2\kappa t) + 9 \underline{\theta}^2 + 4 \kappa^2 t^2 \underline{\theta}^2 - 12 \kappa t \underline{\theta}^2 \\ &\quad + 8 \kappa^2 \exp(-\kappa t) - 4 \kappa^2 - 24 \exp(-\kappa t) \underline{\theta}^2 + \exp(-4\kappa t) \underline{\theta}^2 \\ &\quad + 22 \exp(-2\kappa t) \underline{\theta}^2 + 16 \kappa t \underline{\theta}^2 \exp(-\kappa t) - 8 \exp(-3\kappa t) \underline{\theta}^2 - 4 \exp(-2\kappa t) \underline{\theta}^2 \kappa t \} \end{aligned}$$

Bond Price Corrections for a Stochastic Volatility Model

We obtain a simple two state stochastic volatility model by setting $\underline{\theta} = 0$. The x_t process follows

$$dx_t = \kappa (\theta - x_t) dt + \sigma(y_t) dW_t$$

with y_t as before. The leading antisymmetric and symmetric correction terms are

$$\begin{aligned} \underline{v}_1 &= \frac{\underline{\sigma}^2}{4\kappa^2} (1 - \exp(-\kappa t))^2 \\ \underline{v}_2 &= \frac{1}{192\kappa^7} \underline{\sigma}^2 \{ 136 \exp(-3\kappa t) \underline{\sigma}^4 - 71 \exp(-4\kappa t) \underline{\sigma}^4 \\ &\quad - 157 \underline{\sigma}^4 \exp(-2\kappa t) - 25 \underline{\sigma}^4 + 22 \underline{\sigma}^4 \exp(-5\kappa t) - 3 \underline{\sigma}^4 \exp(-6\kappa t) \\ &\quad + 96 \exp(-2\kappa t) \kappa^6 + 12 \kappa t \underline{\sigma}^4 \exp(-2\kappa t) + 12 \kappa t \underline{\sigma}^4 + 98 \underline{\sigma}^4 \exp(-\kappa t) \\ &\quad - 96 \exp(-\kappa t) \kappa^6 - 24 \kappa t \underline{\sigma}^4 \exp(-\kappa t) \} \\ \overline{v}_1 &= \frac{1}{96\kappa^5} \underline{\sigma}^4 \{ -3 \exp(-4\kappa t) + 16 \exp(-3\kappa t) - 36 \exp(-2\kappa t) \\ &\quad + 48 \exp(-\kappa t) + 12 \kappa t - 25 \} \end{aligned}$$

The notation $\underline{\sigma}^4$ is a little misleading, but the more correct $(\underline{\sigma}^2)^2$ looks clumsy. Approximations to these correction terms to leading order in maturity

$$\begin{aligned} \underline{v}_1 &\approx \frac{1}{4} \underline{\sigma}^2 t^2 \\ \underline{v}_2 &\approx -\frac{1}{2} \underline{\sigma}^2 t \\ \overline{v}_1 &\approx \frac{1}{40} \underline{\sigma}^4 t^5 \\ \overline{v}_2 &\approx -\frac{1}{16} \underline{\sigma}^4 t^4 \end{aligned}$$

may or may not assist the intuition. Higher order terms were also calculate exactly but are omitted.

3.4.5 Numerical Tests of the Rapid Fluctuation Asymptotics

In the preceding section we did not attempt to state a priori conditions under which the preceding rapid fluctuation asymptotics are valid. We have also been a little vague in defining what is meant by rapid fluctuation as there are more than two time scales involved. There are in fact three time scales of importance. They are the maturity of the contract, the mean reversion time of the x_t process and the mean length of time between regime switches. Therefore the relative sizes of T , $1/\kappa$ and $1/\lambda$ provide heuristics for the applicability of the fast mixing formulas. We examine several cases and drawn up guidelines for the use of the bond price formulas. As stated in the introduction, the potential use of the analytics goes beyond what one can guarantee rigorously.

Methodology

We recall that the bond prices arrived at using the rapid fluctuation asymptotics can be expressed

$$\begin{pmatrix} \text{Regime switching} \\ \text{bond price} \end{pmatrix} = \begin{pmatrix} \text{Vasicek} \\ \text{price} \end{pmatrix} \times \left(1 + \begin{matrix} \text{Symmetric} \\ \text{correction} \end{matrix} \pm \begin{matrix} \text{Antisymmetric} \\ \text{correction} \end{matrix} \right)$$

where the symmetric and antisymmetric corrections comprise series whose coefficients of $\epsilon^k = 1/\lambda^k$ are $\overline{v_k}(t)$ and $\underline{v_k}(t)$ respectively. Both series may be calculated exactly as demonstrated in section 3.4.4. To test the applicability of the corrections we compared the results with simulation, reporting the bond prices and more importantly those measured quantities which correspond to the symmetric and antisymmetric corrections.

The simulation technique used was the method of simulation of arrival times as described in section 3.3.3 with details given in appendix B.3.6. The simulation of arrival times was checked against brute force simulation of the default intensity for a few examples. The results that follow are intended to be illustrative examples conforming to heuristics devised from more numerous trials. We typically used hundreds of thousands of simulations for each example - vastly more than would be used in any

practical setting.

Extreme example with constant volatility

In this first example the level to which the short rate mean reverts switches between thirty five percent and fifteen percent. These large jumps produce symmetric and asymmetric corrections on the order of hundreds of basis points, providing a good test for the formulas.

We assume the mean reversion time of the short rate, determined by κ , is around six months. We switch between two possible regimes approximately five times per year on average, with intensity $\lambda = 5$. The maturity of the bond was set to 15 years. The time scales in increasing order are

$1/\lambda$	$1/\kappa$	T
0.2	0.5	15

Within each regime the short rate follows a Vasicek process with parameters.

Parameter	Regime 1	Regime 2
κ	2	2
θ	0.35	0.15
σ	0.0001	0.0001

In other words, $\kappa = 2$, $\bar{\theta} = 0.25$, $\bar{\sigma}^2 \approx 0$, $\underline{\theta} = 0.10$, $\underline{\sigma}^2 = 0$, $\epsilon = 1/5$. We report the bond prices for the two regimes and the asymmetric and symmetric corrections. The latter are computed by dividing the sum and difference of the bond prices by twice the “mean Vasicek” bond price. (The mean Vasicek is the first term in the expansion $\bar{u}_0(t, x)$ obtained by substituting $\bar{\theta}$ and $\bar{\sigma}$ into the classical formula). Several methods are compared. Using the first term in the slow mixing approximation (Vasicek with a deterministic mean reversion level equal to the conditional expectation of $\theta(y_t)$) we calculate the “Dynamic mean Vasicek” bond prices. Two rapid fluctuation asymptotics are calculated. The first is an approximate solution using seven terms from the basis $P(\tau)$. The second uses seven terms from the exact expansion with coefficients

in $P(t, \tau)$. It was calculated directly using a symbolic package.

Method	Bond 1	Bond 2	Antisymmetric	Symmetric
Mean Vasicek	0.023517	0.023517	0	0
Dynamic mean Vasicek	0.023383	0.023652	-0.0057143	1.6308e-05
Asymptotic series (approx)	0.023368	0.023734	-0.0077734	0.00142
Asymptotic series (exact)	0.023611	0.024067	-0.010281	0.01427
Simulation (200,000)	0.023622	0.024095	-0.010020	0.01447
Sampling Error	0.4%	0.4%	0.0003	0.0003

Around 200,000 sequences of regime switches were used in the simulation. The sampling error in the bond prices is on the order of a hundredth of a basis point. Although seven terms were used in the calculation of the exact rapid fluctuation asymptotics, two would have been more than sufficient. The expansion converges very quickly for both the asymmetric and symmetric corrections.

Due to the size of the jumps in the level of mean reversion a substantial error is introduced if one used simple averaging rather than the formulas derived here. Recall that the “Mean Vasicek” approximation shown in the first row is the first term in the expansion and equal to the traditional vanilla Vasicek with averaged parameters $\bar{\theta}$ and $\bar{\sigma}^2$. One may be tempted to argue that with an average of $T\lambda = 75$ regime switches before maturity there would be little harm done in adopting this approach. However the numbers tell a different story. The higher order terms are crucially important.

Both the asymmetric and symmetric corrections estimated by these higher order terms are in excellent agreement with simulation. The discrepancies are within the sampling error of the simulation as can be seen from the two rows shown in bold. For this example the asymptotic solution converged very quickly and it made very little difference where the series was truncated. The figures shown use seven terms in the expansion.

This example works nicely because the smallest time scale is the typical time between regimes. If we are to err on the side of caution this might constitute the definition of “rapid fluctuation”. We speculate the rule of thumb:

First Heuristic

If $1/\lambda < 1/\kappa < T$ then the rapid fluctuation asymptotics is justified and the asymptotic series converges quickly.

Example where the rapid fluctuation asymptotics requires truncation

Unfortunately the first heuristic is quite restrictive in many practical settings. In this example we test whether the regime switching time must really be less than the mean reversion time.

We use the same model parameters as in the previous example. Again volatility is kept constant and only the mean reversion level changes. The regime switches occur with intensity 1. The mean reversion parameter κ is increased to 25 and the maturity is reduced to only five years. In this situation it is difficult to justify the rapid fluctuation asymptotics since both the maturity and the mean reversion time are small relative to the mean time spend in one regime.

Parameter	Regime 1	Regime 2
κ	25	25
θ	0.35	0.15
σ	0.0001	0.0001

Some separation is maintained between the regime switching time and the maturity so that we are still in a “rapid fluctuation” world. However the ordering of time scales is modified:

$1/\kappa$	$1/\lambda$	T
0.5	1	15

and it turns out that this does make a difference. We find that the rapid fluctuation expansion fails to converge whether we use the exact solution or the approximate method. The first eight terms in the exact asymmetric and symmetric correction are

listed below

	Asymmetric	Symmetric
1	0.050000000	0.0247000
2	0.002470000	-0.0018899100
3	-.6889910000e-3	-.6285928720e-1
4	-.6310628720e-2	-1.565533822
5	-.1565414921	-39.13840040
6	-3.913776688	-978.4822810
7	-97.84666285	-24461.50472
8	-2446.111334	-611336.5805

From this table it would appear that we have already pushed the rapid fluctuation envelope too far. Curiously however a judicious truncation of the series gives a very accurate estimate of the simulated corrections. The symmetric and asymmetric corrections for various truncations of the series are tabulated below.

	Asymmetric	Symmetric
Dynamic mean Vasicek	0.0956	0.0046
Asymptotic series (1 term)	0.0500	0.0247
Asymptotic series (2 terms)	0.0525	0.0228
Simulation (75,000)	0.0516	0.0226
Sampling error	0.0004	0.0004

For this example the simple slow mixing approximation (“Dynamic mean Vasicek”) is not at all accurate. The simulated asymmetric correction is only half the predicted one whereas the simulated symmetric correction is four or five times it’s theoretical counterpart.

An excellent approximation is achieved by truncating the asymptotic solution after the first two terms. The agreement (numbers in bold) between the formulas and simulation is all the more surprising given that the series is divergent. We are not truly in a rapid fluctuation world as $1/\lambda$ is an intermediate rather than a smallest time scale. Nonetheless we speculate once again:

Second Heuristic

If $1/\kappa < 1/\lambda \ll T$ then the rapid fluctuation asymptotics is not necessarily justified and the asymptotic series may not converge quickly or at all. However truncation of the series after one or two terms is likely to give good results.

Example with slower mixing and stochastic volatility

In the previous two examples volatility was fixed and only the level of mean reversion $\theta(y)$ was state dependent. We test here a stochastic volatility example, again testing the limits of the rapid fluctuation approximation. We aim for a little more realism in the numbers and demonstrate some of the flexibility of the model. A sample path and histogram of the bimodal unconditional distribution for this process is included in the introductory chapter.

The parameters are $\kappa = 2, \theta_1 = 15\%, \theta_2 = 5\%, x_0 = 0.08, \sigma_1 = \sqrt{0.0035}, \sigma_2 = \sqrt{0.0005}$. We $\lambda = 0.5$ so that regime switches occur only once every two years. The maturity T was 25 years. In summary, the time scales in increasing order are

$1/\kappa$	$1/\lambda$	T
0.5	2	25

The time scale of regime switches is again “intermediate” (though considerable shorter than the maturity) and again the rapid fluctuation expansion fails to converge whether we use the exact solution or the approximate method. The first seven terms in the exact asymmetric and symmetric correction are listed below

	Asymmetric	Symmetric
1	.5018750000e-1	.6107740480e-1
2	.6130644504e-2	.1211664219e-2
3	-.3840253241e-3	-.1081383350e-1
4	-.1147205644e-2	-.4124629859e-1
5	-.4131133651e-2	-.1631200618
6	-.1636099564e-1	-.6400725594
7	-.6420580301e-1	-2.453931996

However as with the previous example, truncation of the series gives very good results:

Method	Asymmetric	Symmetric
Dynamic mean Vasicek	0.0793	-0.0029
Rapid fluctuation (1 term)	0.05019	0.06107
Rapid fluctuation (2 terms)	0.0563	0.0623
Simulation (21,000)	0.0524	0.0597
Sampling error	0.002	0.002

These results are very encouraging. Regime switches occur only once every two years in this case and yet the rapid fluctuation approximation, appropriately truncated, does very well. Moreover only one or at most two correction terms are needed. We consider this further support for the second heuristic. Perhaps “one and a half” terms in the expansion should be used as a general rule of thumb.

Example where the rapid fluctuation approximation converges but truncation gives better results

In this example we return to the “rapid fluctuation” ordering $1/\lambda < 1/\kappa$. The regime switches are quite large and both the volatility and level of mean reversion are stochastic. We stress test the formulas along another dimension by reducing the maturity to one year. In short we put $\kappa = 2, \theta_1 = 45\%, \theta_2 = 5\%, \sigma_1 = \sqrt{0.0035}, \sigma_2 = \sqrt{0.0005}, \lambda = 3, T = 1, x_0 = 0.08$. In increasing order we have:

$1/\lambda$	$1/\kappa$	T
0.33	0.5	1

There is very little difference between the time scales so it is not clear a priori how the formulas will perform.

The results are tabulated below. In keeping with the first heuristic the expansion converges. However we notice something interesting. As can be seen from the correction terms in bold it appears wiser to truncate the asymptotic solution *even though the series converges*. The correct answer, given by simulation, is somewhere between

the first and the first plus second correction terms.

Method	Asymmetric	Symmetric
Dynamic mean Vasicek	0.0236	0.0002
rapid fluctuation (1 term only)	0.0288	0.00254
rapid fluctuation (2 terms)	0.0260	0.00172
rapid fluctuation (4 terms)	0.0225	0.00787
rapid fluctuation (6 terms)	0.0210	0.00376
Simulation (120,000)	0.0266	0.0021
Sampling error	0.0001	0.0001

One might suggest:

Third Heuristic

To play it safe, use “one and a half” symmetric and antisymmetric corrections.

Conclusions

The analytic formulas derived using the rapid switching asymptotics perform rather well but further work is required to sharpen the preceding three heuristics. We have not investigated the possibility of extrapolation or using the series in combination with other analytic approximations. Work in progress on the slow mixing asymptotics may prove enlightening in this regard.

3.5 Analysis of US Corporate Spread Data

Beginning with a data set consisting of thousands of US corporate spread time series we show that spreads exhibit bimodality and predictability inconsistent with the hypothesis that they follow simple diffusive processes such as Ornstein-Uhlenbeck. On the other hand our simulation results demonstrate that both qualitative features can be accommodated by regime switching models. Our analysis focuses on global

features of spread dynamics, in contrast to He [39], Duffie and Singleton [24], Duffie and Lando[22] and other treatments of the term structure of credit spreads.

3.5.1 Corporate Spread Data

We are grateful to Robert Fuhrman⁶ for providing a large data set of US Corporate spreads. The initial data set consisted of several thousand time series of varying length. To this collection we applied a very stringent filter intended to avoid problems associated with stale quotes. Each time series consisted of up to sixty data points representing five years of monthly data. If we found two consecutive months with the same spread the entire series was discarded. Due to price quantization this reduced significantly the number of spreads we had to work with and in particular culled a disproportionate number of high grade issues. This was unfortunate but did have the positive side effect of biasing our population toward the more liquid securities.

In all, 307 spread time series passed the test. A random selection of names from the list is found in table 3.3. We are not privy to the details of the data collection, cleaning process or precise algorithms by which the spread is calculated from default swaps whose maturities varied.

3.5.2 Eyeballing the Spread Time Series

Given the large number of times series in the data set we shall not attempt to persuade the reader that the examples in figures 3.4 and 3.5 were randomly chosen. Our tests however are intended to demonstrate that they are not atypical.

Though stop short of arguing, however tempting, that in these and other examples regime switches are visible to the naked eye. However, the persistence of small spreads punctuated by periods of substantially greater credit risk is an appealing story to keep in mind while looking at figure 3.5.

As suggested to the author by Zhifeng Zhang⁷, this pattern ought to show up in the spread histograms such as 3.4. To quantify these suspicions and demonstrate

⁶Morgan Stanley Dean Witter Fixed Income Index Group

⁷Morgan Stanley Dean Witter Derivative Products Group

that bimodality in spread histograms cannot be attributed to chance we apply kernel smoothing tests developed by Silverman[56].

3.5.3 Tests for Bimodality in Spreads

Critical Smoothing as a Test for Bimodality

We use a kernel test for modality as described in Silverman [56]. For each time series $x_i = x_{t_i}$ and smoothing parameter $h > 0$ we calculate a kernel estimate $\hat{f}$ of the unconditional density. In other words, we smooth the empirical distribution using a kernel with width h :

$$\hat{f}(y; h) = \frac{1}{nh} \sum_{i=1}^n \phi\left(\frac{1}{h}(y - X_i)\right)$$

where ϕ is the standard normal density. Modes (maxima) of the smoothed density $\hat{f}$ are then counted (in practice we achieve this by discretizing the range of $\hat{f}$ into approximately fifty points). A theorem in Silverman [56] guarantees that the number of maxima in $\hat{f}$ is monotonic in the kernel parameter h . Therefore it is easy to search for the **critical value** $h_{critical}$ where the smoothed density $\hat{f}(x; h)$ goes from being unimodal to bimodal. One can also search for critical values corresponding to higher modality in the data. We define the n 'th critical value to be

$$h_{critical}^n = \inf \left\{ h : \hat{f}(\cdot; h) \text{ has } n \text{ maxima} \right\}$$

and use it as a relative measure of unimodality. If a data set is roughly unimodal then only a small width h is required to smooth $\hat{f}$ until it indeed has precisely one local maxima. We expect this to be true for data x_i from long time series simulated using a constant parameter Vasicek model since the unconditional distribution is gaussian. By contrast a data set which is strongly bimodal requires a larger width kernel in order that the smoothed density be unimodal.

Bump Hunting in Spread Data

For the selected 307 spread time series we obtained the critical parameters h_{critical} , the first few reported in table 3.3 along with the means μ and standard deviations σ of the spread time series. There is, not surprisingly, a very strong linear relationship between the critical smoothing width and the standard deviation of the time series. Therefore we report the normalized critical smoothing parameters $\frac{h_{\text{critical}}}{\sigma}$. A histogram of these relative critical values is shown in figure 3.7. After taking the variance of the time series into account in this manner we found no further relationship between the mean of the spread time series and the critical width. Given the scatter plot 3.8 we are not able to say whether high grade asset spreads are “more bimodal” than low grade spreads or vice versa.

The relative smoothing numbers cannot be interpreted in isolation as our time series, consisting of only sixty data points, are relatively short. To put the relative smoothing results in some perspective, we generated time series, also of sixty points, using independent gaussian increments. Smoothed densities were then obtained using the median normalized critical value $h = 0.53$.

We hypothesize that spreads are generated by this simple process and the bimodality arises from sampling over such short lengths of time. Under this hypothesis we expect roughly fifty percent of the smoothed densities to exhibit more than one hump, but in fact, the actual percentage is closer to one in four. Furthermore, in order to obtain at least two humps in fifty percent of the population we needed a significantly smaller smoothing window, closer to forty percent of the standard deviation.

We reject the “Brownian motion” hypothesis and are tempted to cast doubt on the ability of *any* simple diffusion model to replicate the degree of bimodality present in the data.

Bootstrapped P -Values as a Test for Higher Modality

Next we attempted to determine whether more than two modes were present in the data. All smoothed density estimates for various values of h can be considered candidates for the “true” distribution and used for the purpose of bootstrapping P values

defined as the fraction of bootstrapped data sets which (when smoothed by the critical parameter h_{critical}), display three or more maxima. The critically smoothed density $\hat{f}(\cdot, h_{\text{critical}})$ itself is, as argued by Silverman, a conservative estimate of the true distribution for use in this context. At the first critical value (hereafter *the* critical value) we sit on the cusp between a unimodal and bimodal distribution and the critically smoothed density is the “most bimodal” distribution amongst those smoothed density estimates with only one maxima. An example of a critically smoothed density corresponding to the time series in 3.4 is shown in figure 3.6 along with two examples of bimodal estimates corresponding to smaller h values.

We calculate P values by bootstrapping as described above using the critical distribution. As the smoothing parameter is sensitive to variance we adjust the sampling somewhat using a suggestion of Efron [25]. Efron points out, as mentioned in [56], that sampling from this critically smoothed density with jack-knife variance adjustment is equivalent to sampling the random variable

$$Z_j = \left(1 + \frac{h_{\text{crit}}^2}{\sigma^2}\right)^{-1/2} (X_{I(j)} + \zeta_j h_{\text{crit}})$$

where σ is the sample variance of the data $\{X_i\}$, ζ_j are independent standard normal random variables, and $I(j)$ selects an index uniformly from those in the data set. The P values (for a subset of the 307 spread time series) are reported in table 3.3. In general, large P -values indicate the possibility of higher modality in the data and might suggest the possibility of three or more regimes. However, as can be seen from the histogram of P values in figure 3.7, the occurrences of large P -values were rare. Therefore we do not claim evidence of multi-modality in spread data.

Bump Hunting in Simulated Time Series

For each of the 307 time series we estimated, subject to prior restrictions on parameters, a two state regime switching model. A variety of methods were explored for fitting the model but not surprisingly the modality results were fairly robust to the exact method employed. A single realization of the model was simulated and the

Issuer	Mean Spread	Std Dev σ	h_{critical}	$h_{\text{critical}}/\sigma$	P value
PEERLESS TUBE CO	0.0551	0.0140	0.0066	0.4723	0.00
SYNERGY BRANDS	0.0317	0.0298	0.0177	0.5917	0.00
PHC INC/MA -CL A	0.0399	0.0297	0.0216	0.7292	0.00
REUTER MANUFACTURING	0.0877	0.0350	0.0180	0.5150	0.25
ANDERSEN GROUP INC	0.0162	0.0122	0.0069	0.5696	0.05
PHYAMERICA PHYSICIAN	0.0958	0.0586	0.0440	0.7510	0.05
WORKSAFE INDUSTRIES	0.0513	0.0127	0.0051	0.4048	0.10
ACTV INC	0.0297	0.0101	0.0061	0.6046	0.00
ARGOSY GAMING CORP	0.0298	0.0250	0.0108	0.4343	0.40
TREEV INC	0.0436	0.0138	0.0078	0.5688	0.00
CHEMTRAK INC	0.0543	0.0521	0.0333	0.6387	0.00
ELECTROSOURCE INC	0.0621	0.0175	0.0097	0.5516	0.05
INTELEFILM CORP	0.0351	0.0226	0.0167	0.7409	0.00
BETHLEHEM CORP	0.0464	0.0062	0.0022	0.3532	0.45
NATIONAL DATACOMPUTER	0.0309	0.0135	0.0069	0.5139	0.15
RECORDLAB CORP	0.0313	0.0176	0.0104	0.5895	0.05
HEMACARE CORP	0.0194	0.0136	0.0082	0.6019	0.00
ALPHA MICROSYSTEMS	0.0192	0.0148	0.0063	0.4299	0.45
TECHNICLONE CORP	0.0415	0.0261	0.0145	0.5540	0.10
NANTUCKET INDUSTRIES	0.0665	0.0465	0.0269	0.5784	0.05
INTELLICORP INC	0.0263	0.0131	0.0079	0.6053	0.00
NATIONAL DATACOMPUTER	0.0309	0.0135	0.0069	0.5139	0.05
INFONOW CORP	0.0260	0.0279	0.0268	0.9616	0.00
STAR SCIENTIFIC INC	0.0468	0.0401	0.0192	0.4786	0.40
METAL RECOVERY TECHN	0.0727	0.0385	0.0158	0.4102	0.35
NAPRO BIOTHERAPEUTICS	0.0291	0.0402	0.0328	0.8140	0.00
WASTE RECOVERY	0.0671	0.0435	0.0339	0.7787	0.00

Table 3.3: A sample of the 307 corporate bond issues whose spread time series were tested for bimodality using a critical smoothing test. For each name we examined the five years of monthly data ending around July 2000. The first column is the mean of the time series, the second the standard deviation. Column three contains the critical parameter h_{critical} describing the width of the gaussian kernel which was required to smooth the empirical distribution of spread values to the point where it was unimodal. The fourth column reports the critical value as a fraction of the standard deviation, and the final column is the bootstrapped P statistic explained in section 3.5.3. Large P values suggest more than two modes in the data.

Model	θ_{high}	θ_{low}	κ	λ	σ_{high}	σ_{low}
Standard Vasicek	fit	$= \theta_{high}$	fit	1	fit	$= \sigma_{high}$
Regime Switching Vasicek I	$2\bar{x}$	0	1	1	fit	$= \sigma_{high}$
Regime Switching Vasicek II	$1.5\bar{x}$	$0.5\bar{x}$	1	0.5	fit	$= \sigma_{high}$
Regime Switching Vasicek III	$2\bar{x}$	0	6	0.5	fit	$= \sigma_{high}$
Regime Switching Vasicek IV	$2\bar{x}$	0	1	1	fit	$= \frac{1}{\sqrt{3}}\sigma_{high}$

Table 3.4: Description of models used to study multimodality of simulated time series. Models were fitted to each of the 307 time series individually and a single realization was simulated. A smoothed estimate of the unconditional distribution was obtained by smoothing with a window width equal to the critical value obtained for the spread time series itself. For the purpose of fitting the models rough and ready estimates of the time series parameters were used subject to parameter restrictions tabulated here. The variable $\bar{x}$ in the table indicates the sample mean of the spread series. In Regime Switching Vasicek model IV the two values taken by σ were chosen so that the mean squared value $\frac{1}{2}\sigma_{high}^2 + \frac{1}{2}\sigma_{low}^2$ was commensurate with value obtained from the variance of spread changes. Likelihood and moment methods were used to estimate κ with no qualitative differences in the results. Moment methods were preferred as they were significantly faster and more robust.

smoothed density was calculated using the empirical critical value previously estimated for the spread in question. The number of humps in the smoothed density was recorded, and the results summarized in table 3.5.3

Head to Head Model Comparison

Since the methods used to fit the regime switching models to spread data are somewhat ad hoc, we performed another numerical experiment to determine whether variance effects were interfering with the results. We began by simulating a long time series (50,000 years of monthly data) using the regime switching model with parameters given in table 3.5.3. Next we used the long time series to estimate a standard Vasicek model, with parameters reported also in table 3.5.3. More precisely, for an estimate of θ in the Vasicek model we use the mean of the time series generated by the regime switching model. We estimate the parameter σ for the Vasicek model using the variance of monthly changes in the time series generated by the regime switching model. Next we estimate κ by the variance of the unconditional distribution of

Model	Samples	One Hump	Two Humps	Three +
Spread Data	307	153(49.8%)	154(50.2%)	0
Standard Vasicek	307	232(75.6%)	56(18.2%)	19(6.2%)
Regime Switching Vasicek I	307	198 (64.5%)	80 (26.1%)	29 (9.4%)
Regime Switching Vasicek II	307	246 (80.1%)	46(15.0%)	15 (4.9%)
Regime Switching Vasicek III	307	86(28.0%)	162(52.8%)	59(19.2%)
Regime Switching Vasicek IV	307	206(67.1%)	79(25.7%)	22(7.17%)

Table 3.5: Number of humps in densities estimated from simulated time series by smoothing with an empirically estimated critical window. Models are described in table 3.5.3. We include the “results” for the data set itself. By definition of the critical window width this same test (applied with infinitesimal symmetric noise added to the critical value rather than the critical value itself) results in close to a fifty fifty split on the spread data set - half the time returning one hump and the other half returning two humps. Comparing this with bump counts for the densities of the simulations, smoothed by the same amount, we are able to quantify a relative degree of multimodality amongst models and determine whether any can match the data. We find that the fourth regime switching model replicates multimodality to the same or greater degree that we see in the data. Other choices of parameters give rise to more unimodal processes.

the same, and finally, after removing the predictable component of monthly changes implied by these estimates, we recalculate σ and κ .

This estimation procedure is close to a maximum likelihood estimate and ensures that the two models have the same mean and unconditional variance. The variance of monthly innovations is greater for the standard Vasicek model because the mean reversion adjustment fails to capture fully the predictability in the regime switching model.

The estimated Vasicek model was used to simulate a long time series (also 50,000 years of monthly data). We randomly selected subsequences of length sixty from each of the simulated time series. The sample standard deviation $\sigma_{\text{subsample}}$, critical smoothing parameter h_{critical} and normalized critical smoothing parameter $\frac{h_{\text{critical}}}{\sigma_{\text{subsample}}}$ were recorded. In the interests of time and because we were not overly interested in higher modality the bootstrap procedure for calculating P -values was omitted.

Figure 3.9 is a scatter plot of critical smoothing values and normalized critical smoothing parameters. The mean and sample variances of these quantities are reported in table 3.7.

Although the unconditional variances for the two time series are similar, we see from table 3.7 that the variance for a randomly drawn five year subsequence is larger for the regime switching process than the standard Ornstein-Uhlenbeck process. However it is unlikely that this variance effect alone is dominant because, as can be seen from the normalized critical values in table 3.7 and scatter plot 3.9, the regime switching model critical values deviate by far more than variance alone would suggest.

We conclude that the apparent superiority of the regime switching model over the constant parameter Vasicek model, in the task of replicating of spread bimodality, cannot be attributed to variance effects or artifacts of the estimation procedure. This conclusion is not masked by the relative shortness of spread time series with which we worked.

Model	θ_{high}	θ_{low}	κ	λ	σ_{high}	σ_{low}
Standard Vasicek	0.0944	0.0944	0.528		0.0555	0.0555
Regime Switching Vasicek	0.040	0.150	6	0.5	0.02	0.04

Table 3.6: Model parameters for a regime switching Vasicek model and standard Vasicek model with similar unconditional moments. A long simulated regime switching time series is used to estimate the standard Vasicek model. For an estimate of θ in the standard Vasicek model we use the mean of the time series. We estimate the parameter σ based on the variance of monthly changes and κ by the variance of the unconditional distribution. Removing the predictable component of monthly changes implied by these parameters and θ we then recalculate σ and κ .

Critical Smoothing Statistic	Regime Switching Vasicek	Standard Standard
Samples	20	20
Mean $h_{critical}$	0.0323(0.0095)	0.0168(0.0097)
Mean $\sigma_{subsample}$	0.0450(0.0094)	0.0394(0.0132)
Mean $h_{critical}/\sigma_{subsample}$	0.6969(0.1283)	0.4098(0.1189)

Table 3.7: Results of the head to head comparison between regime switching and standard Vasicek models whose parameters are reported in table 3.5.3. Subsamples of five year length were randomly selected from very long simulated time series. The subsample standard deviation $\sigma_{subsample}$, critical smoothing parameter $h_{critical}$ for the subsample and normalized critical smoothing parameter $h_{critical}/\sigma_{subsample}$ were recorded. The standard deviation across the twenty records for each statistic is shown in parentheses. We notice that the variance for a randomly drawn five year subsequence is larger for the regime switching process than the standard Ornstein-Uhlenbeck process, but even accounting for this the former displays much greater tendency for bimodality.

3.5.4 Trends and Predictability of Spread Data

From the point of view of simple diffusion models, the spreads we examined display a very alarming degree of predictability. In particular, in a given month spreads are roughly twice as likely to continue to move in the same direction as reverse direction. No simple mean reverting process mimic this behavior and in fact, as can be deduced from elementary considerations, they must exhibit the opposite property. For example, the Ornstein-Uhlenbeck process is more likely to move towards its mean than away from its mean because of the drift term. However it is also more likely to have just come from a position closer to the mean than one further away, so the probability of taking two consecutive steps in the same direction is strictly less than one half. We find that the empirical estimate of spread predictability is $66.1\% \pm 0.35\%$, forty five standard deviations away from this upper bound.

There is no arbitrage argument or iron clad law preventing spread movements from displaying this high degree of predictability, but it is surprising nonetheless. Not having been privy to the collection and cleaning of the data we must express concern that the (66.1%) figure is partially a result of interpolation schemes used in the cleaning process.

Nonetheless it is interesting to examine whether the predictability can be replicated in a regime switching model. As numerical tests tabulated in table 3.5.4 demonstrate, regime switching models fall on both sides of the fifty percent line.

Model	θ_{high}	θ_{low}	κ	λ	σ_{low}	σ_{high}	Sample	Trending
Spread Data							18,113	66.1%
Vasicek (regimes)	0.0758	0	1	2	0.008	0.008	18,000	66.0%
Vasicek (regimes)	0.0758	0	1	0.5	0.02	0.02	18,000	52.5%
Vasicek (regimes)	0.0758	0	1	5	0.02	0.02	18,000	51.0%
Vasicek (regimes)	0.0758	0.0458	1	5	0.02	0.02	180,000	49.15%
Vasicek (standard)	0.0758	0.0758	1		0.02	0.02	180,000	48.6%
Brownian Motion	0	0	0		1	1	180,000	50.03%

Table 3.8: Percentage of occasions on which spreads, real and simulated, on a monthly time scale, “trend” (move in the same direction, either up or down, as they did the previous month). The errors in these estimates (as estimated by standard means and confirmed by simulation) are 0.35% where 18,000 simulations have been used and 0.11% for those rows with 180,000 simulations. The empirical results for spread predictability lie forty five standard deviations away from an upper bound of fifty percent applying to simple mean reverting diffusions. The regime switching model can, for carefully chosen parameters, replicate the same degree of one step ahead predictability we observe in the data. For other choices of parameters the trending property disappears entirely.

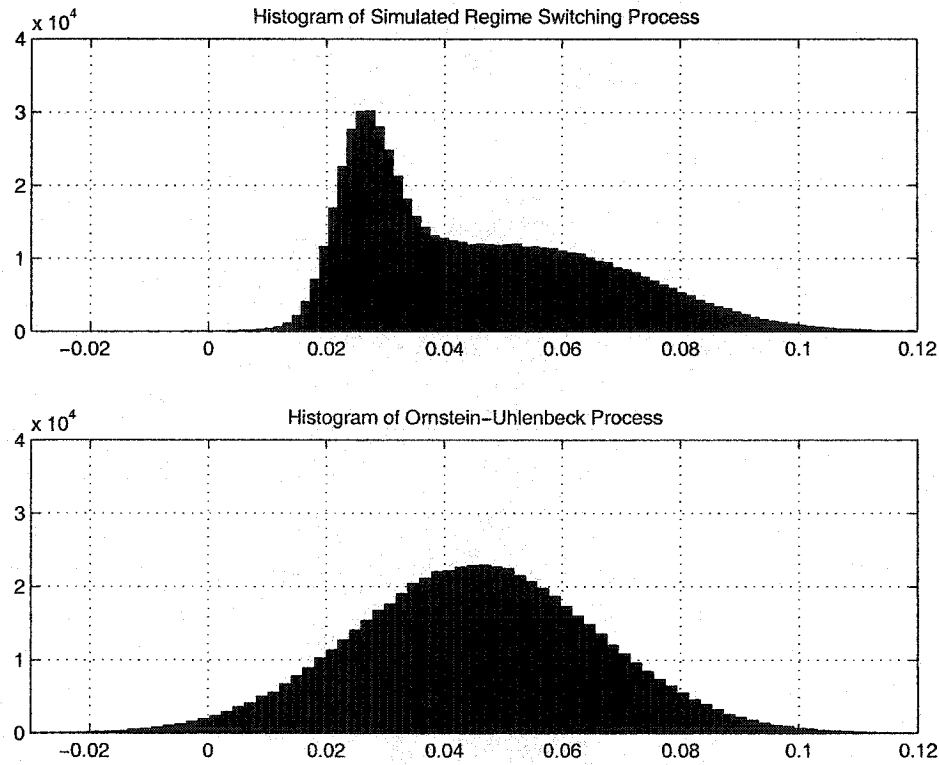


Figure 3.3: Histogram for simulated regime switching process (top) and Ornstein-Uhlenbeck process (bottom). Parameters for the regime switching model: $\theta_{low} = 0.025$, $\theta_{high} = 0.065$, $\kappa = 3$, $\sigma_{low} = 0.01$, $\sigma_{high} = 0.04$, $\lambda = 1$. Parameters for the Ornstein-Uhlenbeck process: $\theta = 0.0451$, $\kappa = 1.321$, $\sigma = 0.0327$. The sample means from a 50,000 month simulation used to construct the histogram are 0.0451 and 0.0454 respectively. The standard deviations are 0.0204 and 0.0207. The regime switching process has a minimal chance of being negative (approximately 1 in 5000) compared with the Ornstein-Uhlenbeck process (approximately 1 in 70).

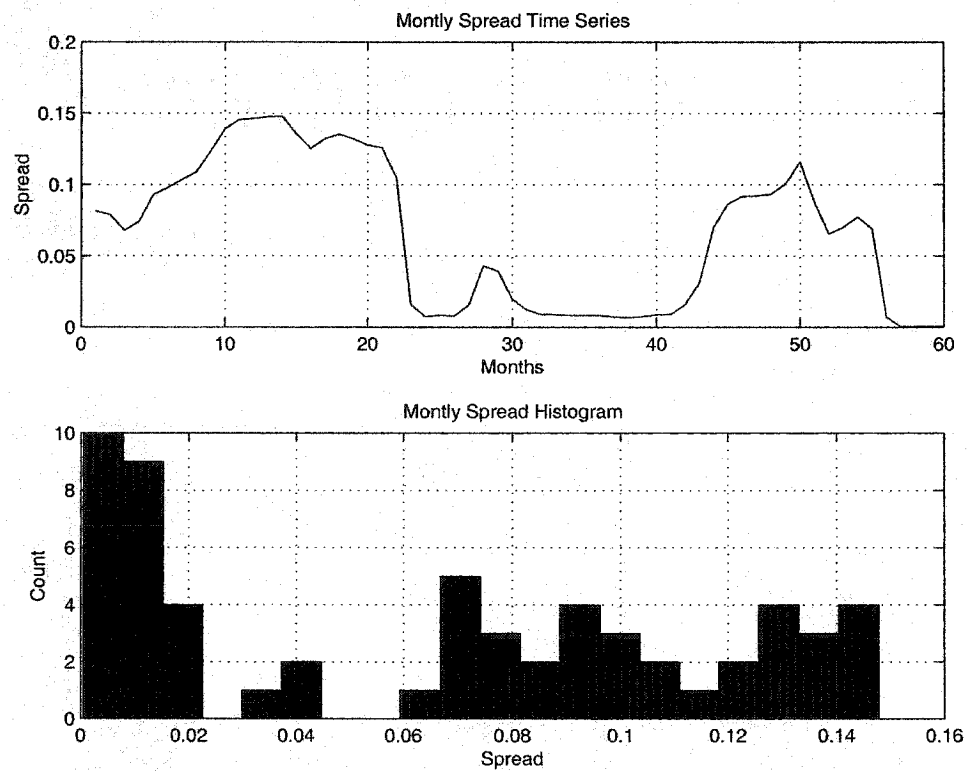


Figure 3.4: Anonymous but not atypical spread time series and histogram. Data is sampled monthly. Regime switches are, it could be argued, visible to the naked eye. The histogram of spread values appears to exhibit bimodality.

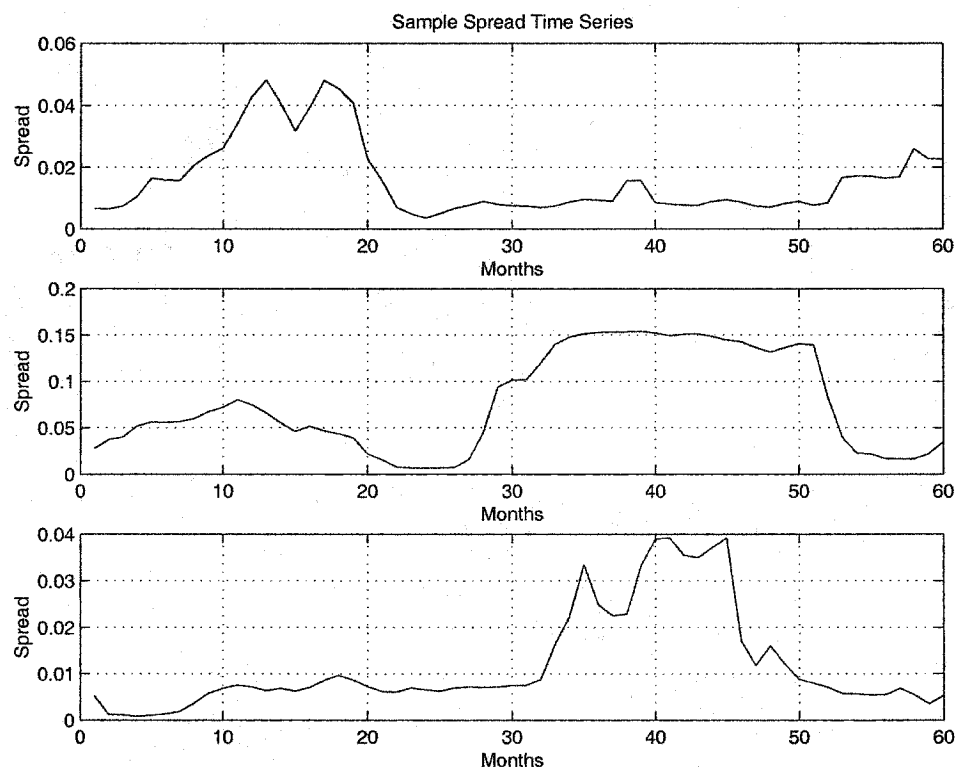


Figure 3.5: Three further examples of spread times series presenting a challenge to simple jump diffusion models. All three assets appear to rather suddenly enter periods of bad credit where the size and (with the exception of the second series) fluctuations of spreads are significantly larger than at other times. At the end of these periods the spread makes a “soft jump” back down to a very low level as uncertainty concerning the firm or agency’s ability to meet obligations evaporates. The analytical difficulty here for jump diffusions arises because the spreads “know” how far to jump, introducing awkward (nonlinear) state dependence on jump sizes and jump intensity. Regime switching models are better suited to this task.

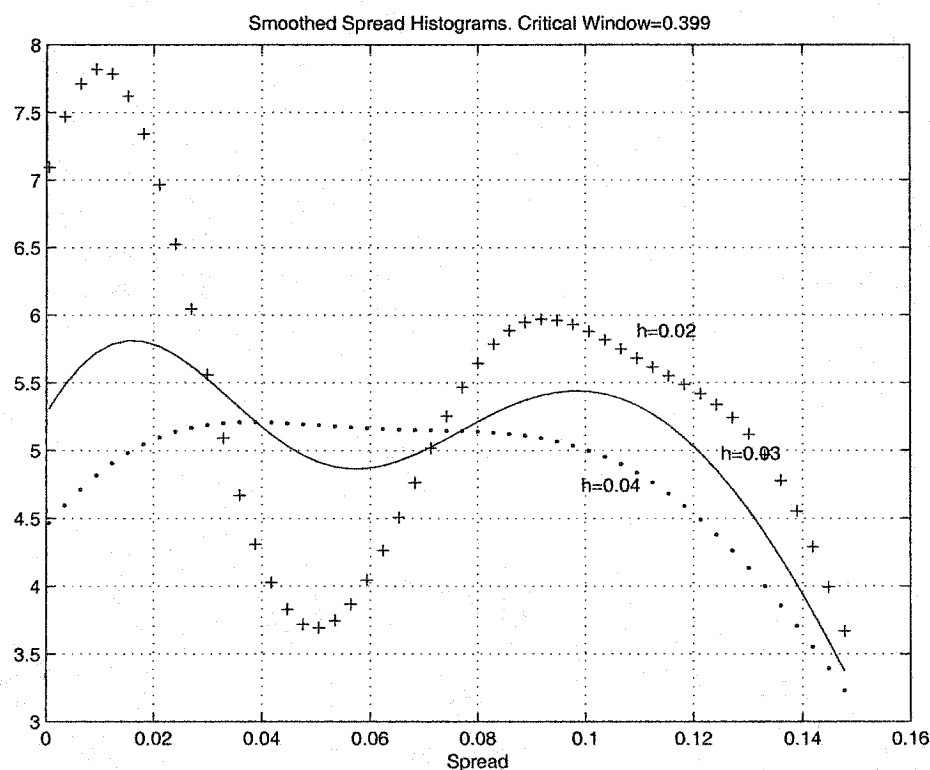


Figure 3.6: Three examples of smoothed histograms corresponding for the spread in figure 3.4. The critical window width is very close to $h = 0.04$. At this value, the smoothed histogram is unimodal. For smaller values at least two humps are evident. The critically smoothed density may be used as a conservative estimate of the true distribution for the purpose of bootstrapping.

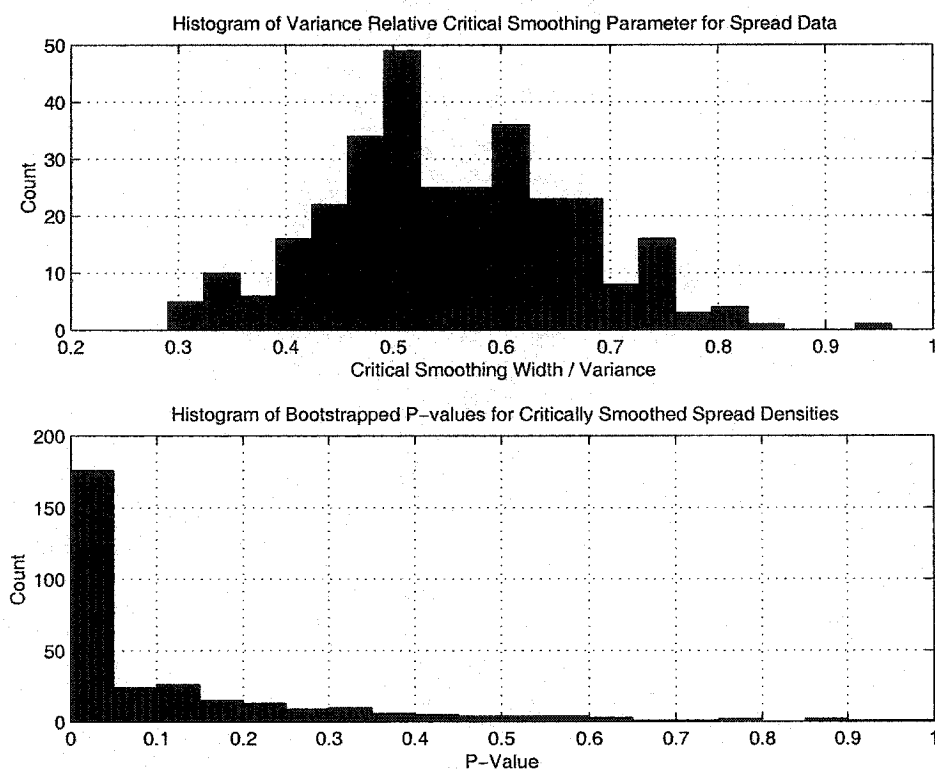


Figure 3.7: Histogram of variance adjusted critical smoothing parameters $h_{\text{critical}}/\sigma$ obtained from 307 time series of spread data each consisting of sixty data points. A kernel width equal to just over half the sample variance of the time series was typically required to smooth the empirical distribution into a unimodal density.

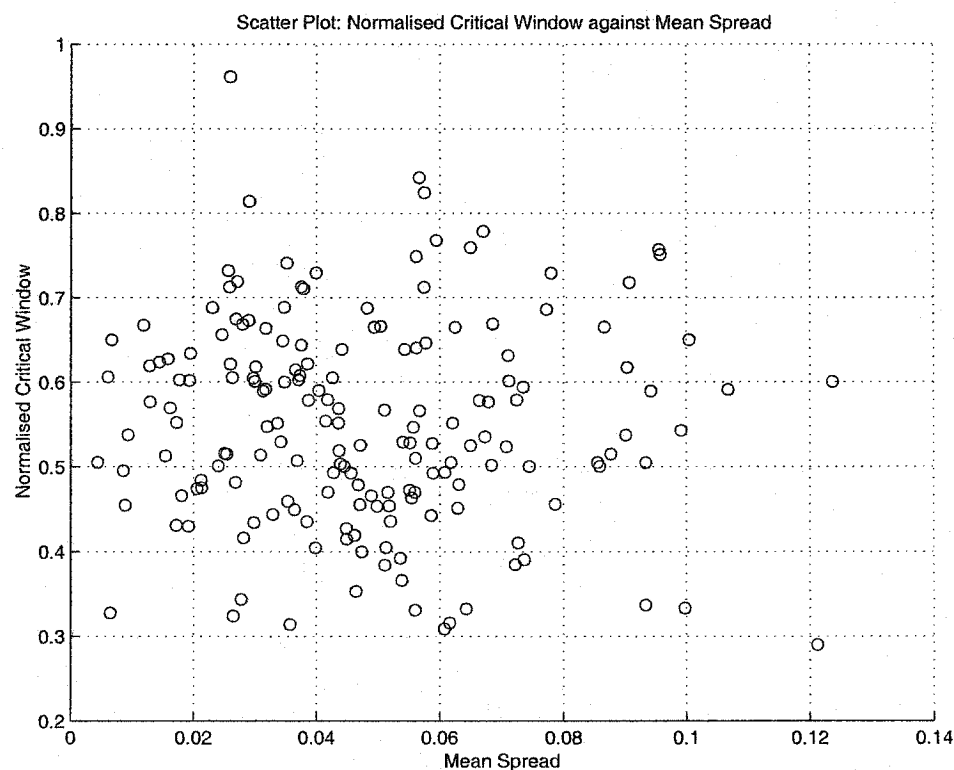


Figure 3.8: Scatter plot showing the normalized critical window width against the mean spread for 307 spread time series. We expect the critical window for a given spread time series to be proportional to the standard deviation of the same. After this normalization we are unable to find a relationship between the mean spread and the critical window. The mean normalized critical window across all 307 spread time series is 0.55. The median is 0.53. These high numbers cannot be reproduced with a simple Vasicek or CIR diffusion.

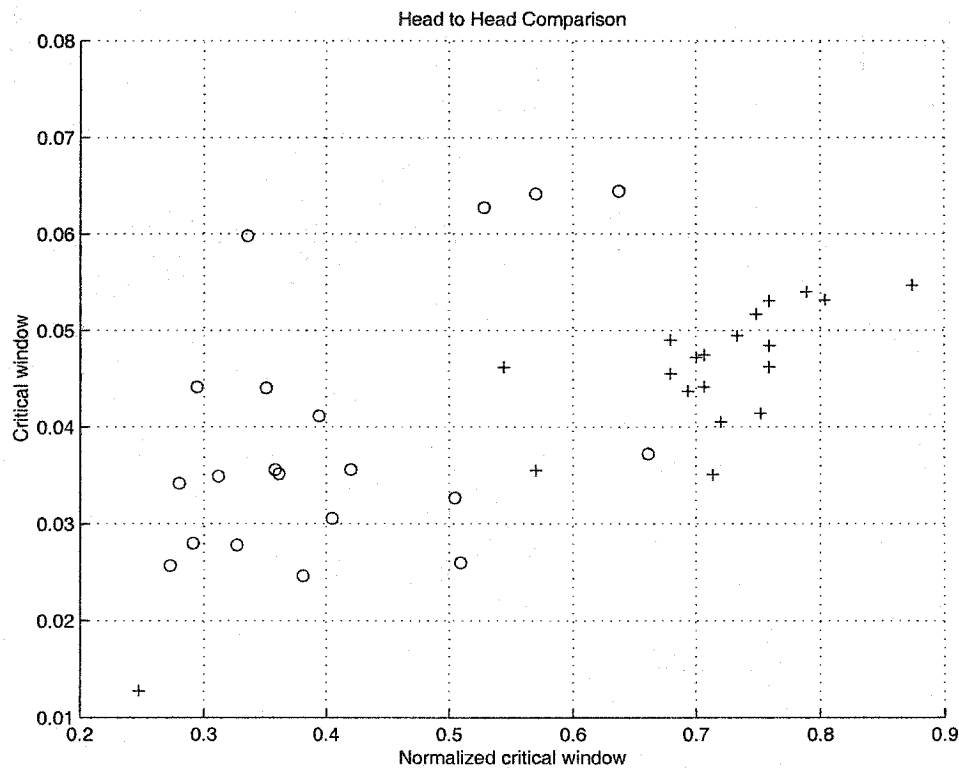


Figure 3.9: Scatter plot of the head to head comparison between regime switching and standard Vasicek models. Each circle represents the critical parameter and normalized critical parameter obtained from a simulated traditional Vasicek model whose parameters are reported in table 3.5.3. Each cross represents the critical parameter and normalized critical smoothing parameter obtained from a simulated regime switching model, with parameters also in table 3.5.3. Statistics are reported in table 3.7. It is clear from only twenty data points that differences between these models in the amount of smoothing necessary to bring the sample unconditional distribution of these models to unimodality can not be attributed to differences in variance alone.

Appendix A

The Vasicek Model

We describe the constant parameter one-factor Vasicek model [57] (**vanilla Vasicek**) into which we incorporate stochastic volatility and regime switches.

In the Vasicek model, the short-rate is modeled as a mean-reverting Gaussian stochastic process $(x_t)_{t \geq 0}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P}^*)$ equipped with an increasing filtration $(\mathcal{F}_t)_{t \geq 0}$. Under the real world probability measure $\mathbb{P}^*$, it satisfies the linear stochastic differential equation (SDE)

$$dx_t = \kappa(x_\infty - x_t)dt + \sigma dW_t^*, \quad (\text{A.1})$$

where $(W_t^*)_{t \geq 0}$ is a standard Brownian motion. Here σ is its constant volatility. Under an equivalent martingale measure $\mathbb{P}^*$ (risk neutral measure) it also follows a linear SDE

$$dx_t = \kappa(\theta - x_t)dt + \sigma dW_t, \quad (\text{A.2})$$

where W_t is a standard $\mathbb{P}$ -Brownian motion, if we assume the market price of interest rate risk, denoted by λ , to be constant. The price of risk is absorbed into $\theta = x_\infty - \frac{\lambda\sigma}{\kappa}$. In other words, in the risk-neutral world $\mathbb{P}$, the short rate x_t is an Ornstein-Uhlenbeck process fluctuating around its mean level θ with a rate of mean-reversion κ . It has the undesirable feature that it can become negative, but it is the simplest model in a larger family of models known as *affine* for which the computation of bond prices

and derivatives is relatively easy.

A discussion of our choice of market price of risk in a slightly more general setting is deferred to appendix B.

A.0.5 Bond Prices

The no-arbitrage price $B(t, x; T)$ at time t of a zero-coupon bond maturing at time T given a short rate $x_t = x$ is given by

$$B(t, x; T) = \mathbb{E} \left[e^{-\int_t^T x_s ds} | \mathcal{F}_t \right], \quad (A.3)$$

Its discounted value is the $\mathbb{P}$ -martingale given by

$$e^{-\int_0^t x_s ds} B(t, T) = \mathbb{E} \left[e^{-\int_0^T x_s ds} | \mathcal{F}_t \right].$$

Using the Markov property, the bond price $B(t, T)$ is a function of the current short rate x_t . According to the Feynman-Kac formula, $\bar{B}(t, x; T)$ is also the solution of the partial differential equation

$$\frac{\partial B}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 B}{\partial x^2} + \kappa(\theta - x) \frac{\partial B}{\partial x} - xB = 0, \quad (A.4)$$

with the terminal condition $B(t, x; T) = 1$. One can find a solution of the form

$$B(t, x; T) = e^{-a(T-t, x) - \tau(T-t; \kappa)x}, \quad (A.5)$$

where

$$\tau(s; \kappa) = \frac{1 - e^{-\kappa s}}{\kappa} \quad (A.6)$$

$$a(s) = \tilde{\theta}(s - \tau(s; \kappa)) + \frac{\sigma^2}{4\kappa} \tau(s; \kappa)^2, \quad (A.7)$$

where we have set

$$\tilde{\theta} = \theta - \frac{\sigma^2}{2\kappa^2} = \bar{r}_\infty - \frac{\lambda\sigma}{\kappa} - \frac{\sigma^2}{2\kappa^2}.$$

This leads to an explicit formula for the zero-coupon bond price $B(t, x; T)$ with

$$B(t, x; T) = \exp \left\{ - \left[\tilde{\theta}(T-t) - (\tilde{\theta} - x) \frac{1 - e^{-\kappa(T-t)}}{\kappa} + \frac{\sigma^2}{4\kappa^3} (1 - e^{-\kappa(T-t)})^2 \right] \right\}, \quad (\text{A.8})$$

which could have been obtained also by computing the expectation in (A.3) using the normal distribution of $\int_t^T x_s ds$. Applying Itô's formula to $B(t, x; T)$ given by (A.8) at $x = x_t$, we deduce

$$dB(t, T) = B(t, T) \left(x_t dt + \sigma \frac{1 - e^{-\kappa(T-t)}}{\kappa} dW_t \right),$$

which shows that the *bond price volatility* is given by $\frac{\sigma}{\kappa}(1 - e^{-\kappa(T-t)})$ independent of θ .

The *yield curve*, defined by

$$R(t, x; T) = -\frac{1}{T-t} \log(B(t, x; T)),$$

as a function of T , is given explicitly by

$$R(t, x; T) = \tilde{\theta} - (\tilde{\theta} - x_t) \frac{1 - e^{-\kappa(T-t)}}{\kappa(T-t)} + \frac{\sigma^2}{4\kappa^3(T-t)} (1 - e^{-\kappa(T-t)})^2,$$

which shows that $R(t, x; T)$ is an affine function of the current rate $x = x_t$ and justifies the name *affine model of term structure*. Observe that for all t , $R(t, x; T)$ converges to $\tilde{\theta} = \theta - \frac{\sigma^2}{2\kappa^2}$ as $T - t \rightarrow +\infty$. Figure A.1 shows a typical bond pricing function (of maturity) and the corresponding yield curve obtained under a Vasicek model. The yield curve (bottom) is a classical increasing curve obtained with this model when the current short rate is lower than the long term rate.

A.0.6 Bond Option Prices

We continue to denote the maturity of the bond by T and the maturity of a European option written on that bond is denoted by T_0 with $t \leq T_0 < T$. The payoff of the

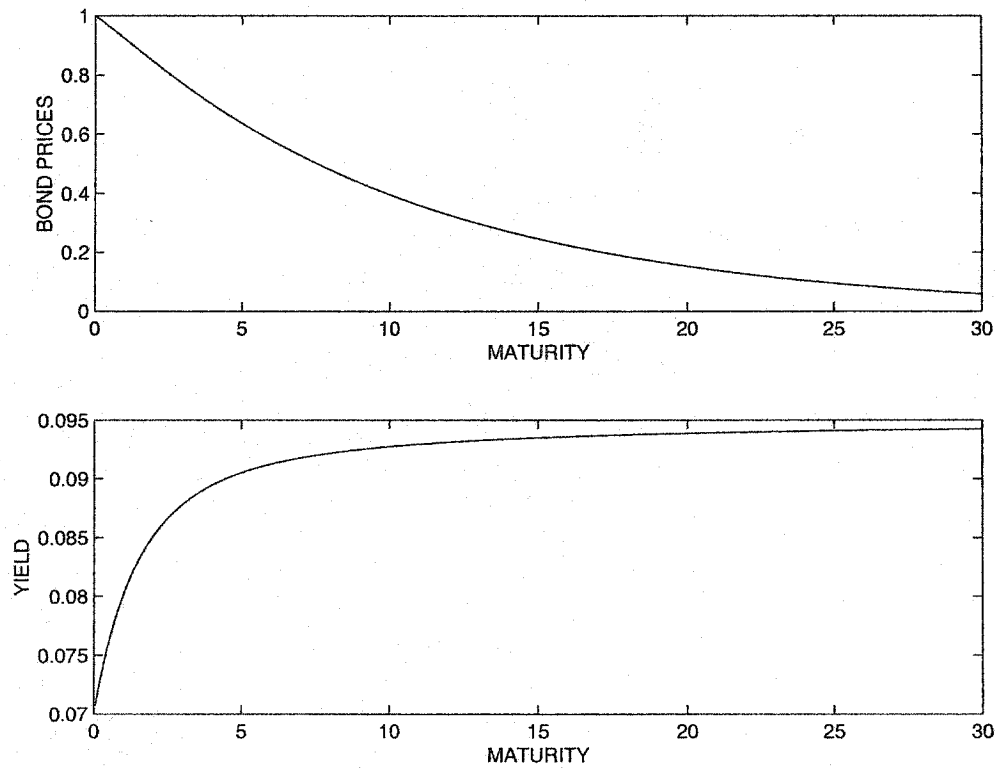


Figure A.1: Bond prices (top) and Yield curve (bottom) in the Vasicek model with $\kappa = 1$, $\theta = 0.1$ and $\sigma = 0.1$. Maturity $T - t$ runs from 0 to 30 years. $\tilde{\theta} = 0.095$ and the initial rate is $x = 0.07$.

option at time T_0 is a function $h(B(T_0, T))$ of the bond price at the expiration time of the option. For instance, a call option is a contract which gives its owner the right but not the obligation to buy the bond at time T_0 at a predetermined strike price K . Its payoff, at maturity T_0 , is then $h(B(T_0, T)) = (B(T_0, T) - K)^+$.

The no-arbitrage price of this option is obtained by computing the expectation of the discounted payoff with respect to the martingale measure $\mathbb{P}$ for the bond. The payoff $h(B(T_0, T))$ is a function $\tilde{h}(x_{T_0})$ of the rate at time T_0 since the bond price $B(T_0, T)$ is a function of the rate x_{T_0} as can be seen from the bond price formula (A.8). Using the Markovian structure, we denote by $\bar{Q}(t, x; T, T_0)$ the price of this

option at time t for an observed short rate $x_t = x$ and we compute it according to

$$\begin{aligned}\bar{Q}(t, x; T, T_0) &= \mathbb{E} \left[e^{-\int_t^{T_0} x_s ds} h(B(T_0, T)) | x_t = x \right] \\ &= \mathbb{E} \left[e^{-\int_t^{T_0} x_s ds} \tilde{h}(x_{T_0}) | x_t = x \right].\end{aligned}\quad (\text{A.9})$$

By the Feynman-Kac formula, it is also the solution of the partial differential equation (A.4) with the terminal condition $\bar{Q}(T_0, x; T, T_0) = \tilde{h}(x) = h(B(T_0, x; T))$ where $B(T_0, x; T)$ is given by (A.8) at $t = T_0$.

Under $\mathbb{P}$ and conditional on $x_t = x$, the pair of random variables $(x_{T_0}, \int_t^{T_0} x_s ds)$ is normally distributed and therefore the price of the option given by (A.9) reduces to the computation of a Gaussian integral.

In the particular case of a call option, denoting the price of the option by $\bar{C}(t, x)$, we obtain

$$\bar{C}(t, x) = B(t, x; T)N(h_1) - KB(t, x; T_0)N(h_2), \quad (\text{A.10})$$

where N is the $\mathcal{N}(0, 1)$ cumulative distribution function. The quantities $h_{1,2}$ are given explicitly by

$$h_{1,2} = \frac{\log \left(\frac{B(t, x; T)}{B(t, x; T_0)} \right) - \log K \pm \frac{1}{2}v^2}{v}, \quad (\text{A.11})$$

$$v^2 = \frac{\sigma^2}{2\kappa^3} (1 - e^{-2\kappa(T_0-t)}) (1 - e^{-\kappa(T-t)})^2, \quad (\text{A.12})$$

with the bond prices $B(t, x; T)$ and $B(t, x; T_0)$ being given by (A.8).

Appendix B

Pricing with Almost Affine Models

Throughout the text we have emphasized the centrality of discount factors:

$$u(x, t; T) = \mathbb{E}^{x,t} \left[e^{-\int_t^T x_s ds} \right]$$

in methodology of pricing various fixed income securities. In this appendix we seek to bolster that claim by describing as best we can the details of pricing bonds and bond options, swaps, caps, floors, forwards and futures within an affine or almost affine framework. By this we refer to a class of models which are either affine or approximately affine in the sense that solutions to B are approximated by exponential affine functions of the factor or factors. Several examples of these approximations are provided in the text.

This chapter naturally reflects the bias of the author and to some extent the members of quantitative research group headed by Joseph Langsam ¹, to whom the author owes a debt of gratitude. In particular we thank Zhifeng Zhang, John Uglum, and Shing Hoi Lee for detailed discussions, notes and assistance. In an attempt to provide a brief and clean introduction to pricing in an affine or almost affine setting we must apologize in advance for omissions and oversimplifications. The reader is referred to [49],[18], [43], [19],[23],[53],[15] for more detailed expositions.

¹formally Morgan Stanley Dean Witter Derivative Product Group

B.1 Multifactor Models

We imagine that a low dimensional stochastic process X_t characterises everything anyone could know about an economy or at least those parts of an economy relevant to a restricted class of financial products which we wish to analyse. Typically diffusions or jump diffusions are used. A very simple example, the Vasicek model itself, assumes that the only thing influencing the price of bonds is the overnight interest rate or short rate x_t following a one dimensional Ornstein-Uhlenbeck process

$$dx_t = \kappa(\theta - x)dt + \sigma dW_t.$$

In this economy if one further makes assumptions on risk preferences as discussed in section B.2 then all uncertainty in bond prices is attributed to uncertainty in the short rate itself. The entire term structure of interest rates shifts up and down from day to day but there is no additional uncertainty in the slope or curvature of the yield curve which cannot be accounted for by changes in the short rate. Put another way, for hedging risk in this simple economy, one bond is as good as ten.

The notion that short maturity and long maturity bonds move together in this manner is counterfactual. To relax the unrealistic assumption one can add more factors (i.e., increase the dimension of X_t), and in this manner multifactor term structure models are born. Each coordinate of X_t is called a factor and under some circumstances may attain economic meaning either before or after estimation. The simplest possible case is that of independent factors (coordinates of X_t literally independent). For example, we can take for a short rate x_t the sum of two factors

$$x(t) = X_1(t) + X_2(t)$$

where $X_1(t)$ and $X_2(t)$ satisfy

$$\begin{aligned} dX_1(t) &= \kappa_1 (\theta_1 - X_1(t)) dt + \sigma_1 dW_1(t) \\ dX_2(t) &= \kappa_2 (\theta_2 - X_2(t)) dt + \sigma_2 dW_2(t) \end{aligned}$$

and $W_1(t)$, $W_2(t)$ are independent Brownian motions. In applications one might wish to interpret these factors (inflation or real interest rates for example). Estimating the model is now more difficult but pricing is not, given the elementary observation

$$\mathbb{E} \left[e^{-\int_t^T X_1(s) + X_2(s) ds} \right] = \mathbb{E} \left[e^{-\int_t^T X_1(s) ds} \right] \mathbb{E} \left[e^{-\int_t^T X_2(s) ds} \right]$$

arising from the independence of $X_1(t)$ and $X_2(t)$. In other words, if the pricing theory can be worked out in the single factor case then multifactor models *with independent factors* can be readily constructed. We contribute to the family of tractible multifactor models by to enlarge the library of one dimensional processes for which quantities such as B can be quickly calculated. An obvious future direction is the application of the techniques presented in this thesis to multifactor models with non trivial dependencies between factors. A rich family of multifactor models is provided by the **affine class** which we now review.

B.1.1 The Affine Class

Since Vasicek there has been, to say the least, significant advances in our ability to model bond prices and related financial products. Cox, Ingersoll and Ross (CIR) [13] established results analogous to Vasicek in an equilibrium model where the endogenous short rate follows a square root process

$$dx_t = \kappa (\theta - x_t) dt + \sigma \sqrt{x_t} dW_t.$$

Their model was extended by Hull and White [42] and Jamshidian[44]. We shall not attempt to catalog the work on other single and multiple factor models, leaving the task to Duffie [18], James and Webber[43] and others. Many fall within a class of tractible models termed “affine” in which bond prices are exponential affine function of one or more state variables. In some of these models, for example Chen and Scott[11]), the short rate is a sum of state variables as in the simple example we gave in the previous section. This need not be the case however. The categorisation of the affine class is attributed to Brown and Schwartz [10] and the pricing and estimation

theory has been developed by many authors, notably Duffie, Kan and Singleton. Affine models are attractive because solving the important PDE's reduces to solution of well understood ordinary differential equations as in [19] and transform techniques can be profitably applied to the problem of pricing options. Soon to be published work by Damir Filipovic² provides a clean characterization of affine models as solutions to partial differential equations.

Intuitively, the affine class models a finite dimensional jump diffusion

$$dX_t = \mu(X_t) dt + \Sigma(X_t) dW_t + \text{Jumps}$$

where the drift $\mu(X_t)$ and squared diffusion coefficient $\Sigma^2(X_t)$ are affine functions of the state X_t . Bond prices are given by

$$B(X_t, t, T) = \exp(-a(X_t, T - t) - b(X_t, T - t) \cdot X_t)$$

for some deterministic vector valued functions $a(Z, s)$ and $b(Z, s)$. The pricing PDE, which we turn to in the next section, can be typically be reduced to ordinary differential equations by postulating a solution of the form B.1.1.

General affine models provide for substantially greater flexibility than the ancestors they nest. Dai and Singleton [14] perform a specification analysis of affine processes used as econometric models for the term structure. They establish canonical representatives of equivalence classes of affine models determined by factor transformations. These "maximal" representatives are compared with each other and with those models they nest using moment criteria. They conclude that when it comes to representing the dynamic behaviour of historical bond prices some families perform better than others. A reasonably complicated interplay between factors is desirable in this context.

²ETH, Zurich <http://www.math.ethz.ch/filipo/>

B.1.2 Almost Affine Models

There exists an impressive and growing number of fixed income tools outside the affine class, including forward rate models (especially Heath, Jarrow and Morton [40]), market models (Brace, Gararek, Musiela [9]), positive interest rate models (Flesaker [27]), and random field models (including Goldstein, [35]). Many of these models have benefited from work on affine models and build upon the structure therein. It makes little sense to speak of direct competition between these approaches and the affine class. Rather it is a case of horses for courses. Whilst some market applications demand a very direct modeling of forward rates free of prior assumptions on the relationship between yields of different maturities implied by the affine structure, others benefit from the parsimony and tractability obtained by using the affine class. Affine models can be used in a self consistent manner and represent a powerful tool for investigating the relationship between implied risk neutral and observed real world price dynamics. In contrast, introducing high dimensional or non Markovian state space is useful for matching a broad range of contemporaneous market prices but standard market practice dictates periodic recalibration - an acknowledgement of internal inconsistency and likely loss of forecasting ability.

In this work we have stayed quite close to the affine family, sympathetic to the philosophy of tractability and parsimony. The models with fluctuating parameters we introduce are approximated by exponential affine formulas taking the form of B.1.1 and this chapter reviews some general pricing benefits of these **almost affine** models. With econometric modeling in mind we turn first to risk premiums and the partial differential equation governing bond prices.

B.2 Market Price of Risk

We present a plausibility argument for the form taken by the partial differential equation for a discount bond

$$\frac{\partial P}{\partial t} + \mathcal{L}_X P - rP + \sum_k \lambda_k \sum_j \sigma_{j,k} \frac{\partial P}{\partial X_j} = 0$$

in an economy where all information is described by one or more diffusions.

There is evidence (for example [17]) that this PDE, or rather the relationship between risk neutral and real world measures which might be implied from it, is too simple. The task boils down to specification of a somewhat arbitrary change of measure and a parameterization of the drift adjustment. The popular choice we follow allows us to express the drift adjustment as a transformation of parameters which leaves the form of the SDE for the short rate unchanged.

Our results are dependent on market price of risk considerations only when we are tempted to treat the Vasicek type models as econometric models in a manner requiring simultaneous use of the real and risk neutral measures. This is however an important application and as can be seen in the stochastic volatility corrections of section 2.1.2, the risk parameters play a crucial role in pricing.

Applications to credit derivatives are relatively insensitive as we model under the risk neutral measure. On the credit side only our empirical results relating to spreads, if taken as a guide for risk neutral models of default, imply a connection to real world probabilities and in this only a very mild one. Our tacit assumption is that the market price of risk is not so pathological that the risk neutral process is regime free and simultaneously the real world dynamics exhibit regime switches³.

For simplicity we have ignored jumps in parameters (such as regime shifts) and the short rate. More securities or risk parameters must be introduced in this case. See for example [37], [23].

We begin with an n -dimensional diffusion representing multiple factors:

$$dX_t = \mu(X, t)dt + \sigma(X, t)dW_t \quad (\text{B.1})$$

where dW_t is an n -dimensional Brownian motion. The dimensions of $\mu(r, t)$ and $\sigma(r, t)$ are $(n \times 1)$ and $(n \times n)$ respectively. We have in mind the specialization $X = (x, y)$ where x is the short rate (or spot rate) and y is a stable Markov process.

Constructing an argument in the style of Hull and White [41] we consider a portfolio of $n + 1$ contingent claims. We can assume that these are bonds with distinct

³One can no doubt devise regime switching market prices of risk canceling real world regime switches, an avenue we have not explored.

maturity dates $\{T_1, \dots, T_{n+1}\}$ although the argument goes through for European options on these bonds and similar securities whose cashflows depend only on the values of X_t at maturity. The process X_t encapsulates all information about the state of the economy and with mild technical assumptions the prices $P^i(X, t) = P(X, t, T_i)$ of these discount bonds must, by Ito's Lemma, satisfy

$$dP_t^i = P_t^i \mu_{P^i}(X_t, t) dt - P_t^i \sigma_{P^i}(X_t, t) dW_t$$

where

$$\mu_{P^i}(X_t, t) = -\frac{1}{P^i(X_t, t)} \left(\frac{\partial P_t^i}{\partial t} + \mathcal{L}_X P_t^i \right)$$

and

$$\sigma_{P^i}(X_t, t)_k = \frac{1}{P^i(X_t, t)} \sum_j \frac{\partial P^i}{\partial X_j}(X_t, t) \sigma_{j,k}(X_t, t).$$

and the operator $\mathcal{L}_X$ is defined by

$$\mathcal{L}_X = \sum_j \mu_j \frac{\partial P^i}{\partial X_j} + \frac{1}{2} \sum_{j,k} (\sigma \sigma')_{jk} \frac{\partial^2 P^i}{\partial X_j \partial X_k}.$$

We assume σ is rank n and that the matrix formed by the vectors $\sigma_{P^i}(X_t, t)$ is also rank n for almost every t . We can almost surely form a riskless self financing portfolio $\gamma \cdot P = \sum_i \gamma_i P^i$ since there are $n+1$ bonds and n risk factors. The requirement is

$$\sum_i \gamma_i P_i \sigma_{P^i} = 0. \quad (\text{B.2})$$

In our economy risk free borrowing at an interest rate of r is available and if there is to be no arbitrage this determines the drift of our riskless portfolio $\gamma \cdot P$. Namely,

$$\sum_i \gamma_i P^i \mu_{P^i}(X_t, t) = r \sum_i \gamma_i P^i(X_t, t). \quad (\text{B.3})$$

If the drift were less than the risk free rate one could sell short the portfolio and buy

the bonds. The above considerations can be summarized by

$$\begin{bmatrix} \sum_j \sigma_{j,1} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,1} \frac{\partial P^{n+1}}{\partial X_j} \\ \dots & \dots & \dots & \dots \\ \sum_j \sigma_{j,n} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,n} \frac{\partial P^{n+1}}{\partial X_j} \\ \frac{\partial P^1}{\partial t} + \mathcal{L}_X P^1 - r P^1 & \dots & \dots & \frac{\partial P^{n+1}}{\partial t} + \mathcal{L}_X P^{n+1} - r P^{n+1} \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \dots \\ \dots \\ \gamma_{n+1} \end{bmatrix} = \begin{bmatrix} 0 \\ \dots \\ \dots \\ 0 \end{bmatrix}.$$

In particular the $(n+1)$ 'th row of the matrix

$$\begin{bmatrix} \sum_j \sigma_{j,1} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,1} \frac{\partial P^{n+1}}{\partial X_j} \\ \dots & \dots & \dots & \dots \\ \sum_j \sigma_{j,n} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,n} \frac{\partial P^{n+1}}{\partial X_j} \\ \frac{\partial P^1}{\partial t} + \mathcal{L}_X P^1 - r P^1 & \dots & \dots & \frac{\partial P^{n+1}}{\partial t} + \mathcal{L}_X P^{n+1} - r P^{n+1} \end{bmatrix}$$

is a linear combination of the first n rows. However the i 'th column is a function of the i 'th maturity date T_i and depends on no other maturity dates. Therefore the vector of coefficients in the linear combination of the first n rows comprising the $(n+1)$ 'th row must be independent of the maturity dates. We may write

$$\begin{bmatrix} \sum_j \sigma_{j,1} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,1} \frac{\partial P^{n+1}}{\partial X_j} \\ \dots & \dots & \dots & \dots \\ \sum_j \sigma_{j,n} \frac{\partial P^1}{\partial X_j} & \dots & \dots & \sum_j \sigma_{j,n} \frac{\partial P^{n+1}}{\partial X_j} \\ \frac{\partial P^1}{\partial t} + \mathcal{L}_X P^1 - r P^1 & \dots & \dots & \frac{\partial P^{n+1}}{\partial t} + \mathcal{L}_X P^{n+1} - r P^{n+1} \end{bmatrix}' \begin{bmatrix} \lambda_1 \\ \dots \\ \lambda_n \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ \dots \\ \dots \\ 0 \end{bmatrix}. \quad (\text{B.4})$$

The function $\lambda = \{\lambda_k = \lambda_k(X_t, t)\}$ will be called the **market price of risk**. This terminology is motivated by the observation

$$\mu_{P^i} - r = \sum_{k=1}^n \lambda_k \sigma_{P^i, k}$$

which expresses the excess return on the bond as a linear combination of the n risks to which it is subject. In terms of this market price of risk this relation or equation B.4 yields a PDE for the bond price process

$$\frac{\partial P}{\partial t} + \mathcal{L}_X P - rP + \sum_k \lambda_k \sum_j \sigma_{j,k} \frac{\partial P}{\partial X_j} = 0 \quad (\text{B.5})$$

From the definition of $\mathcal{L}_X$ we therefore have

$$\frac{\partial P}{\partial t} + \frac{\partial P}{\partial X}(\mu + \sigma\lambda) + \frac{1}{2} \text{Tr} \left((\sigma\sigma')' \frac{\partial^2 P}{\partial X^2} \right) - rP = 0 \quad (\text{B.6})$$

where the Hessian $\frac{\partial^2 P}{\partial X^2}$ is dimension $(n \times n)$, $\frac{\partial P}{\partial X}$ is $(1 \times n)$ and primes indicate transposition. The solution to equation B.6 with boundary condition $P(X, T) = g(X)$ has a Feynman-Kac representation

$$P(X, t; T) = \mathbb{E}^{X,t} \left[\exp \left(- \int_t^T \bar{r}(s) ds \right) g(\bar{X}(T)) \right] \quad (\text{B.7})$$

where $\bar{X}$ is the **risk neutral** interest rate process defined by

$$d\bar{X}_t = \bar{\mu}(\bar{X}, t)dt + \sigma(\bar{X}, t)dW_t$$

where

$$\bar{\mu} = \mu + \sigma\lambda.$$

and $\bar{x}$ is the first component of $\bar{X}$. This demonstrates that a simple choice for the market prices of risk λ leaves the stochastic differential equation for X_t essentially unchanged. Hereafter we shall drop the bar notation and assume unless otherwise stated that the process in question is risk neutral.

B.3 Useful Pricing Abstractions

To avoid repetition of tricks we describe a layer of abstraction just above the level of specific, idealized financial products. Of course below the idealised products treated

in the next section lies the devil and all his myriad details.

B.3.1 Fitting Contemporaneous Prices

Some noneconometric applications require us to add sufficient flexibility to our model that we might fit a specified set of financial products *exactly* at a given instant in time. Therefore we may wish to model the short rate:

$$x^\Lambda(t) = fudge(t) + \beta \cdot X(t)$$

where β is fixed and $X(t)$ is an n -dimensional markov process and $fudge(t)$ is a deterministic fudge factor which enables us to match the initial yield curve. It is however notationally clumsy to slide weakly interacting factors such as

$$\Lambda(t, T) := \exp \left(- \int_t^T fudge(u) du \right)$$

through the calculations. It is simpler to add the fudge factor in afterwards. Therefore in what follows we shall take a short rate

$$x(t) = \beta \cdot X(t)$$

and comment on the corresponding “fudgey formulas” as necessary. One may view fudge factors as a necessary evil. They are likely to be highly unstable over time but if one doesn’t insert them one cannot match all prices exactly and may be picked off immediately. The task of modeling viewed this way is to provide a rich but stable model which minimises the need for these deterministic adjustments.

B.3.2 Generating Function

We assume the existence of a closed form or rapidly computable moment generating function (or briefly, **generating function**),

$$g(t, x, T; \alpha, \beta) := \mathbb{E}_t \left[\exp \left(-\alpha \cdot X(T) - \beta \cdot \int_t^T X(u) du \right) \right] \quad (\text{B.8})$$

This quantity satisfies the same partial differential equation as the bond price, only the initial conditions $g(T, T, x) = e^{-\beta x}$ changing. The calculation proceeds as for the bond price. We will assume all expectations are with respect to the risk neutral measure unless otherwise denoted. In this chapter subscripts on the expectation operator denote the filtration time. Superscripts will indicate a forward measure, a concept we return to shortly.

Keeping in mind an affine process $X(t)$ or a process with fluctuating parameters leading to an almost affine generating function we assume an affine solution or approximate solution

$$g(t, T, x; \alpha, \beta; \Omega) = \exp \left(-a(t, T; \alpha, \beta; \Omega) - b(t, T; \alpha, \beta; \Omega)x \right). \quad (\text{B.9})$$

Here Ω represents the parameters in the process X which starts at $X(t) = x$. In a one factor CIR or Vasicek model we might have $\Omega = (\kappa, \theta, \lambda, \sigma)$ respectively representing the rate of mean reversion, the level to which we mean revert, the market price of risk and the diffusion or volatility parameter.

B.3.3 Generalized Bonds

It will be useful to define

$$\begin{aligned} B_m(t, T) &:= \mathbb{E}_t \left[\exp(m \cdot X(T)) \exp \left(- \int_t^T x(u) du \right) \right] \\ &= g(t, T, x; -m, \beta), \end{aligned}$$

a bond with an additional discount factor. We'll often write $B(t, T)$ for $B_0(t, T)$, and $B(t, T, \tau)$ for $B_0(t, \tau)/B_0(t, T)$.

B.3.4 European Options

We're often interested in European calls on an exponential affine underlying. After normalization these pay $e^{d \cdot X(T)} - K$ contingent on $e^{d \cdot X(T)}$ exceeding the strike K and can be priced using

$$C_{m,d}(t, T, x; K, \Omega) := \mathbb{E}_t \left[\exp(m \cdot X(T)) \exp \left(- \int_t^T x(u) du \right) (e^{d \cdot X(T)} - K)^+ \right]$$

We have once again included an exponential affine term for generality. This may often arise from a change of measure or numeraire where the asset we choose for denomination is affine in the state variables. The function $C_{m,d}(t, T, x; d, K, \Omega)$, can be calculated using the auxiliary function

$$G_{p,q}(y) := \mathbb{E}_t \left[\exp \left(- \int_t^T \beta \cdot X(u) du \right) e^{p \cdot X(T)} \mathbf{1}_{q \cdot X(T) \leq y} \right]$$

which can be thought of as the price of a contract which pays $e^{p \cdot X(T)}$ if $q \cdot X(T) \leq y$. A little algebra gives

$$C_{m,d}(t, T, x; K, \Omega) = G_{m+d, -d}(-\ln(K)) - K G_{m, -d}(-\ln(K)) \quad (\text{B.10})$$

The analogous put function may be calculated by parity

$$\begin{aligned} P_{m,d}(t, T, x; K, \Omega) &:= \mathbb{E}_t \left[\exp(m \cdot X(T)) \exp \left(- \int_t^T x(u) du \right) (K - e^{d \cdot X(T)})^+ \right] \\ &= \mathbb{E}_t \left[\exp(m \cdot X(T)) \exp \left(- \int_t^T x(u) du \right) \right. \\ &\quad \left. \left((e^{d \cdot X(T)} - K)^+ - (e^{d \cdot X(T)} - K) \right) \right] \\ &= C_{m,d}(t, T, x; K) + K B_m(t, T) - B_{m+d}(t, T) \end{aligned}$$

or directly from the definition of $G_{p,q}(y)$,

$$P_{m,d}(t, T, x; d, K, \Omega) = KG_{m,d}(\ln(K)) - G_{m+d,d}(\ln(K))$$

B.3.5 Pricing Transform

The point of this approach is that the auxiliary function $G_{p,q}(y)$ is related to the generating function by the inverse Levy transform[23]

$$G_{p,q}(y) = \frac{g(t, T, x; -p, \beta)}{2} - \frac{1}{\pi} \int_0^\infty \text{Im} [g(t, T, x; -p - ivq, \beta) e^{-ivy}] \frac{dv}{v}$$

This reduces most European type options to a one dimensional quadrature. For more details on this promising approach the reader is referred to Duffie, Pan and Singleton [23].

B.3.6 Fuwards

A **fuward** is motivated in part by the occurrence in nature of particular rolling sequences of forward contracts often collectively termed future contracts. We are as yet too pure for this street terminology however. A future contract for us is marked to market continuously and modeled as a martingale under the risk neutral measure. Actual futures are not, although in practice the difference is almost certainly negligible. The interest arises not from a desire to quantify this tiny effect but from other applications including the simulation scheme for pricing regime switching models described in chapter 3 and the pricing of bonds where parameters are piecewise constant functions of maturity.

We assume a sequence of dates $t = T_0 < T_1 < \dots < T_N = T$, deterministic p and deterministic l_i, m_i associated with the intervals $[T_i, T_{i+1}]$. The fuward is defined recursively, making obvious abbreviations including $f_i := f(T_i; \beta, p; \{T_i, l_i, m_i\})$,

$X_i := X(T_i)$, by the recursive formulas:

$$\begin{aligned} f_N &:= e^{p \cdot X(T)} \\ f_i &:= e^{l_i \cdot X_i} \mathbb{E}_i \left[\exp \left(- \int_{T_i}^{T_{i+1}} \beta \cdot X(u) du \right) e^{m_{i+1} \cdot X_{i+1}} f_{i+1} \right], \quad 0 \leq i \leq N-1 \end{aligned}$$

With $l_i = \beta$ this is, up to a deterministic rescaling, a futures price in the usual meaning. The underlying is exponential affine and worth $e^{p \cdot X(T)}$ on the last settlement date. One can readily check that the fuward is exponential affine :

$$\begin{aligned} f_i &= e^{a_i + b_i \cdot X_i}, \quad 1 \leq i \leq N, \text{ where} \\ a_N &= 0, \\ a_{i-1} &= a_i - a(T_{i-1}, T_i; -m_i - b_i, \beta), \quad 1 \leq i \leq N, \\ b_N &= p, \\ b_{i-1} &= l_{i-1} - b(T_{i-1}, T_i; -m_i - b_i, \beta), \quad 1 \leq i \leq N. \end{aligned}$$

Here β and the model parameters entering $a(T_{i-1}, \dots)$ and $b(T_{i-1}, \dots)$ are most generally piecewise with respect to the settlement times.

A particular example of a fuward is the generating function when the model parameters are piecewise constant with respect to a partition $t = T_0 < T_1 < \dots < T_N = T$. If we write $g_i := g(T_i, T, X(T_i); \alpha, \beta)$ then from the definition of the generating function and iterated expectations :

$$\begin{aligned} g_N &= e^{-\alpha \cdot X(T)}, \\ g_i &:= \mathbb{E}_{T_i} \left[\exp \left(- \int_{T_{i-1}}^{T_i} \beta \cdot X(u) du \right) g_{i+1} \right]. \end{aligned}$$

The affine solution may be expressed

$$\begin{aligned} g_i &= e^{-a_i - b_i \cdot X_i}, \\ a_N &= 0 \\ a_{i-1} &= a_i + a(T_{i-1}, T_i; b_i, \beta) \\ b_N &= \alpha, \\ b_{i-1} &= b(T_{i-1}, T_i; b_i, \beta). \end{aligned}$$

which at time $t = T_0$ is

$$g(t, T, x; \alpha, \beta) = \exp \left(-b_0 \cdot x - \sum_{i=0}^{N-1} a(T_i, T_{i+1}; b_i, \beta) \right).$$

This idea may be applied to regime switching processes where the regime driving process is independent of the primary process. One simulates a sequence at which regime switches occur and then applies the recursive formulas above to evaluate the conditional expectation. A bond price estimate obtained by averaging the conditional expectation over sequences of regime switch times converges much faster than the straightforward monte carlo using direct simulation of the short rate. The procedure is referred to as “smart simulation” in Hansen and Poulsen[37].

B.4 Contracts

Using the ideas of the previous section we may price idealized fixed income securities in a very efficient manner. We mention *some* of the contract subtleties which arise when applying the techniques to a few realistic examples, although many of the real world issues are outside the scope of this thesis. Examples of practical issues we are all too happy to avoid include the modeling of treasury specials and figuring out which bond in a basket is the cheapest to deliver.

B.4.1 Notation

The short rate $x(t)$ is modeled once again as

$$x^\Lambda(t) = fudge(t) + \beta \cdot X(t)$$

for some deterministic fudge factor $fudge(t)$. We keep the shorthand for forward discount factors

$$B(t, T, \tau) = B(t, \tau) / B(t, T)$$

and their corresponding yields $R(t, T, \tau)$. In this section we do *not* use τ to denote the function $\frac{1-e^{-\kappa t}}{\kappa}$, as we have done in the body of the thesis.

B.4.2 A Simple Cap Paid in Arrears

A cap is a sum of caplets. For a caplet we specify four dates $t < T \leq \tau < \tau + \Delta\tau$.

t : Today. That is, time at which option is priced

T : Rate quote day

τ : Rate start day

$\tau + \Delta\tau$: Rate end day. Coincides with the payment date.

A caplet pays the receiver, at time $\tau + \Delta\tau$, $\Delta\tau$ times the amount by which the forward rate $R(T, \tau, \tau + \Delta\tau)$ exceeds a preassigned strike K , or nothing if it does not. Typically the rate quote day T is two business days before the rate start day. Since payment will be made in arrears at time $\tau + \Delta\tau$ the value of the contract at time T

is

$$\begin{aligned}
\text{caplet}(T, T, \tau, \tau + \Delta\tau; K) &= B(T, \tau + \Delta\tau) \Delta\tau (R(T, \tau, \tau + \Delta\tau) - K)^+ \\
&= B(T, \tau + \Delta\tau) \Delta\tau \left(\frac{1}{\Delta\tau} \left(\frac{B(T, \tau)}{B(T, \tau + \Delta\tau)} - 1 \right) - K \right)^+ \\
&= B(T, \tau + \Delta\tau) \left(\frac{B(T, \tau)}{B(T, \tau + \Delta\tau)} - (1 + \Delta\tau K) \right)^+ \\
&= (1 + \Delta\tau K) B(T, \tau) \left(\frac{1}{1 + \Delta\tau K} - B(T, \tau, \tau + \Delta\tau) \right)^+
\end{aligned}$$

where the forward bond price is denoted

$$\begin{aligned}
B(T, \tau, \tau + \Delta\tau) &:= B(T, \tau + \Delta\tau) / B(T, \tau) \\
&= \frac{g(T, \tau + \Delta\tau, x; 0, \beta)}{g(T, \tau, x; 0, \beta)} \\
&= e^{-a(T, \tau, \tau + \Delta\tau; 0, \beta) - b(T, \tau, \tau + \Delta\tau; 0, \beta) \cdot X(T)}, \\
a(T, \tau, \tau + \Delta\tau; \alpha, \beta) &:= a(T, \tau + \Delta\tau; \alpha, \beta) - a(T, \tau; \alpha, \beta) \\
b(T, \tau, \tau + \Delta\tau; \alpha, \beta) &:= b(T, \tau + \Delta\tau; \alpha, \beta) - b(T, \tau; \alpha, \beta).
\end{aligned}$$

Therefore if we define

$$\tilde{K} = \frac{1}{1 + \Delta\tau K} e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)}.$$

the price of the caplet at time t is

$$\text{caplet}(t, T, \tau, \tau + \Delta\tau; K) = (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) B(T, \tau) \right] \quad (\text{B.11})$$

$$\begin{aligned} & \left(\frac{1}{1 + \Delta\tau K} - B(T, \tau, \tau + \Delta\tau) \right)^+ \\ &= \frac{1}{\tilde{K}} \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) B(T, \tau) \right. \\ & \quad \left. \left(\tilde{K} - e^{-b(T, \tau, \tau + \Delta\tau) \cdot X(T)} \right)^+ \right] \end{aligned} \quad (\text{B.12})$$

$$\begin{aligned} &= \frac{1}{\tilde{K}} e^{-a(T, \tau; 0, \beta)} \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) e^{-b(T, \tau; 0, \beta) \cdot X(T)} \right. \\ & \quad \left. \left(\tilde{K} - e^{-b(T, \tau, \tau + \Delta\tau) \cdot X(T)} \right)^+ \right] \\ &= \frac{1}{\tilde{K}} e^{-a(T, \tau; 0, \beta)} P_{-b(T, \tau; 0, \beta), -b(T, \tau, \tau + \Delta\tau; 0, \beta)}(t, T, x; \tilde{K}) \quad (\text{B.13}) \end{aligned}$$

showing that the caplet with payment in arrears is a generalized put. When we put the fudge factor back in this becomes

$$\text{caplet}^\Lambda(t, T, \tau, \tau + \Delta\tau; K) = \frac{\Lambda(T, \tau)}{\tilde{K}^\Lambda} e^{-a(T, \tau; 0, \beta)} P_{-b(T, \tau; 0, \beta), -b(T, \tau, \tau + \Delta\tau; 0, \beta)}(t, T, x; \tilde{K}^\Lambda)$$

where

$$\tilde{K}^\Lambda := \frac{\tilde{K}}{\Lambda(\tau, \tau + \Delta\tau)}.$$

is the new strike.

B.4.3 A Floor Paid in Arrears

A floor is a sum of floorlets. We assume a floorlet with the same dates as the caplet :

t : Today. That is, time at which option is priced

T : Rate quote day

τ : Rate start day

$\tau + \Delta\tau$: Rate end day. Coincides with the payment date.

By comparison with the caplet the value at time t of the floorlet is

$$\begin{aligned} \text{floorlet}(t, T, \tau, \tau + \Delta\tau; K) &= \frac{1}{\tilde{K}} \mathbb{E}_t \left[-\exp \left(-\int_t^T r(u) du \right) \right. \\ &\quad \left. B(T, \tau) \left(e^{-b(T, \tau, \tau + \Delta\tau)} - \tilde{K} \right)^+ \right] \\ &= \frac{1}{\tilde{K}} e^{-a(T, \tau; 0, \beta)} C_{-b(T, \tau; 0, \beta), -b(T, \tau, \tau + \Delta\tau; 0, \beta)}(t, T, x; \tilde{K}) \end{aligned}$$

where as before

$$\tilde{K} = \frac{1}{1 + \Delta\tau K} e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)}.$$

is the strike. With the fudge factor :

$$\text{floorlet}^\Lambda(t, T, \tau, \tau + \Delta\tau; K) = \frac{\Lambda(T, \tau)}{\tilde{K}^\Lambda} e^{-a(T, \tau; 0, \beta)} C_{-b(T, \tau; 0, \beta), -b(T, \tau, \tau + \Delta\tau; 0, \beta)}(t, T, x; \tilde{K}^\Lambda)$$

where again

$$\tilde{K}^\Lambda := \frac{\tilde{K}}{\Lambda(T, \tau, \tau + \Delta\tau)}.$$

is the new strike.

B.4.4 A Vanilla Swap Paid in Arrears

[Say what it is] A swaplet paid in arrears whose defining dates are the same as those of the caplet and floorlet above may be priced by parity :

$$\begin{aligned}
 \text{swaplet}(t, T, \tau, \tau + \Delta\tau; K) &= (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) \right. \\
 &\quad \left. B(T, \tau) \left(\frac{1}{1 + \Delta\tau K} - B(T, \tau, \tau + \Delta\tau) \right) \right] \\
 &= (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) \right. \\
 &\quad \left. B(T, \tau) \left(\frac{1}{1 + \Delta\tau K} - B(T, \tau, \tau + \Delta\tau) \right)^+ \right] \\
 &\quad - (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) \right. \\
 &\quad \left. B(T, \tau) \left(B(T, \tau, \tau + \Delta\tau) - \frac{1}{1 + \Delta\tau K} \right)^+ \right] \\
 &= \text{caplet}(t, T, \tau, \tau + \Delta\tau; K) \\
 &\quad - \text{floorlet}(t, T, \tau, \tau + \Delta\tau; K).
 \end{aligned}$$

or directly :

$$\begin{aligned}
 \text{swaplet}(t, T, \tau, \tau + \Delta\tau; K) &= (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) \right. \\
 &\quad \left. B(T, \tau) \left(\frac{1}{1 + \Delta\tau K} - B(T, \tau, \tau + \Delta\tau) \right) \right] \\
 &= \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) B(T, \tau) \right] \\
 &\quad - (1 + \Delta\tau K) \mathbb{E}_t \left[\exp \left(- \int_t^T r(u) du \right) B(T, \tau + \Delta\tau) \right] \\
 &= B(t, \tau) - (1 + \Delta\tau K) B(t, \tau + \Delta\tau).
 \end{aligned}$$

This is modified :

$$\text{swaplet}^\Lambda(t, T, \tau, \tau + \Delta\tau) = \Lambda(t, \tau)B(t, \tau) - (1 + \Delta\tau K)\Lambda(t, \tau)B(t, \tau + \Delta\tau)$$

when the fudge factor is reinstated.

Swap Rates

Consider a vanilla swap paid in arrears with settlement rate set dates $\{T_j\}_{j=i}^{N-1}$, rate start dates $\{\tau_j\}_{j=i}^{N-1}$ and rate end dates $\{\tau_j\}_{j=i+1}^N$. We assume $\tau_i - T_i$ is constant (usually two business days) and that $T_{i+1} - T_i = \tau_{i+1} - \tau_i = \Delta\tau$. The final payment takes place at time τ_N , typically two business days after $T = T_N$. This swap is a sum of swaplets.

$$\begin{aligned} \text{swap}(t, T_j, T) &= \sum_{i=j}^{N-1} \text{swaplet}(t, T_j, \tau_j, \tau_j + \Delta\tau) \\ &= \sum_{i=j}^{N-1} [B(t, \tau_j) - (1 + \Delta\tau K)B(t, \tau_{j+1})] \\ &= B(t, \tau_j) - B(t, \tau_N) - \Delta\tau K \sum_{i=j+1}^N B(t, \tau_i) \end{aligned}$$

The swap rate (i.e., the value of K making the swap worthless) is therefore

$$\kappa(t, T_j, T) = \frac{B(t, \tau_j) - B(t, \tau_N)}{\Delta\tau \sum_{i=j+1}^N B(t, \tau_i)},$$

which unfortunately is not too pretty even with affine bond prices.

B.4.5 Futures

Fix dates $t = T_0 < T_1 < \dots < T_N = T \leq \tau < \tau + \Delta\tau$. The $\{T_i\}$ are the settlement dates with the final mark to market at $T = T_N$. The underlying is a linear function of the forward bond price $B(\tau, \tau + \Delta\tau)$. The final futures price at time T_N is given

by

$$\Delta\tau (1 - F_N/100) + 1 = \frac{B(T, \tau)}{B(T, \tau + \Delta\tau)}.$$

The price at time $t = T_0$ can be estimated or solved exactly.

Approximate Solution

We make the (incorrect) assumption that the futures price is a martingale under the risk neutral measure. That is,

$$F_i = \mathbb{E}_{T^i} \{F_{i+1}\}$$

Rolling back using the (approximate) martingality of $\{F_n\}$ under the risk neutral measure we have

$$\begin{aligned} \Delta\tau (1 - F_t/100) + 1 &= \mathbb{E}_t \left[\frac{B(T, \tau)}{B(T, \tau + \Delta\tau)} \right] \\ &= e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)} \mathbb{E}_t [e^{b(T, \tau, \tau + \Delta\tau; 0, \beta) \cdot X(T)}] \\ &= e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)} g(t, T, x; -b(T, \tau, \tau + \Delta\tau; 0, \beta), 0) \end{aligned}$$

where as before

$$\begin{aligned} a(T, \tau, \tau + \Delta\tau; \alpha, \beta) &:= a(T, \tau + \Delta\tau; \alpha, \beta) - a(T, \tau; \alpha, \beta) \\ b(T, \tau, \tau + \Delta\tau; \alpha, \beta) &:= b(T, \tau + \Delta\tau; \alpha, \beta) - b(T, \tau; \alpha, \beta). \end{aligned}$$

Thus

$$F_t = 100 \left(1 - \frac{1}{\Delta\tau} (e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)} g(t, T, x; -b(T, \tau, \tau + \Delta\tau; 0, \beta), 0) - 1) \right)$$

is the quoted futures price at time $t = T_0$. It becomes

$$F_t^\Lambda = 100 \left(1 - \frac{1}{\Delta\tau} (\Lambda(\tau, \tau + \Delta\tau)^{-1} e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)} g(t, T, x; -b(T, \tau, \tau + \Delta\tau; 0, \beta), 0) - 1) \right)$$

if the fudge factor is reinserted.

Exact Solution

Let

$$f_i := \Delta\tau (1 - F_i/100) + 1$$

Then the final contract price and martingality under the one day forward measures imply

$$\begin{aligned} f_N &= \frac{B(T, \tau)}{B(T, \tau + \Delta\tau)} \\ &= e^{a(T, \tau, \tau + \Delta\tau; 0, \beta)} e^{b(T, \tau, \tau + \Delta\tau; 0, \beta) \cdot X(T)} \\ f_{i-1} &= B(T_{i-1}, T_i)^{-1} \mathbb{E} \left[\exp \left(\int_{T_{i-1}}^{T_i} r(u) du \right) f_i \right] \\ &= e^{a(T_{i-1}, T_i; 0, \beta)} e^{b(T_{i-1}, T_i; 0, \beta) \cdot X_{i-1}} \mathbb{E} \left[\exp \left(\int_{T_{i-1}}^{T_i} r(u) du \right) f_i \right] \end{aligned}$$

Thus

$$\tilde{f}_i := e^{-a(T, \tau, \tau + \Delta\tau; 0, \beta)} e^{-a(T_{i-1}, T_i; 0, \beta)} f_i$$

is a forward :

$$\begin{aligned} \tilde{f}_N &= e^{p \cdot X(T)} \\ \tilde{f}_{i-1} &= e^{l_{i-1} \cdot X_{i-1}} \mathbb{E} \left[\exp \left(\int_{T_{i-1}}^{T_i} r(u) du \right) e^{m_i \cdot X_i} \tilde{f}_i \right] \end{aligned}$$

where

$$\begin{aligned} p &= b(T, \tau, \tau + \Delta\tau; 0, \beta) \\ l_i &= b(T_i, T_{i+1}, 0, \beta) \\ m_i &= 0. \end{aligned}$$

The solution is therefore affine :

$$\begin{aligned}\tilde{f}_i &= e^{\tilde{a}_i + \tilde{b}_i \cdot X_i} \\ \tilde{a}_N &= 0, \\ \tilde{a}_{i-1} &= \tilde{a}_i - a(T_{i-1}, T_i; -\tilde{b}_i, \beta), \quad 1 \leq i \leq N, \\ \tilde{b}_N &= p, \\ \tilde{b}_{i-1} &= \tilde{b}_i - b(T_{i-1}, T_i; -\tilde{b}_i, \beta), \quad 1 \leq i \leq N.\end{aligned}$$

and using the time additive properties of $a(t_1, t_2; \alpha, \beta)$ it is not hard to see that the function f can be expressed

$$\begin{aligned}f_i &= e^{a_i + b_i \cdot X_i} \\ a_N &= -a(T, \tau, \tau + \Delta\tau; 0, \beta), \\ a_{i-1} &= a_i + a(T_{i-1}, T_i; 0, \beta) - a(T_{i-1}, T_i; -b_i, \beta), \quad 1 \leq i \leq N, \\ b_N &= b(T, \tau, \tau + \Delta\tau; 0, \beta), \\ b_{i-1} &= b(T_{i-1}, T_i; 0, \beta) - b(T_{i-1}, T_i; -b_i, \beta), \quad 1 \leq i \leq N.\end{aligned}$$

Rearranging, the futures price at time $t = T_0$ with state vector $x = X(T_0)$ is given by

$$F_t = 100 \left(1 - \frac{1}{\Delta\tau} (e^{a_0 + b_0 \cdot x} - 1) \right),$$

or

$$F_t = 100 \left(1 - \frac{1}{\Delta\tau} (\Lambda(\tau, \tau + \Delta\tau)^{-1} e^{a_0 + b_0 \cdot x} - 1) \right)$$

with the fudge factor.

Appendix C

On Adding Jumps

In section 2.2.6 we added exponentially distributed jumps to the diffusion process

$$dx_t = \kappa (\theta - x_t) dt + \sigma dW_t.$$

The moment generating function, as described in section 2.2.6, requires evaluation of the integral

$$I_{\text{vas+j}} = \int_0^t \frac{1}{1 + \mu (\alpha \tau'(s) + \beta \tau(s))} ds$$

and similar integrals involving rational functions of $\tau(t) = \frac{1-e^{-\kappa t}}{\kappa}$. Closed form solutions are available and were used section 2.2.6 and also in the numerical tests in section 2.3. We present the solutions here.

C.1 Integrals involving a certain mononotic function of maturity

$$\begin{aligned}
\int_0^t \frac{1}{1 + \mu\tau(s)} ds &= \frac{\kappa}{\kappa + \mu} [t + \log(1 + \mu\tau(t))] \\
\int_0^t \frac{1}{(1 + \mu\tau(s))^n} ds &= \left(\frac{\kappa}{\kappa + \mu} \right)^n [t + \log(1 + \mu\tau(t))] \\
&\quad + \sum_{j=1}^{n-1} \left(\frac{\kappa}{\kappa + \mu} \right)^{n-j} \frac{1}{j(\kappa + \mu)} \left[1 - \frac{1}{(1 + \mu\tau(t))^j} \right] \\
\int_0^t \left(\frac{a + b\tau(s)}{c + d\tau(s)} \right)^{-1} ds &= \frac{a\kappa + b}{c\kappa + d} t + \frac{1}{b} \frac{ad - bc}{a\kappa + b} \log \left(1 + \frac{b}{a} \tau(t) \right) \\
\int_0^t \left(\frac{a + b\tau(s)}{c + d\tau(s)} \right)^{-n} ds &= \sum_{k=1}^{n-1} f_{k+1} \frac{1}{\kappa(n-k)} \left(\left(\frac{a + b\tau}{c + d\tau} \right)^{-(n-k)} - \left(\frac{a}{c} \right)^{-(n-k)} \right) \\
&\quad + \frac{\kappa(ad + bc) + 2bd}{a\kappa + b} \left(\frac{c\kappa + d}{a\kappa + b} \right)^n t - \frac{1}{\kappa} f_n \log \left(1 + \frac{b}{a} \tau(t) \right),
\end{aligned}$$

where

$$f_k := \left(\frac{d}{b} \right)^k - \left(\frac{c\kappa + d}{a\kappa + b} \right)^k$$

Some of these integrals are calculated by first finding the appropriate recursion relation. For example, (with thanks to Dimitri Kachin for performing similar integrals and saving the author a lot of work) if we let

$$g_n := \left(\frac{a + b\tau}{c + d\tau} \right)^{-n}$$

then

$$I_n := \int_0^t g_n(s) ds = \int_0^t \left(\frac{a + b\tau(s)}{c + d\tau(s)} \right)^{-1} ds$$

satisfies a recursion relation :

$$I_n = \frac{1}{b} \frac{ad - bc}{a\kappa + b} \frac{1}{n-1} \left(\left(\frac{a + b\tau}{c + d\tau} \right)^{-(n-1)} - \left(\frac{a}{c} \right)^{-(n-1)} \right) \\ + \frac{1}{b} \frac{\kappa(ad + bc) + 2bd}{a\kappa + b} I_{n-1} - \frac{d}{b} \frac{c\kappa + d}{a\kappa + b} I_{n-2}$$

with solution

$$I_n = \sum_{k=1}^{n-1} f_{k+1} \frac{1}{\kappa(n-k)} \left(\left(\frac{a + b\tau}{c + d\tau} \right)^{-(n-k)} - \left(\frac{a}{c} \right)^{-(n-k)} \right) \\ + \frac{\kappa(ad + bc) + 2bd}{a\kappa + b} \left(\frac{c\kappa + d}{a\kappa + b} \right)^n t - \frac{1}{\kappa} f_n \log \left(1 - \left(\kappa + \frac{b}{a} \right) \tau \right),$$

where

$$f_k := \left(\frac{d}{b} \right)^k - \left(\frac{c\kappa + d}{a\kappa + d} \right)^k.$$

C.2 The Integral $I_{\text{vas}+j}(t; \mu)$

Using a result above we have

$$\begin{aligned} I(t; \mu) &= \int_0^t \frac{1}{1 + \mu b_{\text{vas}}(t; \alpha, \beta)} ds. \\ &= \int_0^t \frac{1}{1 + \mu (\alpha \tau'(s) + \beta \tau(s))} ds \\ &= \int_0^t \frac{1}{1 + \mu (\alpha (1 - \kappa \tau(s)) + \beta \tau(s))} ds \\ &= \frac{1}{1 + \mu \alpha} \int_0^t \frac{1}{1 + \tilde{\mu} \tau(s)} ds \\ &= \frac{1}{1 + \mu \alpha} \frac{\kappa}{\kappa + \tilde{\mu}} [t + \log(1 + \tilde{\mu} \tau(t))] \\ &= \frac{\kappa}{\kappa + \beta \mu} \left[t + \log \left(1 + \mu \frac{\beta - \kappa \alpha}{1 + \mu \alpha} \tau(t) \right) \right] \end{aligned}$$

where we've made use of

$$\tilde{\mu} = \mu \frac{\beta - \kappa\alpha}{1 + \mu\alpha}.$$

Bibliography

- [1] Yacine Ait-Sahalia. Testing continuous-time models of the spot interest rate. *Review of Financial Studies*, 9(385):42, 1996.
- [2] Yacine Ait-Sahalia and Andrew Lo. Transition densities for interest rate and other nonlinear diffusions. *Journal of Finance*, 1999.
- [3] E. I. Altman. Financial ratio, discriminant analysis and the prediction of bankruptcy. *Journal of Finance*, 23:589–609, 1968.
- [4] Andrew Ang and Geert Bekaert. Regime switches in interest rates. Working paper, 2000.
- [5] Ravi Bansal and Hao Zhou. Term structure of interest rates with regime shifts. Working paper, 2000.
- [6] D. Bates. The Crash of '87: Was it expected? The evidence from options markets. *J. Fin.*, 46(3):1009–1044, July 1991.
- [7] D. Bates. Testing option pricing models. In G. Maddala and C. Rao, editors, *Statistical Methods in Finance*, volume 14 of *Handbook of Statistics*, chapter 20, pages 567–611. North Holland, Amsterdam, 1996.
- [8] Geert Bekaert, Robert J. Hodrick, and David A. Marshall. “peso problem” explanations for term structure anomalies. Working Paper, 1997.
- [9] A. Brace, D. Gararek, and M. Musiela. The market model of interest rate dynamics. *Mathematical Finance*, 7:127–155, 1997.

- [10] R. H. Brown and S. M. Schaefer. Interest rate volatility and the shape of the term structure. *Philosophical Transactions of the Royal Society of London A*, 347:563–576, 1994.
- [11] Ren-Raw Chen and Louis Scott. Pricing interest rate options in a two-factor Cox-Ingersoll-Ross model of the term structure. *The Review of Financial Studies*, 5(4):613–636, 1992.
- [12] Peter Cotton, Jean-Pierre Fouque, George Papanicolaou, and K. Ronnie Sircar. Stochastic volatility corrections for interest rate derivatives. Working Paper.
- [13] J. Cox, J. Ingersoll, and S. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985.
- [14] Qiang Dai and Kenneth Singleton. Specification analysis of affine term structure models. *Journal of Finance*, LV:1943–1978, 2000.
- [15] Satyajit Das. *Credit Derivatives*. Wiley, 1998.
- [16] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the em algorithm. *Proceedings of the Royal Statistical Society*, 1976.
- [17] Jefferson Duarte. The relevance of the market price of risk in affine term structure models. Working Paper, 2000.
- [18] D. Duffie. *Dynamic Asset Pricing Theory*. Princeton University Press, 2nd edition, 1996.
- [19] D. Duffie and R. Kan. A yield-factor model of interest rates. *Math. Finance*, 6(4):379–406, 1996.
- [20] D. Duffie, J. Pan, and K. Singleton. Transform analysis and option pricing for affine jump-diffusions. Technical report, Graduate School of Business, Stanford University, 1998. In preparation.
- [21] Darrell Duffie. Credit swap valuation. *Financial Analyst's Journal*, Forthcoming.

- [22] Darrell Duffie and David Lando. Term structure of credit spreads with incomplete accounting information. Forthcoming, *Econometrica*.
- [23] Darrell Duffie, Jun Pan, and Kenneth Singleton. Option pricing and transform analysis for affine jump diffusions. Working paper, Graduate School of Business, Stanford University, 1998.
- [24] Darrell Duffie and Kenneth Singleton. An econometric model of the term structure of interest rate swap yields. *Journal of Finance*, 52:1287–1321, 1997.
- [25] W. Efron. Bootstrap methods - another look at the jack-knife. *Ann. Statist.*, 7:1–26, 1979.
- [26] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, 1991.
- [27] B. Flesaker and L. P. Hughston. Positive interest. *Risk*, 9(1):46–49, 1996.
- [28] J.-P. Fouque, G. Papanicolaou, and K.R. Sircar. Financial modeling in a fast mean-reverting stochastic volatility environment. *Asia-Pacific Financial Markets*, 6(1):37–48, 1999.
- [29] J.-P. Fouque, G. Papanicolaou, and K.R. Sircar. From the implied volatility skew to a robust correction to Black-Scholes American option prices. August 1999. Preprint.
- [30] J.-P. Fouque, G. Papanicolaou, and K.R. Sircar. Mean-Reverting Stochastic Volatility. *International Journal of Theoretical and Applied Finance*, 1999. To appear.
- [31] Jean-Pierre Fouque, George Papanicolaou, and K. Ronnie Sircar. *Derivative in Financial Markets with Stochastic Volatility*. Cambridge University Press, 2000.
- [32] Jean-Pierre Fouque, George Papanicolaou, and K. Ronnie Sircar. Stochastic volatility correction to black-scholes. *Risk*, 13(2), 2000.

- [33] Jean-Pierre Fouque, George Papanicolaou, K. Ronnie Sircar, and Knut Solna. Mean-reversion of s&p 500 volatility. Working Paper.
- [34] E. Ghysels, A. Harvey, and E. Renault. Stochastic volatility. In G. Maddala and C. Rao, editors, *Statistical Methods in Finance*, volume 14 of *Handbook of Statistics*, chapter 5, pages 119–191. North Holland, Amsterdam, 1996.
- [35] R. Goldstein. The term structure of interest rates as a random field. Technical report, Fisher College of Business, Ohio State University, 1997.
- [36] J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57:357–384, 1989.
- [37] Asbjorn T. Hansen and Rolf Poulsen. A simple regime switching term structure model. *Finance and Stochastics*, 4:409–200, 2000.
- [38] Lars Peter Hansen and Jose Alexandre Scheinkman. Back to the future: Generating moment implications for continuous-time markov processes. Technical Working Paper No. 141, 1993.
- [39] Hau He. Modeling term structures of swap spreads. Working Paper, Yale School of Management.
- [40] D. Heath, R. A. Jarrow, and A. J. Morton. Bond pricing and the term structure of interest rates: A new methodology for contingent claims valuation. *Econometrica*, 60:77–105, 1992.
- [41] J. Hull and A. White. The Pricing of Options on Assets with Stochastic Volatilities. *J. Finance*, XLII(2):281–300, June 1987.
- [42] J. Hull and A. White. Pricing interest rate derivative securities. *J. Finance*, 3, 1990.
- [43] Jessica James and Nick Webber. *Interest Rate Modelling*. Wiley, 2000.
- [44] F. Jamshidian. A simple class of square root interest rate models. *Applied Mathematical Finance*, 2:61–72, 1995.

- [45] R.A. Jarrow and S. M. Turnbull. Pricing derivatives on financial securities subject to credit risk. *Journal of Finance*, 50:53–85, 1995.
- [46] David Lando. *Three Essays on Contingent Claims Pricing*. PhD thesis, Cornell University, 1994.
- [47] R. Lee. Local volatilities under stochastic volatility. *International Journal of Theoretical and Applied Finance*, 1999. To appear.
- [48] K. Lewis. Was there a “peso problem” in the u.s. term structure of interest rates: 1979-1982? *International Economic Review*, 32:159–173, 1991.
- [49] M. Musiela and M. Rutkowski. *Martingale Methods in Financial Modelling*. Springer-Verlag, 1997.
- [50] George Papanicolaou and K. Ronnie Sircar. Stochastic volatility, smile and asymptotics. *Applied Mathematical Finance*, 1999.
- [51] Monika Piazzesi. An econometric model of the term structure with macroeconomic jump effects. Working paper.
- [52] Mark A Pinsky. *Lectures on Random Evolution*. World Scientific, 1991.
- [53] Riccardo Rebonato. *Interest-Rate Option Models*. Wiley, 1996.
- [54] Fan Yu Robert A. Jarrow, David Lando. Default risk and diversification: Theory and applications. Working Paper, 2001.
- [55] E. Schlögl and D. Sommer. Factor models and the shape of the term structure. *University of Bonn discussion paper*, (395), 1997.
- [56] Silverman. Using kernel density estimates to investigate multimodality. *J. Royal Statistical Soc. B*, 43:97–99, 1981.
- [57] O. Vasicek. An equilibrium characterization of the term structure. *J. Financial Econ.*, 5:177–188, 1977.